

No. 25-13631

**IN THE UNITED STATES COURT OF APPEALS
FOR THE ELEVENTH CIRCUIT**

CITADEL SECURITIES LLC,
Petitioner,

v.

SECURITIES AND EXCHANGE COMMISSION,
Respondent,

INVESTORS EXCHANGE, LLC,
Intervenor.

On Petition for Review of an Order
of the Securities and Exchange Commission

SUPPLEMENTAL APPENDIX

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both NYSE and FINRA or NYSE Amex and FINRA. Accordingly, the Plan promotes efficiency by reducing costs to Common Members. Furthermore, because FINRA, NYSE, and NYSE Amex will coordinate their regulatory functions in accordance with the Plan, the Plan should promote investor protection and the public interest.

In particular, the Commission notes that, under the proposed Plan, FINRA, NYSE, and NYSE Amex have allocated regulatory responsibility for Common Rules to the extent that such responsibilities involve member firm regulation. The Plan also sets forth those areas for which NYSE and NYSE Amex will retain regulatory responsibility, including: examinations of conduct or actions by a Common Member covered by NYSE-only or NYSE Amex-only rules and/or by related Federal laws or regulations; surveillance, investigation, and enforcement with respect to conduct or action relating to trading on or through the systems and facilities of NYSE or NYSE Amex and conduct otherwise covered by NYSE-only or NYSE Amex-only rules, as well as whether such conduct may constitute a violation of Federal laws or regulations; processing of applications for trading licenses or other membership in NYSE or NYSE Amex; and qualification and registration of member firm personnel to effect transactions or work on the floor of NYSE or NYSE Amex pursuant to such SRO's rules.²⁷

In addition, the proposed Plan provides that NYSE and NYSE Amex will retain regulatory responsibility for the application of any Common Rule as it pertains to matters other than member firm regulation, including matters relating to such SRO's retained responsibilities as set forth in the Plan (the "Non-Exclusive Common Rules"). The Non-Exclusive Common Rules are specifically annotated in the List of Common Rules and include those rules for which FINRA, NYSE, and NYSE Amex will each bear their respective regulatory responsibilities, consistent with the scope of the 17d-2 Plan. Such rules are "non-exclusive" in the sense that they may relate to member firm regulation (for which FINRA would assume regulatory responsibility) as well as matters other than member firm regulation (for which NYSE or NYSE Amex would retain regulatory responsibility). Accordingly, NYSE and NYSE Amex will each bear responsibility for the application of their Non-Exclusive Common Rules

concerning their particular regulatory responsibilities.

According to the Plan, whenever any Party seeks to make a change to any of its rules that are Common Rules, before filing a proposed rule change with the Commission, it will inform the other Parties of the intended change to determine whether the other Parties will propose a conforming change to its version of the Common Rule. If the Parties do not agree to propose conforming changes, the Parties agree to file with the Commission an amendment to the 17d-2 Plan to delete such rule from the list of Common Rules.²⁸ Finally, the proposed Plan requires the Parties annually (or more frequently if required by changes in the rules of a Party) to confirm in writing the accuracy of the list of Common Rules.²⁹ This provision ensures that the Parties keep the Common Rules up-to-date vis-à-vis the other Parties and should facilitate the ability of the Parties to accurately administer their responsibilities under the proposed Plan consistent with the scope of the Plan declared effective by the Commission herein.

The proposed Plan also requires the Parties to share information on a number of matters, including, for example, financial and operational matters of Common Members, third-party complaints, and disciplinary actions.³⁰ The Commission believes that the information-sharing provisions contained in the proposed Plan fosters cooperation and coordination among the Parties, thereby promoting investor protection and removing impediments to the development of a national market system.

Finally, the Plan permits any Party to terminate the Plan at any time, subject to 180 days written notice to the other Parties and subject to Commission approval.³¹

V. Conclusion

This Order gives effect to the Plan filed with the Commission in File No. 4-587. The Parties shall notify all members affected by the Plan of their rights and obligations under the Plan.

It is therefore ordered, pursuant to Section 17(d) of the Act,³² that the Plan, made by and among NYSE, NYSE Regulation, NYSE Amex, and FINRA, that is contained in File No. 4-587 and

filed pursuant to Rule 17d-2, is hereby approved and declared effective.

It is therefore ordered that NYSE and NYSE Amex are relieved of those responsibilities allocated to FINRA under the Plan in File No. 4-587.

For the Commission, by the Division of Trading and Markets, pursuant to delegated authority:³³

Florence E. Harmon,
Deputy Secretary.

[FR Doc. E9-18762 Filed 8-5-09; 8:45 am]
BILLING CODE 8010-01-P

SECURITIES AND EXCHANGE COMMISSION

[Release No. 34-60405; File No. 4-546]

Joint Industry Plan; Order Approving the National Market System Plan Relating to Options Order Protection and Locked/Crossed Markets Submitted by the Chicago Board Options Exchange, Incorporated, International Securities Exchange, LLC, The NASDAQ Stock Market LLC, NASDAQ OMX BX, Inc., NASDAQ OMX PHLX, Inc., NYSE Amex LLC, and NYSE Arca, Inc.

July 30, 2009.

I. Introduction

The proposed Options Order Protection and Locked/Crossed Market Plan ("Proposed Plan") was filed jointly, pursuant to Rule 608 of Regulation NMS under the Securities Exchange Act of 1934 ("Act") ("Regulation NMS") ("Rule 608"),¹ by the International Securities Exchange, LLC ("ISE") and NYSE Arca, Inc. ("NYSE Arca") on September 13, 2007 and September 18, 2007, respectively, with the Securities and Exchange Commission ("Commission").² On December 11, 2007, ISE and NYSE Arca separately filed Amendment No. 1 to the Proposed Plan.³ On April 24, 2008, and April 17, 2008, ISE and NYSE Arca, respectively, filed Amendment No. 2 to the Proposed Plan.⁴ On November 10,

¹ 17 CFR 200.30-3(a)(34).

² 17 CFR 242.608.

³ See letter from Michael J. Simon, General Counsel, ISE, to Nancy M. Morris, Secretary, Commission, dated September 12, 2007 ("ISE Letter 1"); and letter from Peter G. Armstrong, Managing Director, Options, NYSE Arca, to Nancy M. Morris, Secretary, Commission, dated September 14, 2007 ("NYSE Arca Letter 1").

⁴ See letter from Michael J. Simon, General Counsel, ISE, to Nancy M. Morris, Secretary, Commission, dated December 10, 2007; and letter from Peter G. Armstrong, Managing Director, Options, NYSE Arca, to Nancy M. Morris, Secretary, Commission, dated December 10, 2007.

⁵ Amendment No. 2 superseded Amendment No. 1 and replaced it in its entirety. See letter from

²⁷ See paragraphs 2(d)(i)-(iv) and (e)(i)-(iv) of the proposed 17d-2 Plan.

²⁸ See paragraph 2(a) of the proposed 17d-2 Plan.

²⁹ See paragraph 2(c) of the proposed 17d-2 Plan.

³⁰ See paragraph 5 of the proposed 17d-2 Plan.

³¹ See paragraph 14 of the proposed 17d-2 Plan. The Commission notes that, as reflected in paragraph 14, Commission approval is required for any Party to terminate its participation in the Plan.

³² 15 U.S.C. 78q(d).

2008 and October 31, 2008, ISE and NYSE Arca, respectively, filed Amendment No. 3 to the Proposed Plan.⁵ On April 30, 2008, May 8, 2008, June 18, 2008, June 18, 2008, and July 9, 2008, respectively, Chicago Board Options Exchange, Incorporated ("CBOE"), The NASDAQ Stock Market LLC ("Nasdaq"), American Stock Exchange LLC ("Amex") (f/k/a NYSE Alternext US LLC, "NYSE Alternext," n/k/a NYSE Amex LLC, "NYSE Amex"), Philadelphia Stock Exchange, Incorporated (n/k/a NASDAQ OMX PHILX, Inc., "Phlx"), and Boston Stock Exchange, Inc. ("BSE") (n/k/a NASDAQ OMX BX, Inc., "BX" and together with ISE, NYSE Arca, CBOE, Nasdaq, Amex, and Phlx, the "Proposing Exchanges") filed with the Commission the Proposed Plan.⁶ On November 25, 2008, November 26, 2008, December 2, 2008, December 4, 2008, and December 5, 2008, CBOE, NYSE Alternext, BSE, Phlx, and Nasdaq, respectively, filed Amendment No. 1 to the Proposed Plan.⁷ On April 2, 2009, a detailed

Michael J. Simon, General Counsel, ISE, to Nancy M. Morris, Secretary, Commission, dated April 16, 2008; and letter from Peter G. Armstrong, Managing Director, Options, NYSE Arca, to Nancy M. Morris, Secretary, Commission, dated April 16, 2008.

⁵ See letter from Michael J. Simon, General Counsel, ISE, to Florence Harmon, Acting Secretary, Commission, dated November 7, 2008 ("ISE Letter 2"); and letter from Peter G. Armstrong, Managing Director, Options, NYSE Arca, to Florence Harmon, Acting Secretary, Commission, dated October 30, 2008 ("NYSE Arca Letter 2").

⁶ In their respective filings of the Proposed Plan, Amex, BSE, CBOE, Nasdaq, and Phlx incorporated the changes made by ISE and NYSE Arca in Amendment No. 2. See letters from Jeffrey P. Burns, Vice President and Associate General Counsel, Amex, to Nancy M. Morris, Secretary, Commission, dated June 17, 2008 ("Amex Letter 1"); Bruce Goodhue, Chief Regulatory Officer, BSE, to Florence Harmon, Acting Secretary, Commission, dated July 8, 2008 ("BSE Letter 1"); Edward J. Joyce, President and Chief Operating Officer, CBOE, to Nancy M. Morris, Secretary, Commission, dated April 29, 2008 ("CBOE Letter 1"); Jeffrey S. Davis, Vice President and Deputy General Counsel, The NASDAQ OMX Group, Inc., to Nancy M. Morris, Secretary, Commission, dated May 7, 2008 ("Nasdaq Letter 1"); and Richard S. Rudolph, Vice President and Counsel, Phlx, to Nancy M. Morris, Secretary, Commission, dated June 17, 2008 ("Phlx Letter 1").

⁷ In their respective Amendment No. 1 to the Proposed Plan, BSE, CBOE, NYSE Alternext, Phlx, and Nasdaq made changes identical to those made by ISE and NYSE Arca in Amendment No. 3. See letters from Edward J. Joyce, President and Chief Operating Officer, CBOE, to Florence Harmon, Acting Secretary, Commission, dated November 25, 2008 ("CBOE Letter 2"); Jeffrey P. Burns, Managing Director, NYSE Alternext, to Florence Harmon, Acting Secretary, Commission, dated November 25, 2008 ("Amex Letter 2"); John Katovich, Vice President, BSE, to Florence Harmon, Acting Secretary, Commission, dated December 1, 2008 ("BSE Letter 2"); Richard S. Rudolph, Vice President and Counsel, Phlx, to Florence Harmon, Acting Secretary, Commission, dated December 3, 2008 ("Phlx Letter 2"); and Jeffrey S. Davis, Vice President and Deputy General Counsel, The

summary of the Proposed Plan was published for comment in the **Federal Register**.⁸

The Commission received one comment on the Proposed Plan.⁹

This order approves the Proposed Plan, with changes as the Commission deems necessary or appropriate, thus authorizing CBOE, ISE, Nasdaq, BX, Phlx, Amex, and NYSE Arca to act jointly to implement the Proposed Plan, as modified herein, as a means of facilitating a national market system in accordance with the requirements of Section 11A of the Act.¹⁰

II. Background

A. Section 11A of the Act

In 1975, Congress directed the Commission, through the enactment of Section 11A of the Act,¹¹ to facilitate the establishment of a national market system to link together the individual markets that trade securities. Congress found the development of a national market system to be in the public interest and appropriate for the protection of investors and the maintenance of fair and orderly markets to assure fair competition among the exchange markets.¹² Section 11A(a)(3)(B) of the Act directs the Commission, "by rule or order, to authorize or require self-regulatory organizations to act jointly with respect to matters as to which they share authority under this title in planning, developing, operating, or regulating a national market system (or a subsystem thereof) or one or more facilities."¹³ The Commission's approval of a national market system plan is conditioned upon

NASDAQ OMX Group, Inc., to Florence Harmon, Acting Secretary, Commission, dated December 4, 2008 ("Nasdaq Letter 2").

⁸ Securities Exchange Act Release No. 59647 (March 30, 2009), 74 FR 15010 (File No. 4-546) ("Proposed Plan Notice"). The full text of the Proposed Plan submitted by the Proposing Exchanges, is available on the Commission's Web site at <http://sec.gov/rules/sro/nms/nmsarchive/nms2007.shtml#4-546>, at each Proposing Exchange, and at the Commission's Public Reference Room.

⁹ Letter from John C. Nagel, Managing Director & Deputy General Counsel, Citadel Investment Group L.L.C. ("Citadel") to Nancy M. Morris, Secretary, Commission, dated July 18, 2008 ("Citadel Letter"). The Citadel Letter cited to Citadel's comments made in a letter from John C. Nagel, Managing Director & Deputy General Counsel, Citadel to Nancy M. Morris, Secretary, Commission, dated July 15, 2008 (Petition for Rulemaking to Address Excessive Access Fees in the Options Markets) ("Petition for Rulemaking").

¹⁰ 15 U.S.C. 78k-1. See also 17 CFR 242.608(b)(2). The approved Options Order Protection and Locked/Crossed Market Plan, which incorporates the changes the Commissions deems necessary or appropriate, is attached here as **Appendix A** and is referred to herein as the "Options Linkage Plan."

¹¹ 15 U.S.C. 78k-1.

¹² 15 U.S.C. 78k-1(a)(1)(C).

¹³ 15 U.S.C. 78k-1(a)(3)(B).

a finding that the proposed plan is "necessary or appropriate in the public interest, for the protection of investors and the maintenance of fair and orderly markets, to remove impediments to, and perfect the mechanism of, a national market system, or otherwise in furtherance of the purposes of the Act."¹⁴

B. Current Plan

Currently, the Proposing Exchanges are signatories to the Plan for the Purpose of Creating and Operating an Intermarket Option Linkage ("Current Plan"). The Current Plan is a national market system plan linking its participants. The Commission approved the Current Plan on July 28, 2000.¹⁵ Subsequently, both Pacific Exchange, Inc. (n/k/a "NYSE Arca") and Phlx submitted proposed amendments to the Current Plan to become participants to the Current Plan. These proposed amendments were approved on November 16, 2000.¹⁶ On February 5, 2004, BSE's proposed amendment to become a participant to the Current Plan became effective.¹⁷ Further, Nasdaq's proposed amendment to become a participant to the Current Plan became effective on March 21, 2008.¹⁸

The Current Plan requires its participants to avoid, absent reasonable justification and during normal market conditions, trading at a price inferior to that displayed on another market ("trade-through").¹⁹ The Current Plan provides for several exceptions to trade-through liability, including, among other things, systems malfunction, failure of the receiving market to respond to an incoming order within 30 seconds, failure of the market traded through to complain within the specified time period, complex trades, trading rotations, and non-firm quotations on the market that was traded through.²⁰ The Current Plan also provides a mechanism by which a member of a participating exchange

¹⁴ 17 CFR 242.608(b)(2).

¹⁵ See Securities Exchange Act Release No. 43086 (July 28, 2000), 65 FR 48023 (August 4, 2000) (File No. 4-429).

¹⁶ See Securities Exchange Act Release Nos. 43573 (November 16, 2000), 65 FR 70851 (November 28, 2000) (File No. 4-429) and 43574 (November 16, 2000), 65 FR 70850 (November 28, 2000) (File No. 4-429).

¹⁷ See Securities Exchange Act Release No. 49198 (February 5, 2004), 69 FR 7029 (February 12, 2004) (File No. 4-429).

¹⁸ See Securities Exchange Act Release No. 57545 (March 21, 2008), 73 FR 16394 (March 27, 2008) (File No. 4-429).

¹⁹ Section 8(c) of the Current Plan.

²⁰ Section 8(c)(iii) of the Current Plan.

could seek satisfaction if a customer order is traded through.²¹

Under the Current Plan, its participants agree that the dissemination of “locked” or “crossed” markets should be avoided, and, if their members lock or cross a market, they should take remedial actions to unlock or uncross such market.²² Further, the Current Plan contains provisions to address trade comparison, clearing, trading halts, non-firm quotations, and administration of the Current Plan.²³ Except with respect to the addition of new participants and the withdrawal of current participants, any proposed change to the Current Plan must be approved unanimously by its participants.²⁴

The participating exchanges comply with the requirements of the Current Plan, including the prohibition against trade-throughs, by utilizing a stand alone system (“Linkage Hub”) to send and receive specific order types. The Linkage Hub is a centralized data communications network that electronically links the options exchanges to one another. The Options Clearing Corporation (“OCC”) operates the Linkage Hub.²⁵

There are three defined order types under the Current Plan that its participants could route through the Linkage Hub to limit trade-throughs: orders represented by eligible market makers on behalf of customers (“Principal Acting as Agent Orders” or “P/A Orders”);²⁶ orders for the principal accounts of market makers and specialists (“Principal Orders”);²⁷ and orders intended to satisfy trade-through liabilities (“Satisfaction Orders”).²⁸ Non-market-maker broker-dealers do not have access to the Linkage Hub.

C. Proposed Plan

The Proposing Exchanges are now seeking approval of an alternative linkage plan, the Proposed Plan. As described in more detail below, the Proposed Plan would not require a central linkage mechanism akin to the Current Plan’s Linkage Hub, and would introduce certain new features to

linkages between options markets, including an Intermarket Sweep Order (“ISO”) similar to that available for NMS stocks under Regulation NMS.²⁹

III. Discussion

As discussed above, in 1975, Congress directed the Commission, through the enactment of Section 11A of the Act,³⁰ to facilitate the development of a national market system consistent with the objectives of the Act. In particular, Section 11A(a)(3)(B) of the Act³¹ authorizes the Commission “by rule or order, to authorize or require self-regulatory organizations to act jointly with respect to matters as to which they share authority under this title in planning, developing, operating, or regulating a national market system (or a subsystem thereof) or one or more facilities.” Rule 608 establishes the procedures for filing, amending, and approving a national market system plan. Approval of such a plan is conditioned upon a finding that the proposed plan “is necessary or appropriate in the public interest, for the protection of investors and the maintenance of fair and orderly markets, to remove impediments to, and perfect the mechanisms of, a national market system, or otherwise in furtherance of the purposes of the Act.”³²

After careful review, the Commission has determined to approve the Proposed Plan, pursuant to Section 11A(a)(3)(B) of the Act³³ and Rule 608 thereunder,³⁴ with changes set forth herein as the Commission has deemed necessary and appropriate.³⁵ Specifically, the Commission finds that changes to the Proposed Plan set forth herein are necessary and appropriate in the public interest. The Commission further finds that the Options Linkage Plan is in furtherance of the purposes of the Act in that it requires the protection of the best priced displayed quotes and avoidance and reconciliation of locked and crossed markets, and thus is necessary and appropriate in the public interest, for the protection of investors and the maintenance of fair and orderly markets, to remove impediments to, and

perfect the mechanisms of, a national market system.³⁶

The Commission believes that Proposed Plan’s decentralized structure will allow the Proposing Exchanges to take advantage of new technology that allow for efficient routing and executions. The Proposed Plan will give the Proposing Exchanges greater flexibility for order handling as it would allow the exchanges to utilize private linkages, instead of requiring each Proposing Exchange to connect to, and participate in the maintenance of, a centralized hub. In addition, the Proposed Plan would permit the use of ISOs in the options markets. As such, the Proposed Plan would allow the Proposing Exchanges to move towards the market structure approved by the Commission for NMS stocks under Regulation NMS.³⁷ The Commission believes that the Options Linkage Plan will allow the Proposing Exchanges to update the way in which they accomplish effective quote protection and locked and crossed market reconciliation. For the reasons described above, the Commission believes that these provisions of the Options Linkage Plan will provide benefits to the options markets, including the Proposing Exchanges and market participants generally.

In its comment letter on the Proposed Plan, Citadel referenced the comments it made with regard to access fees in the options markets in its Petition for Rulemaking.³⁸ There, Citadel encouraged the Commission to institute a rulemaking proceeding to limit the fees that options exchanges may charge non-members to obtain access to quotations.³⁹ Commission staff is currently considering Citadel’s petition.

A. Order Protection

1. Requirement of Reasonable Policies and Procedures

The Options Linkage Plan requires each Participant⁴⁰ to establish,

²¹ 17 CFR 242.608(b)(2).

²² See *supra* note 29.

²³ See Citadel Letter, *supra* note 9.

²⁴ See Petition for Rulemaking, *supra* note 9.

⁴⁰ The Options Linkage Plan defines “Participant” to mean an Eligible Exchange whose participation in the plan has become effective pursuant to Section 3(c) of the Options Linkage Plan. See Section 2(15) of the Options Linkage Plan. The Options Linkage Plan defines “Eligible Exchange” to mean a national securities exchange registered with the Commission in accordance with Section 6(a) of the Act that, among other things, is a Participant Exchange in OCC (as that term is defined in Section VII of the OCC by-laws) and is a party to the OPRA Plan (as that term is described in Section I of the OPRA Plan). “OPRA Plan” means the plan filed by the Options Price Reporting Authority with the Commission pursuant to Section 11A(a)(1)(C)(iii) of the Act and approved by the

²¹ Section 8(c)(ii) of the Current Plan.

²² Section 7(a)(i)(C) of the Current Plan.

²³ Sections 5, 9, and 10 of the Current Plan.

²⁴ Section 5(c)(i) of the Current Plan.

²⁵ See ISE Letter 2 and NYSE Arca Letter 2, *supra* note 5; see also Amex Letter 2, BSE Letter 2, CBOE Letter 2, Nasdaq Letter 2, and Phlx Letter 2, *supra* note 7.

²⁶ Sections 2(16)(a) and 7(a)(ii)(A), (B) of the Current Plan.

²⁷ Sections 2(16)(b) and 7(a)(ii)(C) of the Current Plan.

²⁸ Sections 2(16)(c) and 7(a)(ii)(D) of the Current Plan.

²⁹ See Securities Exchange Act Release No. 51808 (June 9, 2005), 70 FR 37496 (June 29, 2005) (File No. S7-10-04) (“NMS Release”); 17 CFR 242.600 *et seq.*

³⁰ 15 U.S.C. 78k-1.

³¹ 15 U.S.C. 78k-1(a)(3)(B).

³² 17 CFR 242.608.

³³ 15 U.S.C. 78k-1(a)(3)(B).

³⁴ 17 CFR 242.608.

³⁵ The Commission has modified the Proposed Plan to amend Section 7 of the Proposed Plan relating to the implementation date of the plan (see *infra* notes 140–143 and accompanying text).

maintain, and enforce written policies and procedures as approved by the Commission that are reasonably designed to prevent Trade-Throughs in that Participant's market in Eligible Options Classes.⁴¹ A "Trade-Through"⁴² is defined as a transaction in an option series, either as principal or agent, at a price that is lower than a Protected Bid or higher than a Protected Offer. A "Protected Bid" or a "Protected Offer"⁴³ means a bid or offer in an option series that is displayed by an Eligible Exchange, is disseminated pursuant to the OPRA Plan, and is the Best Bid or Best Offer of an Eligible Exchange. A "Best Bid" or "Best Offer"⁴⁴ means the highest bid price or the lowest offer price communicated by a member of an Eligible Exchange to any broker-dealer or to any customer at which such member is willing to buy or sell, either as principal or agent.

The Options Linkage Plan also requires each Participant to agree to conduct surveillance of its market on a regular basis to ascertain the effectiveness of the policies and procedures to prevent Trade-Throughs and to take prompt action to remedy deficiencies in such policies and procedures.⁴⁵

As is the case currently for NMS stocks under Regulation NMS,⁴⁶ the Commission believes the Options Linkage Plan's policies and procedures-based approach to preventing Trade-Throughs in options is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets, and is consistent with Section 11A(a)(1)(C) of the Act.⁴⁷ The requirement in Section 5(a)(i) of the Options Linkage Plan is virtually identical to the requirement in Rule 611(a) of Regulation NMS. The Commission expects the Participants in the Options Linkage Plan will establish, maintain, and enforce written policies and procedures comparable to those established, maintained and enforced by the market centers subject to Rule

611(a). The Commission believes that a policies and procedures-based approach to preventing Trade-Throughs in options is reasonable given the increasingly high volume of trading in options, and the latencies and other discrepancies in the delivery and receipt of quotation data. The requirement of written policies and procedures, as well as the responsibility assigned to Participants to regularly surveil to ascertain the effectiveness of their procedures and take prompt remedial steps, is designed to achieve the objective of eliminating all Trade-Throughs that reasonably can be prevented, while also recognizing the inherent difficulties of eliminating Trade-Through transactions that, despite a Participant's reasonable efforts, may occur.

The Commission believes that each Participant's policies and procedures must enable it to monitor, on a real-time basis, the Protected Quotations displayed by Eligible Exchanges so as to determine the prices at which the Participant can and cannot execute trades. In addition, the Commission believes that a Participant's policies and procedures must establish objective standards and parameters governing its use of the exceptions set forth in Section 5(b) of the Options Linkage Plan, discussed below, and expects each Participant's order-handling and trading systems to be programmed in accordance with these policies and procedures. Finally, the Participant must take such steps as are necessary to enable it to enforce its policies and procedures effectively. For example, the Commission believes that Participants will need to establish procedures such as regular exception reports to evaluate their trading and order-routing practices. The Commission believes that each Participant Exchange will need to examine such reports to affirm that its policies and procedures have been followed by its personnel and properly coded into its systems and, if not, to promptly identify the reasons and take remedial action.⁴⁸

Participants' obligations under the Options Linkage Plan to maintain and enforce policies and procedures reasonably designed to prevent Trade-Throughs is reinforced by the Options Linkage Plan's explicit assignment of responsibility to Participants to surveil to ascertain the effectiveness of their policies and procedures. Participants cannot merely establish policies and procedures that may be reasonable when created and assume that such policies and procedures continue to

satisfy the requirements of the Options Linkage Plan. Rather, the Commission believes that Participants must regularly assess the continuing effectiveness of their procedures and take prompt action when needed to remedy deficiencies. In particular, Participants must engage in regular surveillance to determine whether Trade-Throughs are occurring without an applicable exception and whether they have failed to implement and maintain policies and procedures that would have reasonably prevented such Trade-Throughs. Further, this requirement is an important element of a Participant's obligations under Rule 608(c) of Regulation NMS, which require that each self-regulatory organization, absent reasonable justification or excuse, enforce compliance with any national market system plan by its members and persons associated with its members.⁴⁹

2. Exceptions to Trade-Throughs

The Options Linkage Plan provides exceptions for certain transactions from the prohibition against Trade-Throughs.⁵⁰ The Options Linkage Plan also provides that, if a Participant relies on an exception, it would be required to establish, maintain, and enforce written policies and procedures reasonably designed to assure compliance with the terms of the exception.⁵¹ Except for the proposed exception for stopped orders and price improvement,⁵² the exceptions in the Options Linkage Plan correspond to trade-through exceptions found in either the Current Plan or in Regulation NMS.⁵³ The Options Linkage Plan includes the following exceptions from the prohibition against Trade-Throughs: system issues;⁵⁴ trading rotations;⁵⁵ crossed markets;⁵⁶ intermarket sweep orders;⁵⁷ quote flickering;⁵⁸ non-firm quotes;⁵⁹ complex trades;⁶⁰ customer stopped orders;⁶¹ stopped orders and price

⁴⁹ 17 CFR 242.608(c).

⁵⁰ See Proposed Plan Notice at 15012, *supra* note 8, for a more detailed description of the proposed Trade-Through exceptions.

⁵¹ Section 5(a)(i) of the Options Linkage Plan.

⁵² Section 5(b)(x) of the Options Linkage Plan.

⁵³ Rule 611 of Regulation NMS, known also as the Order Protection Rule, governs trade-through liability for NMS Stocks. See 17 CFR 242.611.

⁵⁴ Section 5(b)(i) of the Options Linkage Plan.

⁵⁵ Section 5(b)(ii) of the Options Linkage Plan.

⁵⁶ Section 5(b)(iii) of the Options Linkage Plan.

For the definition of a "Crossed Market," see *infra* note 119 and accompanying text.

⁵⁷ Section 5(b)(iv)–(v) of the Options Linkage Plan.

⁵⁸ Section 5(b)(vi) of the Options Linkage Plan.

⁵⁹ Section 5(b)(vii) of the Options Linkage Plan.

⁶⁰ Section 5(b)(viii) of the Options Linkage Plan.

⁶¹ Section 5(b)(ix) of the Options Linkage Plan.

Commission and declared effective as of January 22, 1976, as from time to time amended. See Section 2(14) of the Options Linkage Plan. For the definitions of "Trade-Through," "Best Bid" or "Best Offer," "Locked Market," and "Crossed Market," see *infra* notes 42, 44, and 119 and accompanying texts.

⁴¹ Section 5(a)(i) of the Options Linkage Plan.

⁴² Section 2(21) of the Options Linkage Plan.

⁴³ Section 2(17) of the Options Linkage Plan. Protected Bid and Protected Offer, together are referred to herein as "Protected Quotation." See Section 2(18) of the Options Linkage Plan.

⁴⁴ Sections 2(1) and 2(2) of the Options Linkage Plan.

⁴⁵ Section 5(a)(ii) of the Options Linkage Plan.

⁴⁶ See Rule 611(a) of Regulation NMS (17 CFR 242.611(a)).

⁴⁷ 15 U.S.C. 78k–1(a)(1)(C).

⁴⁸ See NMS Release at 37535, *supra* note 29.

improvement;⁶² and benchmark trades.⁶³

The Commission believes these exceptions will permit a workable intermarket price protection structure for the options market, and are consistent with the principle of price protection. As discussed below, the Commission finds that each of these exceptions is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁶⁴ and believes each assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.⁶⁵

System Issues:⁶⁶ This exception, similar to an exception in the Current Plan, permits a Participant to trade through a Protected Quotation if the Eligible Exchange displaying the Protected Quotation that was traded through was experiencing a failure, material delay, or malfunction of its systems or equipment when the Trade-Through occurred. This exception gives Participants a “self-help” remedy if another Eligible Exchange repeatedly fails to provide an immediate response to incoming orders attempting to access its quotes. As the Commission stated in approving a parallel exception for stocks under Regulation NMS, the Eligible Exchange receiving an order can only be held responsible for its own turnaround time (*i.e.*, from the time it first received an order to the time it transmits a response to the order). Accordingly, the routing exchange will be required to develop policies and procedures that allow for any potential delays in transmission not attributable to the receiving exchange. This exception also covers any failure or malfunction of an Eligible Exchange’s systems or equipment, as well as any material delay.⁶⁷

Participants will need to establish specific objective parameters governing their use of this “self-help” exemption as part of their reasonable policies and procedures. The Commission believes, for example, a single failure to respond within one second generally will not justify future bypassing of another Eligible Exchange’s quotations. Many failures to respond within one second in a short time period, in contrast, clearly will warrant use of the exception. The Commission believes that a Participant making use of this exception must notify the non-responding Eligible

Exchange immediately after (or at the same time as) electing this exception pursuant to reasonable and objective standards contained in its policies and procedures in order to alert the non-responding Eligible Exchange that the Participant intends to make use of this exception with respect to the non-responding Eligible Exchange’s quotes.⁶⁸

The Commission believes that a Participant should be entitled to bypass an away market’s quotations if that market fails to respond to incoming orders attempting to access a displayed quote. The Commission believes that this exception will provide Participants with the necessary flexibility for dealing with problems that occur on an away market during the trading day. Further, the Commission finds that this exception is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁶⁹ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.⁷⁰

Trading Rotations:⁷¹ This exception, which is carried over from the Current Plan⁷² and similar to an exception available for NMS stocks under Regulation NMS,⁷³ permits a Participant to trade through a Protected Quotation disseminated by an Eligible Exchange during a trading rotation. Options exchanges use a trading rotation to open an option for trading or reopen an option after a trading halt.

As noted by the Participants, the trading rotation is effectively a single price auction to price the option,⁷⁴ and there are no practical means to include prices on other exchanges in that auction.⁷⁵ As such, the Commission emphasizes that the exception will not permit a Participant to declare a trading halt merely to be able to circumvent the operation of the Options Linkage Plan’s Trade-Through provisions upon reopening; instead, the Commission believes a Participant must conduct, pursuant to its rules, a formalized and transparent process for executing orders during reopening after a trading halt that involves the queuing and ultimate

execution of multiple orders at a single equilibrium price. In addition, a Participant must have formally declared a trading halt pursuant to its rules. Therefore, the Commission finds that it is reasonable to include this as an exception to the general prohibition on Trade-Throughs as it is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁷⁶ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.⁷⁷

Crossed Markets:⁷⁸ This exception permits a Participant to trade through a Protected Quotation when the market is crossed, and corresponds to an exception for NMS stocks under Regulation NMS.⁷⁹ A Crossed Market occurs when a Protected Bid is higher than a Protected Offer in a given options class. The Commission believes that it is appropriate to permit executions without regard to Trade-Throughs in a Crossed Market because allowing such transactions should permit the market to quickly resolve any unintentional crosses. For the foregoing reasons, the Commission finds that this exception is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁸⁰ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.⁸¹

Intermarket Sweep Orders:⁸² The Options Linkage Plan includes two exceptions from the prohibition against Trade-Throughs for certain transactions involving ISOs. These two exceptions correspond to the exceptions relating to ISOs for NMS stocks under Regulation NMS.⁸³ First, the Options Linkage Plan permits a Participant to execute orders marked as ISOs even when the Participant is not at the national best bid or offer (“NBBO”). Second, a Participant is permitted to execute a transaction when such transaction is not at the NBBO, provided it simultaneously “sweeps” all better priced Protected Quotations by routing an ISO to execute against the full displayed size of any Protected Quotation that was traded through.

⁶² *Id.*

⁶³ 17 CFR 242.608(b)(2).

⁶⁴ 15 U.S.C. 78k-1(a)(1)(C).

⁶⁵ Section 5(b)(ii) of the Options Linkage Plan.

⁶⁶ See Section 8(c)(iii)(E) of the Current Plan.

⁶⁷ See Rule 611(b)(3) of Regulation NMS under the Act (17 CFR 242.611(b)(3)).

⁶⁸ See ISE Letter 2 and NYSE Arca Letter 2, *supra* note 5; see also Amex Letter 2, BSE Letter 2, CBOE Letter 2, Nasdaq Letter 2, and Phlx Letter 2, *supra* note 7.

⁶⁹ See ISE Letter 2 and NYSE Arca Letter 2, *supra* note 5; see also Amex Letter 2, BSE Letter 2, Nasdaq Letter 2, and Phlx Letter 2, *supra* note 7.

⁷⁰ 17 CFR 242.608(b)(2).

⁷¹ 15 U.S.C. 78k-1(a)(1)(C).

⁷² Section 5(b)(iii) of the Options Linkage Plan.

⁷³ See Rule 611(b)(4) of Regulation NMS (17 CFR 242.611(b)(4)).

⁷⁴ 17 CFR 242.608(b)(2).

⁷⁵ 15 U.S.C. 78k-1(a)(1)(C).

⁷⁶ Section 5(b)(iv) and (v) of the Options Linkage Plan.

⁷⁷ See Rule 611(b)(5) and (6) of Regulation NMS (17 CFR 242.611(b)(5) and (6)).

⁶² Section 5(b)(x) of the Options Linkage Plan.

⁶³ Section 5(b)(xi) of the Options Linkage Plan.

⁶⁴ 17 CFR 242.608(b)(2).

⁶⁵ 15 U.S.C. 78k-1(a)(1)(C).

⁶⁶ Section 5(b)(i) of the Options Linkage Plan.

⁶⁷ See NMS Release at 37535, *supra* note 29.

An ISO is defined as a limit order for an options series that, when routed to an Eligible Exchange, is identified as an Intermarket Sweep Order and, simultaneously with the routing of the order, one or more additional orders, as necessary, are routed to execute against the full displayed size of any Protected Bid, in the case of a limit order to sell, or any Protected Offer, in the case of a limit order to buy, for the options series with a price that is superior to the limit price of the order.⁸⁴ Any such additional orders would also be marked as ISOs.

The availability of ISOs will allow the Participants to access multiple price levels simultaneously displayed on the same or multiple markets, without violating the prohibition against Trade-Throughs. As the Commission stated with respect to ISOs for stocks under Regulation NMS, the Commission believes that allowing a Participant to immediately execute an order identified as an ISO when that exchange is not at the NBBO is fully consistent with the principle of protecting the best displayed prices because the exception is premised on the condition that the market participant sending the ISO has already attempted to access all better-priced Protected Quotations up to their displayed size. Consequently, there is no reason why a Participant that receives an ISO while displaying an inferior-priced quotation should be required to delay an execution of the order.⁸⁵ This exception should help to ensure more efficient and faster executions.

The second ISO Trade-Through exception, under subparagraph (b)(v) of Section 5 of the Options Linkage Plan, should benefit market participants in their ability to handle orders efficiently. For example, market participants should be able to use this exception to more efficiently execute block trades one or more minimum price increments away from the NBBO. So long as ISOs are simultaneously routed to execute against better-priced Protected Quotation on other markets, the block order could be executed contemporaneously with the routing of the ISOs.

The Commission notes that Section 5(c) of the Options Linkage Plan requires Participants to take reasonable steps to establish that ISOs are properly routed in an attempt to execute against all applicable Protected Quotations.

For the reasons stated above, the Commission finds that the exception from Trade-Through liability when an

exchange or market participants sends an ISO is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁸⁶ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.⁸⁷

Quote Flickering:⁸⁸ Subparagraph (b)(vi) of Section 5 of the Options Linkage Plan sets forth an exception for flickering quotations, and corresponds to an exception for NMS stocks under Regulation NMS.⁸⁹ It excepts a transaction if the Eligible Exchange displaying the Protected Quotation that was traded through had displayed, within one second prior to execution of the Trade-Through, a Best Bid or Best Offer, as applicable, for the options series with a price that was equal or inferior to the price of the Trade-Through transaction.

As the Commission stated with respect to the similar exception for stocks under Regulation NMS,⁹⁰ this exception thereby provides a "window" to address false indications of Trade-Throughs that in actuality are attributable to rapidly moving quotations. It should also reduce the number of instances in which a Participant must alter its normal trading procedures and route orders to other trading centers to comply with the Options Linkage Plan. The exception is thereby intended to promote more workable intermarket price protection. The Commission finds it is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁹¹ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.⁹²

Non-Firm Quotes:⁹³ This exception, which is carried over from the Current Plan,⁹⁴ permits a Participant to trade through a Protected Quotation that was "Non-Firm."⁹⁵ "Non-Firm" is defined to mean, with respect to Quotations in an Eligible Options Class, that members of a Participant are relieved of their obligations under that Participant's firm quote rule in that Eligible Options Class.⁹⁶

⁸⁶ 17 CFR 242.608(b)(2).

⁸⁷ 15 U.S.C. 78k-1(a)(1)(C).

⁸⁸ Section 5(b)(vi) of the Options Linkage Plan.

⁸⁹ See Rule 611(b)(8) of Regulation NMS (17 CFR 242.611(b)(8)).

⁹⁰ See NMS Release at 37536, *supra* note 29.

⁹¹ 17 CFR 242.608(b)(2).

⁹² 15 U.S.C. 78k-1(a)(1)(C).

⁹³ Section 5(b)(vii) of the Options Linkage Plan.

⁹⁴ See Section 8(c)(iii)(C) of the Current Plan.

⁹⁵ See Section 2(11) of the Options Linkage Plan.

⁹⁶ The Commission notes that, when quotations in an Eligible Options Class are Non-Firm, exchange

The Commission believes that Participants should not be required to protect the price of an away market when that market identifies its quotes as "Non-Firm." The Commission finds that this exception is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,⁹⁷ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C).⁹⁸

Complex Trades:⁹⁹ This exception carries forward the complex trade exception in Section 8(c)(iii)(G) of the Current Plan¹⁰⁰ and permits a Participant to trade through a Protected Quotation if the transaction was part of a "complex trade." The definition of "complex trade" would be implemented through rules adopted by the Participants, which would be subject to notice, comment, and Commission review pursuant to the Section 19(b) rule filing process.

Complex trades, such as those submitted by market participants under the Proposing Exchanges complex order mechanisms,¹⁰¹ are composed of multiple transactions effected at a net price. As the Proposing Exchanges state,¹⁰² it is not always practical to require each leg to be transacted at a price that does not constitute a Trade-Through, and the Commission believes that permitting an exception for transactions effected as a portion of a complex trade is appropriate. By narrowly crafting the definition of complex trades in each Participants' rules,¹⁰³ the Commission believes that this exception will not undercut the general Trade-Through protections of the Options Linkage Plan, and finds it is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,¹⁰⁴ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C).¹⁰⁵

rules require the exchange to provide notice that its quotations are Non-Firm by appending an indicator to its quotations. See, e.g., CBOE Rule 43.14(b) and NYSE Arca Rule 6.86(d)(1)(C).

⁹⁷ 17 CFR 242.608(b)(2).

⁹⁸ 15 U.S.C. 78k-1(a)(1)(C).

⁹⁹ Section 5(b)(viii) of the Options Linkage Plan.

¹⁰⁰ Section 8(c)(iii)(G) of the Current Plan.

¹⁰¹ See, e.g., ISE Rule 722.

¹⁰² See ISE Letter 2 and NYSE Arca Letter 2, *supra* note 5; see also Amex Letter 2, BSE Letter 2, CBOE Letter 2, Nasdaq Letter 2, and Phlx Letter 2, *supra* note 7.

¹⁰³ All changes to rules of national securities exchanges are subject to notice, comment and Commission review pursuant to Section 19(b) of the Act. 15 U.S.C. 78s(b).

¹⁰⁴ 17 CFR 242.608(b)(2).

¹⁰⁵ 15 U.S.C. 78k-1(a)(1)(C).

⁸⁴ Section 2(9) of the Options Linkage Plan.

⁸⁵ See NMS Release at 37523, *supra* note 29.

Customer Stopped Orders:¹⁰⁶ This exception permits a Participant to trade through a Protected Quotation if the trade executed a “stopped order.” The exception requires that the “stopped order” be for the account of a Customer;¹⁰⁷ that the Customer agreed to the specified price on an order-by-order basis; and that the price of the Trade-Through was, for a stopped buy order, lower than the national Best Bid in the options series at the time of execution, or, for a stopped sell order, higher than the national Best Offer in the options series at the time of execution. This exception corresponds to the customer stopped order exception under Regulation NMS.¹⁰⁸

The Commission recognizes that the use of stopped orders is a valuable tool, particularly for the execution of large orders.¹⁰⁹ The Commission believes that this narrowly-drawn exception would give market participants the ability to execute large Customer orders over time at a price agreed upon by a Customer, even though the price of the option may change before the order is executed in its entirety, without undermining the general principles of price protection under the Options Linkage Plan. For these reasons, the Commission finds that this exception is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,¹¹⁰ and assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.¹¹¹

Stopped Orders and Price Improvement:¹¹² This exception permits a Participant to trade through a Protected Quotation if the transaction that constituted the Trade-Through was the execution by a Participant of an order that is stopped at a price that did not constitute a Trade-Through at the time of the stop. This exception allows a Participant to seek price improvement for an order, even if the market moves in the interim, and the transaction ultimately is effected at a price that would trade through the then currently-displayed market. The rules of several of the Proposing Exchanges currently

contain provisions relating to price improvement mechanisms.¹¹³

These price improvement mechanisms offer price improvement to orders received by the exchange during a specified period of time (“auction”). During this auction period, the NBBO could move from where it was when the order was received. However, the exchange is only required to guarantee a price no worse than the NBBO at the time the order was received. Thus, following the auction, an execution could result in a Trade-Through if the NBBO improves from the time the order was received although, had the order been executed at the time of receipt, the execution would not have resulted in a Trade-Through.

This exception would allow a Participant to seek price improvement for an order, even if the market moves in the interim, and the transaction ultimately is effected at a price that would trade through the then currently-displayed market. By allowing this exception, the Commission expects that Participants would be able to continue to use price improvement mechanisms, thereby offering market participants potentially better-priced executions. The Commission finds that this exception is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,¹¹⁴ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.¹¹⁵

Benchmark Trades:¹¹⁶ This exception permits a Participant to trade through a Protected Quotation if the trade was executed at a price not based directly or indirectly on the quoted price of an options series at the time of execution and for which the material terms were not reasonably determinable at the time of the commitment to make the trade.

This exception allows a “benchmark order” and corresponds to an exception for NMS stocks under Rule 611 of Regulation NMS.¹¹⁷ A common example of a benchmark order for NMS stocks is a volume-weighted average price, or “VWAP,” order. The Commission notes that none of the Proposing Exchanges currently permit these types of options trades, and any Participant seeking to make use of this exception would be

required to submit a proposed rule change which would be subject to notice, comment and Commission review under Section 19(b) of the Act. The Commission finds that this exception is in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets, and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C) of the Act.¹¹⁸

B. Locked and Crossed Markets

The Options Linkage Plan also addresses Locked and Crossed Markets.¹¹⁹ The requirements in the Options Linkage Plan relating to Locked and Crossed Markets are virtually identical to those applicable to market centers for NMS stock under Regulation NMS.¹²⁰

Specifically, the Options Linkage Plan requires each Participant to establish, maintain, and enforce written rules that require their members reasonably to avoid displaying Locked and Crossed Markets.¹²¹ Participants would also be required to establish, maintain, and enforce written rules reasonably designed to assure the reconciliation of Locked and Crossed Markets.¹²² Finally, the Options Linkage Plan would provide that Participants must establish, maintain, and enforce written rules that prohibit their members from engaging in a pattern or practice of displaying Locked and Crossed Markets, subject to exceptions as may be contained in the Participants’ rules, as approved by the Commission.¹²³

The Commission recognizes that Section 6 of the Options Linkage Plan, by restricting Locked Markets, can prohibit the display of an order that would otherwise have been displayed and reduced the quoted spread to zero. However, as the Commission stated with respect to locked markets for stocks under Regulation NMS, the

¹¹⁸ 15 U.S.C. 78k-1(a)(1)(C).

¹¹⁹ Section 6 of the Options Linkage Plan. A “Locked Market” is defined as a quoted market in which a Protected Bid is equal to a Protected Offer in a series of an Eligible Options Class. See Section 2(10) of the Options Linkage Plan. A “Crossed Market” is defined as a quoted market in which a Protected Bid is higher than a Protected Offer in a series of an Eligible Options Class. See Section 2(4) of the Options Linkage Plan.

¹²⁰ See Rule 610(d) of Regulation NMS (17 CFR 242.610(d)).

¹²¹ Section 6(a) of the Options Linkage Plan.

¹²² Section 6(b) of the Options Linkage Plan.

¹²³ Section 6(c) of the Options Linkage Plan. The Commission notes that the proposed rule changes relating to all necessary implementing rules of the Participants, including those required by Section 6 of the Options Linkage Plan, would be subject to notice, comment, and Commission review pursuant to Section 19(b) of the Act.

¹⁰⁶ Section 5(b)(ix) of the Options Linkage Plan.

¹⁰⁷ “Customer” would be defined to mean an individual or organization that is not a “Broker/Dealer.” See Section 2(5) of the Options Linkage Plan.

¹⁰⁸ See Rule 611(b)(9) of Regulation NMS (17 CFR 242.611(b)(9)).

¹⁰⁹ See NMS Release at 37527, *supra* note 29.

¹¹⁰ 17 CFR 242.608(b)(2).

¹¹¹ 15 U.S.C. 78k-1(a)(1)(C).

¹¹² Section 5(b)(x) of the Options Linkage Plan.

¹¹³ See, e.g., Securities Exchange Act Release No. 50819 (December 8, 2004), 69 FR 75093 (December 15, 2004) (SR-ISE-2003-06) (approving rules implementing ISE’s Price Improvement Mechanism under ISE Rule 723).

¹¹⁴ 17 CFR 242.608(b)(2).

¹¹⁵ 15 U.S.C. 78k-1(a)(1)(C).

¹¹⁶ Section 5(b)(xi) of the Options Linkage Plan.

¹¹⁷ See Rule 611(b)(7) of Regulation NMS (17 CFR 242.611(b)(7)).

Commission believes that Locked Markets may not actually represent two market participants willing to buy and sell at the same price. Instead, a locking market participant may not truly be willing to trade at the displayed locking price, but chooses to lock rather than execute against the already-displayed quotation to receive a liquidity rebate. The Commission believes that giving priority to the first-displayed Protected Bid or Protected Offer, particularly when it includes a public customer's order, will encourage price discovery and contribute to fair and orderly markets.¹²⁴

The Options Linkage Plan is designed to ensure that the display of locked and crossed markets would be restricted, while also recognizing that locked and crossed markets do occur accidentally and cannot always be avoided. Thus, the Options Linkage Plan requires that the Participants have written rules that are reasonably designed to assure the reconciliation of any lock or cross. Further, the Options Linkage Plan expressly prohibits a pattern or practice of locking or crossing away markets.

In addition, the Options Linkage Plan would allow exceptions to its general Locked and Crossed Markets provision as might be contained in a given Participant's rules. As with all proposed rule changes of national securities exchanges, such rule changes would be subject to notice, comment and Commission review under Section 19(b)(1) of the Act.¹²⁵ The Commission believes that these provisions are designed to ensure that the display of Locked and Crossed Markets will be limited and that any such display will be promptly reconciled.

For the reasons stated above, the Commission finds that the Options Linkage Plan's provisions relating to Locked and Crossed Markets are in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,¹²⁶ and believes they assure fair competition among exchange markets, consistent with Section 11A(a)(1)(C).¹²⁷

C. Joining the Proposed Plan

Any national securities exchange would be eligible to become a Participant by executing a copy of the Options Linkage Plan and providing each Participant with a copy of such executed Options Linkage Plan¹²⁸ if it is: (1) Registered with the Commission

in accordance with Section 6(a) of the Act; (2) a Participant Exchange in OCC;¹²⁹ and (3) a party to the OPRA Plan.¹³⁰ Further, any such national securities exchange wishing to become a Participant would be required to file an amendment to the Options Linkage Plan by executing a copy of the Options Linkage Plan and filing such executed Options Linkage Plan to the Commission.¹³¹ Such amendment would be effective when the amendment is approved by the Commission or otherwise becomes effective pursuant to Section 11A of the Act and Rule 608 thereunder.¹³² The Commission finds that this process for joining the Options Linkage Plan is in the public interest,¹³³ and believes it is consistent with Section 11A(a)(1)(C) because it is designed to ensure that reasonable procedures are in place to permit additional exchanges to also participate in the Options Linkage Plan.¹³⁴

D. Withdrawal From the Proposed Plan

Any Participant would be able to withdraw from the Options Linkage Plan at any time by providing not less than 30 days' prior written notice to each of the other Participants of such intent to withdraw.¹³⁵ To withdraw, such Participant also would be required to effect an amendment to the Options Linkage Plan by submitting such amended Options Linkage Plan to the Commission for approval.¹³⁶ In submitting the amended Options Linkage Plan to the Commission, the Participant proposing to withdraw from the Options Linkage Plan would be required to state how the Participant plans to accomplish, by alternate means, the goal of the Options Linkage Plan regarding limiting Trade-Throughs of prices on other exchanges trading the same options classes.¹³⁷ Such withdrawal from the Options Linkage Plan would be effective when the amendment is approved by the

Commission or otherwise becomes effective pursuant to Section 11A of the Act and Rule 608 thereunder. Upon the effectiveness of such withdrawal, the withdrawing Participant would have no further rights or obligations under the Options Linkage Plan.

The Commission finds that these requirements for withdrawal from the Options Linkage Plan are in the public interest, appropriate for the protection of investors and the maintenance of fair and orderly markets,¹³⁸ and believes it assures fair competition among exchange markets, consistent with Section 11A(a)(1)(C).¹³⁹

E. Implementation

The Proposed Plan states that the "[Participants] shall implement [the plan] * * * no later than February 27, 2009; provided that, unless the [Commission] otherwise authorizes, the [Participants] shall not implement [the plan] until all Eligible Exchanges either (1) have become parties to [the plan] and the [Commission] has approved all necessary implementing rules or (2) have developed the ability to accept and execute incoming Intermarket Sweep Orders."¹⁴⁰

To provide clarity to market participants regarding the implementation date of the plan, the Commission, after consultation with the Proposing Exchanges, has modified the Proposed Plan to change the implementation date in Section 7 from February 27, 2009 to August 31, 2009. In addition, the Commission notes that all seven options exchanges¹⁴¹ have joined in filing the Proposed Plan with the Commission, and each has submitted proposed rule changes pursuant to Section 19(b) of the Act to modify its rules to comply with the Options Linkage Plan.¹⁴² The Commission believes that the provision that would permit the plan to be implemented if an Eligible Exchange "developed the ability to accept and execute incoming Intermarket Sweep Orders," even if such exchange had not become a party to the plan is no longer necessary because all seven options exchanges have joined the Options

¹²⁹ For a definition of a "Participant Exchange," see Section VII of the OCC by-laws.

¹³⁰ For more information on who is a party to the OPRA Plan, see Section I of the OPRA Plan.

¹³¹ Section 4(b) of the Options Linkage Plan.

¹³² *Id.* These requirements are identical to those contained in the Current Plan. See Sections 4(c)(i) and 5(c) of the Current Plan. The Current Plan also requires that an eligible exchange pay a fee to join the Current Plan. See Section 4(c)(i)(iv) of the Current Plan. The Options Linkage Plan does not require an Eligible Exchange to pay a fee to join the Options Linkage Plan.

¹³³ 17 CFR 242.608(b)(2).

¹³⁴ 15 U.S.C. 78k-1(a)(1)(C).

¹³⁵ Section 3(d) of the Options Linkage Plan.

¹³⁶ Section 4(c) of the Options Linkage Plan.

¹³⁷ *Id.* These requirements are identical to those contained in the Current Plan. See Sections 4(d) and 5(c)(iii) of the Current Plan.

¹³⁸ 17 CFR 242.608(b)(2).

¹³⁹ 15 U.S.C. 78k-1(a)(1)(C).

¹⁴⁰ See Section 7 of the Proposed Plan.

¹⁴¹ That is, CBOE, ISE, Nasdaq, BX, Phlx, Amex and NYSE Arca.

¹⁴² See, e.g., Securities Exchange Act Release Nos. 60014 (June 1, 2009); and 74 FR 27224 (June 8, 2009) (SR-ISE-2009-27) and 60015 (June 1, 2009); 74 FR 27375 (June 9, 2009) (SR-NYSEAmex-2009-19) which propose rules such as provisions that contain relevant definitions, an order protection rule, and a locked and crossed market rule, which correspond to the provisions in the Options Linkage Plan.

¹²⁴ See NMS Release at 37547, *supra* note 29.

¹²⁵ 15 U.S.C. 78s(b)(1).

¹²⁶ 17 CFR 242.608(b)(2).

¹²⁷ 15 U.S.C. 78k-1(a)(1)(C).

¹²⁸ Section 3(c) of the Options Linkage Plan.

Linkage Plan and therefore will, upon implementation of the Options Linkage Plan, accept and execute ISOs.

The Commission finds that these modifications to Section 7 of the Proposed Plan are necessary and appropriate and will further the purposes of the Act by providing clarity to market participants regarding the implementation of the plan while providing appropriate time to self-regulatory organizations to prepare for implementation.

With these modifications, unless the Commission otherwise authorizes, the plan may only be implemented by the Proposing Exchanges when all Proposing Exchanges' proposed rule changes containing the necessary implementing rules¹⁴³ have been approved by the Commission.

IV. Conclusion

It is hereby ordered, that pursuant to Section 11A(a)(3)(B) of the Act¹⁴⁴ and Rule 608 thereunder,¹⁴⁵ that the Proposed Plan submitted by CBOE, ISE, Nasdaq, BX, Phlx, Amex, and NYSE Arca, as modified herein, is approved and declared effective,¹⁴⁶ and that CBOE, ISE, Nasdaq, BX, Phlx, Amex, and NYSE Arca are authorized to act jointly to implement the Options Order Protection and Locked/Crossed Market Plan as a means of facilitating a national market system.

By the Commission.

Florence E. Harmon,

Deputy Secretary.

Appendix A

Options Order Protection and Locked/Crossed Market Plan

Section 1—Preamble

The Participants submit to the SEC this Plan providing a framework for order protection and addressing Locked and Crossed Markets in Eligible Options Classes. The purpose of the Plan is to enable the Participants to act jointly in establishing a framework for providing order protection and addressing Locked and Crossed Markets in Eligible Options Classes. In addition, the Plan provides for a non-exclusive method for achieving order protection and addressing Locked and Crossed Markets. The Participants will submit to the SEC for approval their respective rules that will implement the framework of the Plan. The Participants request that the SEC issue an order pursuant to Section 11A(a)(3)(B) of the Exchange Act and Rule 608 thereunder evidencing its approval of the Plan.

Section 2—Definitions

(1) "Best Bid" and "Best Offer" mean the highest priced Bid and the lowest priced Offer.

(2) "Bid" or "Offer" means the bid price or the offer price communicated by a member of an Eligible Exchange to any Broker/Dealer, or to any customer, at which it is willing to buy or sell, as either principal or agent, but shall not include indications of interest.

(3) "Broker/Dealer" means an individual or organization registered with the SEC in accordance with Section 15(b)(1) of the Exchange Act or a foreign broker or dealer exempt from such registration pursuant to Rule 15a-6 under the Exchange Act.

(4) "Crossed Market" means a quoted market in which a Protected Bid is higher than a Protected Offer in a series of an Eligible Class.

(5) "Customer" means an individual or organization that is not a Broker/Dealer.

(6) "Eligible Exchange" means a national securities exchange registered with the SEC in accordance with Section 6(a) of the Exchange Act that: (a) As a Participant Exchange in OCC (as that term is defined in Section VII of the OCC by-laws); (b) is a party to the OPRA Plan (as that term is described in Section I of the OPRA Plan); and (c) if the national securities exchange chooses not to become a party to this Plan, is a participant in another plan approved by the Commission providing for comparable Trade-Through and Locked and Crossed Market protection.

(7) "Eligible Options Class" means all option series overlying a security (as that term is defined in Section 3(a)(10) of the Exchange Act) or group of securities, including both put options and call options, which class is available for trading on two or more Eligible Exchanges.

(8) "Exchange Act" means the Securities Exchange Act of 1934, as amended.

(9) "Intermarket Sweep Order (ISO)" means a limit order for an options series that meets the following requirements:

(a) When routed to an Eligible Exchange, the order is identified as an ISO;

(b) Simultaneously with the routing of the order, one or more additional ISOs, as necessary, are routed to execute against the full displayed size of any Protected Bid, in the case of a limit order to sell, or any Protected Offer, in the case of a limit order to buy, for the options series with a price that is superior to the limit price of the ISO, with such additional orders also marked as ISOs.

(10) "Locked Market" means a quoted market in which a Protected Bid is equal to a Protected Offer in a series of an Eligible Options Class.

(11) "Non-Firm" means, with respect to Quotations in an Eligible Options Class, that members of a Participant are relieved of their obligations under that Participant's firm quote rule in that Eligible Options Class.

(12) "OCC" means The Options Clearing Corporation.

(13) "OPRA" means the Options Price Reporting Authority.

(14) "OPRA Plan" means the plan filed with the SEC pursuant to Section 11Aa(1)(C)(iii) of the Exchange Act, approved by the SEC and declared effective as of January 22, 1976, as from time to time amended.

(15) "Participant" means an Eligible Exchange whose participation in the Plan has become effective pursuant to Section 3(c) of the Plan.

(16) "Plan" means the plan amended and restated in this instrument as from time to time amended in accordance with its provisions.

(17) "Protected Bid" or "Protected Offer" means a Bid or Offer in an options series, respectively, that:

a. Is displayed by an Eligible Exchange;

b. Is disseminated pursuant to the OPRA Plan; and

c. Is the Best Bid or Best Offer, respectively, of an Eligible Exchange.

(18) "Protected Quotation" means a Protected Bid or Protected Offer.

(19) "Quotation" means a Bid or Offer.

(20) "SEC" means the United States Securities and Exchange Commission.

(21) "Trade-Through" means a transaction in an options series, either as principal or agent, at a price that is lower than a Protected Bid or higher than a Protected Offer.

Section 3—Parties to the Plan

(a) List of Parties

The parties to the Plan are as follows:

Boston Stock Exchange, Inc., registered as a national securities exchange under the Exchange Act and having its principal place of business at 100 Franklin Street, Boston, Massachusetts 02110.

Chicago Board Options Exchange, Incorporated, registered as a national securities exchange under the Exchange Act and having its principal place of business at 400 South LaSalle Street, Chicago, Illinois 60605.

International Securities Exchange, LLC, registered as a national securities exchange under the Exchange Act and having its principal place of business at 60 Broad Street, New York, New York 10004.

The NASDAQ Stock Market LLC, registered as a national securities exchange under the Exchange Act and having its principal place of business at One Liberty Plaza, 50th Floor, New York, New York 10006.

NASDAQ OMX PHLX, Inc., registered as a national securities exchange under the Exchange Act and having its principal place of business at 1900 Market Street, Philadelphia, Pennsylvania 19103.

NYSE Alternext US LLC, registered as a national securities exchange under the Exchange Act and having its principal place of business at 11 Wall Street, New York, NY 10005.

NYSE Arca, Inc., registered as a national securities exchange under the Exchange Act and having its principal place of business at 100 South Wacker Drive, Suite 1800, Chicago, IL 60606.

(b) Compliance Undertaking

By subscribing to and submitting the Plan for filing with the SEC, each Participant agrees to enforce compliance by its members with the provisions of the Plan.

(c) Entry of New Participants

The Participants agree that any other Eligible Exchange may become a Participant by: (i) Executing a copy of the Plan, as then in effect; (ii) providing each then-current

¹⁴³ See *id.*

¹⁴⁴ 15 U.S.C. 78k-1(a)(3)(B).

¹⁴⁵ 17 CFR 242.608.

¹⁴⁶ The approved Plan is attached here as Appendix A.

Participant with a copy of such executed Plan; and (iii) effecting an amendment to the Plan as specified in Section 4(b) of the Plan.

(d) Withdrawal from the Plan

Any Participant may withdraw from the Plan at any time by: (i) Providing not less than 30 days' prior written notice to each of the other Participants of such intent to withdraw; and (ii) effecting an amendment to the Plan as specified in Section 4(c) of the Plan. Upon the effectiveness of such withdrawal the withdrawing Participant shall have no further rights or obligations whatsoever under the Plan.

Section 4—Amendments to the Plan

(a) General Amendment Authority

Except with respect to:

(i) the addition of new Participants to the Plan; and

(ii) the withdrawal of a Plan Participant, any proposed change in, addition to, or deletion from the Plan may be effected only by means of a written amendment to the Plan that is unanimously approved by the Participants and that: (A) sets forth the change, addition or deletion; (B) is executed on behalf of each Participant; and (C) is approved by the SEC or otherwise becomes effective pursuant to Section 11A of the Exchange Act and Rule 608 thereunder.

(b) New Participants

With respect to new Participants, an amendment to the Plan may be effected by a new Eligible Exchange executing a copy of the Plan, as then in effect (with the only change being the addition of the new Participant's name in Section 3(a) of the Plan), and submitting such executed Plan to the SEC. Such amendment will be effective when the amendment is approved by the SEC or otherwise becomes effective pursuant to Section 11A of the Exchange Act and Rule 608 thereunder.

(c) Withdrawal from the Plan

A Participant seeking to withdraw from the Plan shall effect an amendment to the Plan as then in effect (with the only change being the deletion of the Participant's name in Section 3(a) of the Plan) by submitting such amended Plan to the SEC for approval. In submitting the amended Plan to the SEC, the Participant proposing to withdraw from the Plan shall state how the Participant plans to accomplish, by alternate means, the goal of the Plan regarding limiting Trade-Throughs of prices on other exchanges trading the same options classes. Such withdrawal from the Plan shall be effective when the amendment is approved by the SEC or otherwise becomes effective pursuant to Section 11A of the Exchange Act and Rule 608 thereunder.

Section 5—Order Protection

(a) Order Protection

(i) **Prevention of Trade-Throughs.** Each Participant agrees that it shall establish, maintain and enforce written policies and procedures as approved by the SEC that are reasonably designed to prevent Trade-Throughs in that Participant's market in Eligible Options Classes that do not fall within an exception set forth in paragraph (b) below, and, if relying on such exception, that are reasonably designed to assure compliance with the terms of the exception.

(ii) **Surveillance.** Each Participant agrees to conduct surveillance of its market on a regular basis to ascertain the effectiveness of the policies and procedures required by paragraph (a)(1) of this section, and to take prompt action to remedy deficiencies in such policies and procedures.

(b) Exceptions.

(i) The transaction that constituted the Trade-Through was effected when the Eligible Exchange displaying the Protected Quotation that was traded through was experiencing a failure, material delay, or malfunction in its systems or equipment;

(ii) The transaction traded through a Protected Quotation being disseminated by an Eligible Exchange during a trading rotation;

(iii) The transaction that constituted the Trade-Through occurred when there was a Crossed Market;

(iv) The transaction that constituted the Trade-Through was the execution of an order identified as an Intermarket Sweep Order;

(v) The transaction that constituted the Trade-Through was effected by a Participant that simultaneously routed an Intermarket Sweep Order to execute against the full displayed size of any Protected Quotation that was traded through;

(vi) The Eligible Exchange displaying the Protected Quotation that was traded through had displayed, within one second prior to execution of the Trade-Through, a Best bid or Best offer, as applicable, for the options series with a price that was equal or inferior to the price of the Trade-Through transaction;

(vii) The Protected Quotation traded through was being disseminated from an Eligible Exchange whose Quotations were Non-Firm with respect to such options series;

(viii) The transaction that constituted the Trade-Through was effected as a portion of a "complex trade," as defined in the rules of a Participant;

(ix) The transaction that constituted the Trade-Through was the execution by a Participant of an order for which, at the time of receipt of the order, a member of the Participant had guaranteed an execution at no worse than a specified price (a "stopped order"), where:

(A) The stopped order was for the account of a Customer;

(B) The Customer agreed to the specified price on an order-by-order basis; and

(C) the price of the Trade-Through was, for a stopped buy order, lower than the national Best Bid in the options series at the time of execution, or, for a stopped sell order, higher than the national Best Offer in the options series at the time of execution;

(x) The transaction that constituted the Trade-Through was the execution by a Participant of an order which was stopped at a price that did not Trade-Through another Eligible Exchange at the time of the stop; or

(xi) The transaction that constituted the Trade-Through was the execution of an order at a price that was not based, directly or indirectly, on the quoted price of the options series at the time of execution and for which the material terms were not reasonably determinable at the time the commitment to execute the order was made.

(c) **Intermarket Sweep Orders.** Participants shall take reasonable steps to establish that

Intermarket Sweep Orders meet the requirements of Section 2(9) of the Plan.

Section 6—Locked and Crossed Markets

The Participants agree that they shall establish, maintain and enforce written rules that:

(a) Require their members reasonably to avoid displaying Locked and Crossed Markets;

(b) Are reasonably designed to assure the reconciliation of Locked and Crossed Markets; and

(c) Prohibit its members from engaging in a pattern or practice of displaying Locked and Crossed Markets;

in all cases subject to such exceptions as may be contained in the rules of a Participant approved by the Commission.

Section 7—Implementation

The Parties shall implement this Plan on a date upon which all Parties agree, but no later than August 31, 2009; provided that, unless the SEC otherwise authorizes, the Parties shall not implement this Plan unless all Eligible Exchanges have become parties to this Plan and the SEC has approved all necessary implementing rules.

Section 8—Counterparts and Signatures

The Plan may be executed in any number of counterparts, no one of which need contain all signatures of all Participants, and as many of such counterparts as shall together contain all such signatures shall constitute one and the same instrument.

In Witness Whereof, this Plan has been executed as of the _____ 2009 by each of the parties hereto.

CHICAGO BOARD OPTIONS EXCHANGE,
INCORPORATED

By: _____
Date: _____

INTERNATIONAL SECURITIES EXCHANGE,
LLC

By: _____
Date: _____

The NASDAQ STOCK MARKET LLC

By: _____
Date: _____

NASDAQ OMX BX, INC.

By: _____
Date: _____

NASDAQ OMX PHLX, Inc.

By: _____
Date: _____

NYSE AMEX LLC

By: _____
Date: _____

NYSE ARCA, INC.

By: _____
Date: _____

[FR Doc. E9-18763 Filed 8-5-09; 8:45 am]

BILLING CODE 8010-01-P

Tab:
Citadel Letter
(D-Limit)

April 23, 2020

Ms. Vanessa A. Countryman
Secretary
Securities and Exchange Commission
100 F Street NE
Washington, DC 20549–1090

Re: IEX D-Limit Proposal (File No. SR-IEX-2019-15)

Dear Ms. Countryman:

Citadel Securities appreciates the opportunity to provide comments to the Securities and Exchange Commission (the “Commission”) on the proposal by Investors Exchange LLC (“IEX”) to introduce a new “Discretionary Limit” (“D-Limit”) order type (the “Proposal”).¹

According to the Proposal, D-Limit orders will be initially ranked in the order book at their limit price, but will be automatically adjusted by IEX to a price that is less aggressive than the NBB (or NBO) if IEX’s “crumbling quote indicator” (“CQI”) determines that the NBBO is likely to change. This means that very few (if any) D-Limit orders will execute when the CQI signal is ON. IEX will bypass its existing speed bump when re-pricing D-Limit orders, transforming IEX’s existing speed bump into a new asymmetric speed bump specifically for D-Limit orders.²

This novel trade avoidance mechanism has profound implications for U.S. equity market structure, creating unfair advantages for both IEX and the trading firms utilizing the D-Limit order type and undermining the integrity of the NBBO. IEX claims the Proposal will “protect liquidity providers from potential adverse selection by latency arbitrage trading strategies,”³ but fails to distinguish purported “latency arbitrage” from a wide range of ordinary trading activities that can trigger the CQI, including sweep orders and bona fide hedging. Thus, as detailed below, the Proposal will broadly and indiscriminately affect myriad liquidity takers, including retail and institutional investors as well as market makers in equities and related asset classes, such as ETFs, options, and futures.

¹ 84 FR 71997 (Dec. 30, 2019), available at: <https://www.govinfo.gov/content/pkg/FR-2019-12-30/pdf/2019-28024.pdf> (the “Proposal”).

² While incoming orders are delayed by the IEX speed bump, D-Limit orders (including displayed orders) will be immediately repriced by IEX based on real-time market data. As a result, rather than being truly predictive, IEX’s CQI model is able to benefit from hindsight, as it can observe trading activity occurring on other venues and use that information to determine whether or not to reprice a resting D-Limit order while incoming orders on IEX are delayed.

³ Proposal at 71997.

We note that the Commission recently disapproved a similar proposal by Cboe EDGA Exchange, Inc. (“EDGA”) to implement an asymmetric speed bump for displayed orders.⁴ As detailed below, the IEX Proposal contains many of the same flaws as the EDGA proposal, and should also be disapproved by the Commission.

Although the IEX Proposal has garnered some support from market participants,⁵ analysis of this support reveals that it cites frequently to, and is heavily reliant on, the data provided by IEX regarding the expected impact of the Proposal. Unfortunately, the data provided by IEX is insufficient and misleading.

Below, we detail:

- I. Our unique perspective in assessing the impact of the Proposal;
- II. Why the IEX data is insufficient and misleading; and
- III. The key questions IEX and supporting letters have failed to answer that impact the Commission’s analysis of whether the Proposal is consistent with the Securities Exchange Act of 1934 (“Exchange Act”).

I. Citadel Securities’ Unique Perspective

In assessing the impact of the IEX Proposal, we are uniquely situated in our ability to incorporate the following diverse perspectives:

- We are a top 5 market maker on IEX (and therefore a key member of the constituency this Proposal is designed to benefit);
- We are one of the top 3 takers of liquidity on IEX when the CQI signal is ON (and therefore have carefully considered the types of liquidity taking orders that will be negatively impacted by the Proposal);
- We are the leading destination for retail order flow and over 50% of our trading activity on IEX is on behalf of retail investors (and therefore we can readily appreciate the negative impact this Proposal will have on retail and institutional investors);
- We are a leading market maker in both equities and related asset classes, such as ETFs, options, and futures (and therefore can assess the impact of the Proposal on bona fide hedging activities by market makers, which often includes liquidity taking activity designed to support the provision of liquidity to market participants); and

⁴ Release No. 34-88261 (Feb. 21, 2020), available at: <https://www.sec.gov/rules/sro/cboeedga/2020/34-88261.pdf> (the “EDGA Disapproval Order”).

⁵ See <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915.htm>.

- We are the largest NYSE Designated Market Maker (“DMM”) by number of listings and care deeply about overall liquidity dynamics in the U.S. equities market (and therefore are well-positioned to assess the broader market impacts of the Proposal).

II. IEX’s Data is Insufficient and Misleading

A. The IEX Data

IEX asserts the Proposal is “designed to protect liquidity providers from potential adverse selection by latency arbitrage trading strategies.”⁶ IEX relies heavily on markout data⁷ to show the existence and associated impact of this purported “latency arbitrage,” comparing markouts when the CQI signal is ON to other executions and concluding that “liquidity-providing orders that are executed while the CQI is on are subject to significant differences in short term markouts.”⁸

IEX supplements the markout data with (a) data showing that the CQI signal will only be ON for seconds per symbol per trading data on average,⁹ and (b) a limited analysis of the trading activity currently executed when the CQI signal is ON purporting to show that “proprietary trading firms” account for the vast majority of trading during these periods.¹⁰ Both of these data points purport to show that other types of market participants will be minimally impacted by the Proposal.

B. Identified Flaws

Based on the data above, IEX has failed to satisfy its burden under the Exchange Act to provide the Commission with sufficient basis to make an affirmative finding that the Proposal is consistent with the Exchange Act.¹¹

IEX’s reliance on markout data to show the existence of “latency arbitrage” is very similar to the EDGA asymmetric speed bump proposal, where the Commission found that markout data was

⁶ Proposal at 71997.

⁷ The term markouts refers to changes in the midpoint of the NBBO measured from the perspective of either the liquidity providing resting order or liquidity removing taking order over a specified period of time following the time of execution. Proposal at 71999, FN 23.

⁸ Proposal at 71999.

⁹ See Proposal at 72001, FN 56.

¹⁰ See Proposal at 72002 (“Within the two millisecond periods following CQI determinations, proprietary trading firms submit 6.8 times as many marketable-to-mid shares (i.e., shares priced at least as aggressively as the midpoint and eligible to trade) compared to full-service and agency firms”).

¹¹ See Release No. 34-88501 (March 27, 2020) at page 8, available at: <https://www.sec.gov/rules/sro/iex/2020/34-88501.pdf>.

insufficient to support an affirmative finding.¹² This is because both the IEX and EDGA markout data simply show that short-term price changes are likely to result from trading activities that can trigger the CQI, such as sweep orders taking out a price level across the market and bona fide hedging activities by market makers. Any markout analysis comparing transactions that are part of a sweep order taking out a price level across the market with transactions that execute against only a portion of posted market-wide liquidity will of course show significant differences in short-term markouts.

IEX's analysis of trading activity currently executed when the CQI signal is ON therefore becomes critical in assessing whether IEX has met its burden under the Exchange Act. Although the CQI is only ON for a limited duration, IEX has acknowledged that 33.7% of marketable orders are received and 24% of displayed volume is executed during these periods.¹³ Therefore, a very significant portion of total trading activity will be negatively impacted by the Proposal, and it is critical to understand exactly what comprises this trading activity. Unfortunately, IEX only performs a cursory analysis purporting to show that "proprietary trading firms" account for the vast majority of trading during these periods.¹⁴ As detailed below, this analysis suffers from a number of flaws and is insufficient to substantiate IEX's broad assertion that "aggressive taking orders while the CQI is on are strongly correlated to latency arbitrage strategies."¹⁵

First, IEX's classification of member firms is imprecise, rendering its conclusions inaccurate regarding the types of firms trading when the CQI is ON.¹⁶ For example, for purposes of its analysis, IEX classified Citadel Securities as a "proprietary trading firm" even though over 50% of our trading activity on IEX is on behalf of retail investors. IEX has made no attempt to separate out, or otherwise distinguish, this trading activity. Therefore, any negative impact from the Proposal on our retail orders is simply being characterized by IEX as a negative impact on a "proprietary trading firm." In this regard, it is important to note that over 80% of the retail orders we route to IEX are liquidity taking orders, and therefore risk being adversely impacted by the proposed trade avoidance mechanism.

Second, IEX fails to conduct a thorough analysis of the liquidity taking orders executed when the CQI is ON. Relevant questions include:

- How many orders/executions are from retail or institutional investors?

¹² See EDGA Disapproval Order at pages 25-26. We note that IEX has also failed to adequately explain the apparent inconsistency between (1) claims regarding the extent to which purported "latency arbitrage" activity is negatively impacting displayed liquidity on IEX throughout the trading day and (2) the limited duration that the CQI is ON during the trading day.

¹³ Proposal at 71999 and 72001.

¹⁴ See Proposal at 72002. See also "A Deliberate Strategy" (Dec. 17, 2019), available at: <https://medium.com/boxes-and-lines/a-deliberate-strategy-bb8b0cff074b>.

¹⁵ Letter from John Ramsay, Chief Market Policy Officer, IEX (Feb. 13, 2020) at page 14, available at: <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915-6808882-208490.pdf> ("IEX Response Letter").

¹⁶ See Proposal at 72002, FN 58.

- How many orders/executions are part of a sweep order taking out a price level across the market which triggers the CQI?
- How many orders/executions are related to bona fide hedging activities by market makers in equities or related asset classes, such as ETFs, options, and futures?
- Is a similar percentage of overall trading volume executed on other exchanges when the CQI is ON (meaning that trading activity generally clusters around times when price levels are changing and is not evidence of purported “latency arbitrage” activity on IEX)?

It is irresponsible for IEX to assert that liquidity taking orders executed when the CQI is ON are “strongly correlated to latency arbitrage strategies”¹⁷ without undertaking this type of analysis. Furthermore, for IEX to be correct, it would mean that approximately 33.7% of marketable orders and 24% of displayed volume executed on IEX related to “latency arbitrage,” even though IEX’s CQI does not incorporate cross-asset signals, and therefore cannot be considered to even capture the full extent of purported “latency arbitrage” activities.¹⁸ Based on our trading activity, we know that a variety of trading strategies and market participants take liquidity when the CQI is ON and therefore will be negatively impacted by the Proposal. However, instead of conducting this analysis, IEX has clearly made a decision to focus on the length of time the CQI is ON (rather than the trading volume that is affected) in order to seek approval of its ill-advised Proposal.¹⁹

C. Unanswered Questions

Without more robust data, the Commission lacks sufficient basis to make an affirmative finding that the Proposal is consistent with the Exchange Act. The data provided by IEX does not demonstrate the existence, or associated impact, of purported “latency arbitrage” on IEX, or that the Proposal is appropriately tailored to address any such problem.²⁰ In addition, as detailed above, the provided data does not allow the Commission to accurately assess the impact on (a) retail and institutional investors (given the imprecise entity classifications),²¹ (b) sweep orders taking out a price level across the market,²² or (c) bona fide hedging activities by market makers in equities or related asset classes, such as ETFs, options, and futures (including the higher spreads that will be

¹⁷ *Supra* note 15.

¹⁸ We note that one supporter of the Proposal identified the lack of cross-asset signals as a “limitation in the likely effectiveness of IEX’s proposed D-Limit order.” Letter from Eric Swanson, CEO, XTX Markets LLC (Americas) (Jan. 17, 2020) at page 4, available at: <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915-6671703-203990.pdf>.

¹⁹ *See*, for example, the additional data provided in the IEX Response Letter at page 16.

²⁰ *See* EDGA Disapproval Order at page 24.

²¹ *See id.* at page 36.

²² *See id.* at page 37.

borne by retail and institutional investors where hedging is disrupted).²³ These are almost identical shortcomings to those identified by the Commission when considering the EDGA asymmetric speed bump proposal.

In addition, as with the EDGA asymmetric speed bump proposal, IEX has not provided any data to substantiate the claimed benefits of the Proposal.²⁴ As noted by commenters, the D-Limit order type appears to lose much of its value if IEX is alone at the NBBO and therefore routed to first, as the CQI signal will not provide added protection in this situation. As a result, liquidity providers on IEX should not generally be expected to narrow prevailing market-wide spreads if the Proposal is adopted; instead, we would expect IEX liquidity providers to often post prices equal to the NBBO set by liquidity providers on other exchanges. Indeed, this is exactly what happened with an asymmetric speed bump implemented in Canada that provides liquidity providers with a similar advantage. Academic research found that liquidity providers on TSX Alpha were almost never alone at the NBBO,²⁵ a conclusion confirmed by our own analysis of recent market data.²⁶ While the asymmetric speed bump reduced adverse selection for TSX Alpha liquidity providers, this financial advantage accrued to the liquidity providers in the form of increased profitability and not to the broader market in the form of narrower bid-ask spreads.²⁷ It is clear that liquidity providers would prefer to opt-out of providing liquidity when advantageous to do so, such as when a sweep order is taking out a price level across the market, but there is no evidence to suggest that IEX's Proposal to provide liquidity providers with such a trade avoidance mechanism will benefit other market participants.

Each of the data-related omissions detailed above prevent the Commission from being able to perform the necessary analysis under the Exchange Act and should lead to the disapproval of the Proposal.

III. Inconsistencies with the Exchange Act

The insufficient and misleading data provided by IEX has also directly impacted the comment file. While some commenters have raised a number of detailed concerns regarding the proposal,²⁸ many commenters have simply relied on the data provided by IEX to form a view. Given the material issues identified above, the IEX data (and related letters) fail to adequately address the concerns raised.

²³ See *id.* at page 36.

²⁴ See *id.* at page 22.

²⁵ Haoming Chen et al., "The value of a Millisecond: Harnessing Information in Fast, Fragmented Markets" (Nov. 18, 2017) at pages 15 and 65 (Figure 4), available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2860359.

²⁶ See Letter from Citadel Securities dated Oct. 21, 2019 on the EDGA asymmetric speed bump proposal at page 16, available at: <https://www.sec.gov/comments/sr-cboe-edga-2019-012/srcboe-edga2019012-6321413-194388.pdf>.

²⁷ *Supra* note 25 at pages 4-5.

²⁸ See <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915.htm>.

Below, we detail the Proposal's key inconsistencies with the Exchange Act: (a) unfairly discriminating against liquidity takers, (b) inappropriately burdening market competition, and (c) impairing fair and efficient access to displayed quotations.

A. The Proposal Unfairly Discriminates Against Liquidity Takers, Including Retail and Institutional Investors

Several commenters raised concerns regarding the impact of the Proposal on liquidity takers, including retail and institutional investors. We agree with these concerns, particularly for orders that are sent to more than one venue for execution, such as intermarket sweep orders. Since IEX will use real-time market data to observe activity on other venues and then bypass its existing speed bump to reprice D-Limit orders while incoming liquidity taking orders are delayed, the Proposal transforms IEX's existing speed bump into a new asymmetric speed bump specifically for the benefit of D-Limit orders. Furthermore, since IEX will reprice D-Limit orders to be less aggressive than the NBBO, this mechanism will result in significant quote fading and a decline in fill rates, adversely impacting all types of liquidity takers seeking to fill larger orders.

It is important to note that, while most supporters of the Proposal did not specifically address concerns from the perspective of a liquidity taker, a majority of commenters who did address this topic acknowledged the legitimacy of concerns around liquidity taking orders that are sent to more than one venue for execution, such as intermarket sweep orders.²⁹ In response, IEX and certain supporters of the Proposal make two arguments:

- First, reference is made to IEX's data regarding the length of time the CQI signal is ON and the purported trading of "proprietary trading firms" during these periods in order to argue that other types of market participants will be generally unaffected by the Proposal.³⁰

However, as detailed in Section II above, despite the CQI signal being ON for a limited duration, a very significant portion of total trading activity will be affected, as 33.7% of marketable orders are received and 24% of displayed volume is executed during these periods. In addition, given IEX's imprecise entity classifications and insufficient analysis of the trading activity occurring during these periods, the data does not establish that liquidity taking orders sent to more than one venue for execution, such as intermarket sweep orders, will not be subject to unfair discrimination under the Proposal.

²⁹ We identified the following 12 comment letters addressing the topic: ACS Execution Services, AJO, Clearpool, DLA Piper, FIA PTG, Healthy Markets, IMC, NASDAQ, Proof Services, Raymond James, SIFMA, and T. Rowe Price. These include 6 in support of the Proposal, 5 against, and 1 neutral. Of the 7 letters either neutral or in support of the Proposal, 4 acknowledge the legitimacy of concerns around sweep orders (ACS Execution Services, AJO, Clearpool, and Healthy Markets).

³⁰ See IEX Response Letter at page 4 and T. Rowe Price letter.

- Second, IEX and certain supporters argue that market participants will be able mitigate quote fading concerns by staging the routing of larger orders so as to route to IEX first and thereby avoid the CQI signal turning ON before the IEX portion of the order is executed.³¹ This line of argument tacitly acknowledges the legitimacy of the concerns raised, and requires retail and institutional investors to modify routing practices with respect to larger orders to route first to IEX before routing to other venues after accounting for the 350 microsecond IEX speed bump.

In response, we find it interesting that IEX has argued that most market participants do not have access to the “sophisticated technology, connectivity, and market data”³² necessary to utilize the proposed 4 millisecond EDGA asymmetric speed bump, yet believes all investors will be able to precisely stage sweep orders across the U.S. equities market in a manner that avoids the CQI turning ON. IEX has not provided any estimate of the costs for market participants to universally adopt this type of capability nor considered the associated operational risks.

More importantly, even if all investors had the ability to stage the routing of larger orders in this manner, the Commission must still assess whether this constitutes unfair discrimination, as noted in the EDGA Disapproval Order.³³ A Proposal that effectively compels market participants to route orders to IEX first in order to successfully access displayed liquidity – potentially at a higher cost of execution – raises a number of serious concerns that must be addressed, including:

- (a) whether broker-dealers intentionally delaying the routing of marketable orders is consistent with best execution and other regulatory requirements;
- (b) whether broker-dealers should be comfortable delaying the routing of marketable orders given the risk of the market moving in the interim (particularly if greater size was displayed on other venues);
- (c) what are the implementation costs associated with market participants updating their routing methodologies in this manner, taking into account the likely proliferation of similar mechanisms on other exchanges;
- (d) what would be the economic and regulatory implications for market participants that do not update their routing methodologies in this manner;
- (e) what are the risks of unnecessarily increasing the complexity of order routing systems, especially during periods of heightened market volatility; and

³¹ See IEX Response Letter at page 8 and Proof Services and Raymond James letters.

³² Proposal at 72001.

³³ See EDGA Disapproval Order at page 37.

- (f) what would happen if other exchanges implemented asymmetric speed bumps, perhaps with differing durations.

In addition to the above, and similar to the EDGA asymmetric speed bump proposal, IEX has also failed to explain why providing a benefit to sophisticated proprietary trading firms acting as liquidity providers without a corresponding obligation is consistent with the Exchange Act when the Proposal permits discrimination against liquidity takers.³⁴

Contrary to assertions by IEX,³⁵ these are fundamentally new considerations for liquidity takers seeking to access displayed liquidity on IEX and there are many reasons why a market participant may not wish to route a portion of an order to one specific venue first (for example, there may be greater size displayed on other venues or lower venue fees). A Proposal that requires such a solution (which will soon be antiquated as other exchanges introduce their own trade avoidance mechanisms) in order to address legitimate concerns about quote fading and declining fill rates constitutes unfair discrimination against liquidity takers. These liquidity taking orders may include (i) orders from retail and institutional investors, (ii) sweep orders taking out a price level across the market, and (iii) bona fide hedging activities by market makers in equities or related asset classes, such as ETFs, options, and futures.

B. The Proposal Inappropriately Burdens Market Competition

Several commenters raised concerns regarding the impact of the Proposal on market competition. Below, we detail three inappropriate burdens on market competition that have not been adequately addressed by IEX or supporters of the Proposal in the comment file.

1. Liquidity Takers

The Proposal advantages liquidity providers over liquidity takers by giving liquidity providers an ability to avoid unfavorable executions. Academic research exploring the impact of an asymmetric speed bump implemented in Canada providing liquidity providers with a similar advantage found that liquidity provider profitability significantly increased as a result.³⁶ IEX asserts that liquidity providers will instead use this advantage to narrow market-wide spreads,³⁷ but has failed to provide any supporting data. As noted in the EDGA Disapproval Order,³⁸ IEX

³⁴ See EDGA Disapproval Order at pages 36-37.

³⁵ See IEX Response Letter at page 8 (“ISOs sent to IEX in the same manner in which they are sent today, accounting for IEX’s existing 350 microsecond speed bump, will result in a comparable fill rate”). We note that to access a displayed order on IEX today, an incoming order must only reach the IEX POP before a message to cancel or amend that displayed order reaches the IEX POP. Under the Proposal, to access a displayed D-Limit order, an incoming order must also traverse the IEX speed bump before the CQI signal turns ON.

³⁶ *Supra* note 25 at pages 1 and 4.

³⁷ See IEX Response Letter at page 4.

³⁸ See EDGA Disapproval Order at page 22.

should provide data regarding (a) the anticipated impact on liquidity provider profitability, and (b) the amount (if any) liquidity providers will return to investors through narrower spreads.

2. Liquidity Providers on Other Venues

The Proposal advantages liquidity providers on IEX over liquidity providers on other venues, as IEX liquidity providers can free-ride on the pricing heuristics and risk taking capabilities of others by posting prices equal to the NBBO and relying on IEX to observe away executions and reprice D-Limit orders to frequently avoid unfavorable executions. Such free-riding discourages innovation and undermines the competitive dynamics that have dramatically reduced spreads and transaction costs for all market participants over the past decade. Contrary to assertions by IEX, the proposed D-Limit order type raises fundamentally different considerations than order types pegged to the NBBO.³⁹ To the extent D-Limit orders are repriced, they will be repriced to be less aggressive than the NBBO, effectively shielding liquidity providers from unwanted executions.

This venue-specific advantage will significantly impact market competition, as there will be significant pressure for other exchanges to adopt a similar mechanism. We note that several commenters have raised concerns about the proliferation of such mechanisms advantaging liquidity providers,⁴⁰ and not a single commenter has agreed with IEX that “[i]f other markets adopted a very similar mechanism, IEX does not believe this would result in more inaccessible quotes.”⁴¹ In our view, IEX has failed to adequately consider the consequences of additional exchanges implementing asymmetric speed bumps. Among others, concerns around the reliability of the NBBO and the resiliency of liquidity, particularly during periods of market volatility, will be amplified to the extent that multiple exchanges adopt mechanisms supporting systematic quote fading.

3. IEX Member Firms

IEX offers a model that “forecasts when the price of a stock is likely about to change”⁴² (i.e. the CQI) that will be in direct competition with the predictive models used by IEX member firms. IEX has stated that “[i]f others are leveraging short term prediction models to anticipate NBBO changes, than [sic] we can build such a model ourselves.”⁴³ While we note that an exchange offering functionality that is similar to a service offered by a broker-dealer does not automatically

³⁹ See IEX Response Letter at page 4.

⁴⁰ We identified the following 7 comment letters addressing the topic: ACS Execution Services, The Bacidore Group, Clearpool, DLA Piper, FIA PTG, NASDAQ, and SIFMA. These include 5 against the Proposal, 1 in support, and 1 neutral.

⁴¹ IEX Response Letter at page 17.

⁴² “What is the IEX Signal?” available at: <https://iextrading.com/trading/signal/>.

⁴³ “The Evolution of the Crumbling Quote Signal,” Allison Bishop, IEX White Paper (2017) at page 3, available at: <https://iextrading.com/docs/The%20Evolution%20of%20the%20Crumbling%20Quote%20Signal.pdf>.

render such functionality an inappropriate burden on competition,⁴⁴ in this case, the Proposal provides the IEX model with unfair competitive advantages over its member firms.

First, IEX will bypass its speed bump to reprice D-Limit orders whereas IEX member firms remain subject to the speed bump when attempting to reprice their own resting displayed orders. This means that resting displayed orders not using the D-Limit order type will be at a material disadvantage, as only the IEX model will be able to observe activity occurring on other venues in order to determine whether or not to reprice a resting order while incoming orders on IEX are delayed. As a result, resting displayed orders not using the D-Limit order type can be expected to experience increased adverse selection and worse execution quality. These negative impacts will affect a range of market participants, including (a) those who submit “held” orders, such as retail investors, which will not be able to utilize the D-Limit order type given the nature of the discretion involved, and (b) liquidity providers who prefer to utilize their own predictive models instead of the IEX CQI model. In turn, the unlevel playing field created by the Proposal provides IEX with an opportunity to commercially profit by charging member firms for the benefit of bypassing the IEX speed bump.⁴⁵

Second, IEX has certain regulatory advantages when employing its CQI model compared to IEX member firms, including broad regulatory immunity and no best execution requirements. We are not aware of other regulatory requirements specific to IEX as an exchange that would address this particular imbalance.

IEX argues that there is precedent for its proposed use of the CQI model, both with the existing IEX D-Peg and P-Peg non-displayed order types, and with displayed order types on other venues that are pegged to the NBBO.⁴⁶

In response, we note that while it is true that the IEX D-Peg and P-Peg order types leverage the CQI model, these order types cannot be used for displayed orders. In approving the D-Peg order type, the Commission expressly noted that it was limited to non-displayed orders and therefore did not raise the unique considerations specific to displayed orders.⁴⁷ In addition, IEX has acknowledged that “D-Peg is also a fundamentally different order type [compared to D-Limit].”⁴⁸ In light of the unique regulatory considerations specific to displayed orders, and their importance in promoting price discovery, the mere existence of the D-Peg and P-Peg order types

⁴⁴ See 85 FR 4726 (Jan. 27, 2020) at 4735, available at: <https://www.govinfo.gov/content/pkg/FR-2020-01-27/pdf/2020-01253.pdf>.

⁴⁵ See, e.g., Letter from Hudson River Trading, available at: <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915-6678383-204977.pdf> (“In fact, IEX’s business model is largely based on having created a speed bump, then charging its members for the ability to bypass it”).

⁴⁶ See IEX Response Letter at pages 6-7.

⁴⁷ 81 FR 41142 (June 23, 2016) at 41156, FN 215 and 216, available at: <https://www.govinfo.gov/content/pkg/FR-2016-06-23/pdf/2016-14875.pdf> (“IEX Approval Order”).

⁴⁸ IEX Response Letter at page 8, FN 20.

does not provide the Commission with a sufficient basis to approve the use of the IEX CQI model for displayed orders.

Displayed order types pegged to the NBBO also do not provide IEX with helpful precedent. These order types do not rely on a model that is in direct competition with the predictive models used by member firms; instead displayed orders are repriced based on an objective fact (i.e. whether the NBBO has changed). This important distinction has been alluded to by IEX⁴⁹ and is reflected in statistics showing that the CQI model is often inaccurate. Based on data provided by IEX, it appears that IEX's CQI model is accurate only approximately 50% of the time the CQI signal turns ON.⁵⁰ This is fundamentally different than pegged order types that accurately reflect the NBBO.

C. Displayed D-Limit Orders Cannot Be Considered Protected Quotations

In no event can displayed D-Limit orders be considered protected quotations under Rule 611 of Regulation NMS, as the D-Limit order type utilizes an intentional delay to frustrate market participant access to resting quotations. The Proposal transforms IEX's existing speed bump into a new asymmetric speed bump specifically for the benefit of D-Limit orders – while incoming orders are delayed by the IEX speed bump, IEX will bypass the speed bump to immediately reprice D-Limit orders to be less aggressive than the NBBO. This asymmetric speed bump created by the D-Limit order type does not satisfy the *de minimis* standard under the Commission's Automated Quotations Interpretive Guidance, which requires any delay to be “so short as to not frustrate the purposes of Rule 611 by impairing fair and efficient access to an exchange's quotations.”⁵¹

In response, IEX argues that the Commission has already determined that its speed bump satisfies the *de minimis* standard, and since the Proposal does not “create any additional delay” then it should continue to be considered *de minimis*.⁵² This argument fails for a number of reasons:

- In determining that IEX's speed bump and related order types satisfied the *de minimis* standard, the Commission specifically noted that order types enabling IEX to bypass

⁴⁹ Proposal at 72004 (“While the D-Limit proposal is novel in that it would provide an exchange with flexibility to reprice a displayed order [. . .]”).

⁵⁰ *Supra* note 43 at page 28.

⁵¹ 81 Fed. Reg. 40785 (June 23, 2016) at 40786, available at: <https://www.gpo.gov/fdsys/pkg/FR-2016-06-23/pdf/2016-14876.pdf> (“Automated Quotations Interpretative Guidance”).

We note that IEX has previously argued that asymmetric speed bumps may not satisfy Rule 611. See Letter from Sophia Lee, General Counsel, IEX (Nov. 13, 2015) at page 8, available at: <https://www.sec.gov/comments/10-222/10222-20.pdf> (“The PSX proposal (as with the Canadian exchanges) **would have changed the accessibility of a protected quote by altering the relative speeds of quotation makers and takers** [emphasis added].”) and Letter from Sophia Lee, General Counsel, IEX (Nov. 23, 2015) at page 5, available at: <https://www.sec.gov/comments/10-222/10222-26.pdf> (“PSX's proposal was **rightfully withdrawn because its 'speed bump' was applied only to a subset of orders and actually may have harmed the accessibility of its quotations** [emphasis added].”)

⁵² IEX Response Letter at page 11.

the speed bump were limited to non-displayed orders and therefore did not require the Commission to assess consistency with Rule 611;⁵³

- The Proposal significantly changes the analysis regarding whether market participants have fair and efficient access to IEX's displayed quotations. At the moment, 24% of displayed volume is executed on IEX when the CQI signal is ON.⁵⁴ The Proposal is designed to ensure that close to zero displayed D-Limit orders execute when the CQI signal is ON by repricing D-Limit orders to be less aggressive than the NBBO, meaning that displayed D-Limit orders will be inaccessible during these periods.

Experience with IEX's non-displayed D-Peg and P-Peg order types demonstrate how successful IEX is in preventing quote accessibility when the CQI signal is ON, with only .17% of D-Peg and P-Peg volume executed during these periods.⁵⁵ IEX has also acknowledged the Proposal's impact on quote accessibility, stating that "D-Limit orders may not be accessible to other market participants" when the CQI signal is ON.⁵⁶

- While the Proposal does not change the duration of the IEX speed bump, it fundamentally alters how it operates for displayed orders, as IEX will now be able to bypass the speed bump and immediately reprice displayed D-Limit orders in order to prevent quote accessibility when the CQI signal is ON.

The Commission's Automated Quotations Interpretive Guidance is clear that the duration of a speed bump is not the only relevant consideration, and instead the *de minimis* standard requires a facts and circumstances analysis regarding whether an exchange has frustrated the purposes of Rule 611 by impairing fair and efficient access to its quotations.⁵⁷ In this case, that analysis must include the combined interaction of the existing IEX speed bump and the new proposed order type for displayed orders. Similarly, nothing in the current staff guidance on Automated Quotations under Regulation NMS⁵⁸ suggests that an exchange with an approved *de minimis* intentional

⁵³ See IEX Approval Order at 41156, FN 216 ("The Commission notes that IEX will only reprice pegged orders, which are non-displayed. Non-displayed orders are not reflected in an exchange's quotations, and Rule 611 applies order protection to publicly displayed quotes only. Accordingly, an access delay that does not allow the repricing of displayed orders does not impact an exchange's displayed quotation, and cannot be said to lead to 'maybe' quotations.").

⁵⁴ Proposal at 71999.

⁵⁵ See "Leveling the Playing Field for Lit Trading" (Dec. 17, 2019), available at: <https://medium.com/boxes-and-lines/leveling-the-playing-field-for-lit-trading-682dc723cefl>.

⁵⁶ Proposal at 72003.

⁵⁷ See Automated Quotations Interpretative Guidance at 40792 ("The Commission believes that, in light of the evolving nature of technology and the markets, and the need to assess the impact of intentional access delays on the markets, establishing a bright line *de minimis* threshold is not appropriate at this time.").

⁵⁸ See Staff Guidance on Automated Quotations under Regulation NMS (June 17, 2016), available at: <https://www.sec.gov/divisions/marketreg/automated-quotations-under-regulation-nms.htm>.

delay can then introduce an order type designed to prevent quote accessibility during periods when 24% of total displayed volume is executed without being subject to a *de novo* review under Rule 611 that takes into account the combined interaction of the intentional delay and the new order type.⁵⁹ While the intentional delay, in isolation, is considered *de minimis* under the current guidance, the D-Limit order type is certainly not given that its express purpose is to frustrate access to resting quotes when the CQI is ON.

IEX and certain commenters also argue that, even though displayed D-Limit orders are not accessible when the CQI is ON, this only occurs for very limited durations and will only impact purported “latency arbitrage” activity, so therefore IEX displayed quotes should still be considered accessible under Rule 611.⁶⁰ These arguments also lack merit.

- First, although the CQI is only ON for a limited duration, IEX has acknowledged that 33.7% of marketable orders are received and 24% of displayed volume is executed during these periods.⁶¹ Therefore, the Proposal will impair fair and efficient access to IEX’s displayed quotations for a very significant portion of total trading activity.
- Second, as detailed above, IEX has failed to perform the necessary data analysis to substantiate claims that only purported “latency arbitrage” activities will be impacted. Specifically, IEX has failed to comprehensively analyze the trading activity occurring when the CQI is ON, including disclosing the portion of orders from (a) retail and institutional investors, (b) sweep orders taking out a price level across the market, and (c) bona fide hedging activities by market makers in equities or related asset classes, such as ETFs, options, and futures.
- Third, even if IEX were to show that 24% of its displayed volume related to purported “latency arbitrage” activity, Rule 611 does not provide that “‘accessibility’ is a relative term,” as claimed by IEX.⁶² The Commission must apply the regulatory framework in a fair and consistent manner and should not entertain suggestions that only certain types of market participants or trading strategies deserve to be afforded fair and efficient access to displayed quotations. Rule 611 does not permit an exchange to prevent quote accessibility during periods when 24% of its displayed volume is executed and yet maintain protected quotation status.

⁵⁹ We note that in due course Commission staff may consider updating its guidance to reflect subsequent market experience with intentional access delay proposals, including additional considerations that may be relevant in assessing whether an intentional access delay impairs fair and efficient access to displayed quotations, such as (a) whether the delay is symmetric or asymmetric, and (b) whether the delay is coupled with an order type that, in combination, may impact fair and efficient access to displayed quotes.

⁶⁰ See IEX Response Letter at pages 10-11 and Letters from Baird, Council of Institutional Investors, Goldman Sachs & Co. LLC, and Ontario Teachers' Pension Plan et al.

⁶¹ Proposal at 71999 and 72001.

⁶² IEX Response Letter at page 10.

* * * * *

Ultimately, introducing an unprecedented asymmetric speed bump for displayed orders into the U.S. equities market should require a much more thorough analysis of the impact on investors, competition, market quality, and resiliency. Similar to the recent EDGA asymmetric speed bump proposal, IEX has provided insufficient data to support the Proposal. In particular, IEX has failed to demonstrate the existence, or associated impact, of purported “latency arbitrage” activity on IEX and has elected not to comprehensively analyze the trading activity currently occurring when the CQI is ON. Therefore, the data does not allow the Commission to accurately assess the adverse impact on (a) retail and institutional investors (given the imprecise entity classifications), (b) sweep orders taking out a price level across the market, or (c) bona fide hedging activities by market makers in equities or related asset classes, such as ETFs, options, and futures.

In addition, the Proposal (i) unfairly discriminates against liquidity takers, (ii) inappropriately burdens market competition, and (iii) impairs fair and efficient access to displayed quotations. IEX will bypass its speed bump to reprice D-Limit orders, enabling the IEX model to observe activity occurring on other venues in order to determine whether or not to reprice a resting order while incoming orders on IEX are delayed. This trade avoidance mechanism will provide IEX liquidity providers with a material advantage when using the D-Limit order type, resulting in significant quote fading and a decline in fill rates, adversely impacting all types of liquidity takers seeking to fill larger orders. It may also cause the relevant products to become more susceptible to manipulation (such as spoofing) to the extent liquidity providers are able to anticipate when posted quotations are likely to be repriced before being honored, and will heighten concerns around the resiliency of liquidity, particularly during periods of market volatility.

We urge the Commission to disapprove this Proposal.

Please feel free to call the undersigned at [REDACTED] with any questions regarding these comments.

Respectfully,

/s/ Stephen John Berger

Managing Director

Global Head of Government & Regulatory Policy

Tab:
85 FR 54438

the United States Postal Service voted unanimously to hold and to close to public observation a special meeting in Washington, DC, via teleconference. The Board determined that no earlier public notice was practicable.

GENERAL COUNSEL CERTIFICATION: The General Counsel of the United States Postal Service has certified that the meeting may be closed under the Government in the Sunshine Act.

CONTACT PERSON FOR MORE INFORMATION: Michael J. Elston, Secretary of the Board, U.S. Postal Service, 475 L'Enfant Plaza SW, Washington, DC 20260-1000. Telephone: (202) 268-4800.

Michael J. Elston,

Secretary.

[FR Doc. 2020-19455 Filed 8-28-20; 4:15 pm]

BILLING CODE 7710-12-P

POSTAL SERVICE

Board of Governors; Sunshine Act Meeting

DATES AND TIMES: September 9, 2020, at 9:00 a.m.

PLACE: Washington, DC

STATUS: Closed.

MATTERS TO BE CONSIDERED:

Wednesday, September 9, 2020, at 9:00 a.m.

1. Strategic Items.
2. Financial and Operational Matters.
3. Compensation and Personnel Matters.
4. Administrative Items.

GENERAL COUNSEL CERTIFICATION: The General Counsel of the United States Postal Service has certified that the meeting may be closed under the Government in the Sunshine Act.

CONTACT PERSON FOR MORE INFORMATION: Katherine Sigler, acting Secretary of the Board, U.S. Postal Service, 475 L'Enfant Plaza SW, Washington, DC 20260-1000. Telephone: (202) 268-4800.

Michael J. Elston,

Secretary.

[FR Doc. 2020-19447 Filed 8-28-20; 4:15 pm]

BILLING CODE 7710-12-P

SECURITIES AND EXCHANGE COMMISSION

[Release No. 34-89686; File No. SR-IEX-2019-15]

Self-Regulatory Organizations; Investors Exchange LLC; Order Approving a Proposed Rule Change To Add a New Discretionary Limit Order Type Called D-Limit

August 26, 2020.

I. Introduction

On December 16, 2019, the Investors Exchange LLC ("IEX" or the "Exchange") filed with the Securities and Exchange Commission ("Commission"), pursuant to Section 19(b)(1) of the Securities Exchange Act of 1934 ("Exchange Act")¹ and Rule 19b-4 thereunder,² a proposed rule change to adopt a new order type, the Discretionary Limit order ("D-Limit"). The proposed rule change was published for comment in the *Federal Register* on December 30, 2019.³ On February 12, 2020, the Commission designated a longer period within which to approve the proposed rule change, disapprove the proposed rule change, or institute proceedings to determine whether to disapprove the proposed rule change.⁴ On March 27, 2020, the Commission instituted proceedings to determine whether to approve or disapprove the proposed rule change ("OIP").⁵ This order approves the proposed rule change.

II. Description of the Proposed Rule Change

A. Latency Arbitrage

IEX explains that its proposal "is designed to protect liquidity providers, institutional investors as well as market makers, from potential adverse selection by latency arbitrage trading strategies in a fair and nondiscriminatory manner. . . ." IEX uses the term "latency arbitrage" to refer to trading strategies that trade based on the market participant's ability to minimize latencies in seeing, and reacting to, quote and trade data through its use of low-latency systems and technology, as

well as connectivity and proprietary market data it purchases from exchanges, which may allow them to react faster to changing market prices than other market participants who have not purchased those same low-latency systems, connectivity, and data sources, which can be relatively expensive.⁷

B. IEX Speed Bump

In the Notice, the Exchange explains how it has designed its market model around "ways to counter or reduce speed advantages that can harm investors by exposing them to execution at stale prices when their orders are traded against by traders with more complete and timely information about market prices."⁸ The primary feature of that market model is the IEX "speed bump," which employs physical path latency to introduce an equivalent 350 microseconds of latency between the network access point (the Point-of-Presence, or "POP") and the Exchange's system at its primary data center.⁹ The speed bump provides time for IEX to update pegged orders resting on its exchange when the national best bid and offer ("NBBO") changes, so that the resting pegged orders are accurately pegged to current market prices.¹⁰ Without this protection, pegged orders resting on IEX have the potential to be subject to latency arbitrage (*i.e.*, executed at disadvantageous "stale" prices because IEX has not yet been able to update the prices of those resting orders in response to changes in the NBBO) by those market participants that can rapidly aggregate market data feeds and react faster than IEX to NBBO updates.¹¹ The IEX speed bump by itself currently provides no protection or benefits for displayed orders or non-displayed orders at fixed limit prices.¹² The speed bump works together with several non-displayed order types on IEX that are "pegged" to a specified

⁷ See, e.g., Notice, *supra* note 3, at 72004 and IEX First Response to Comments, Letter from John Ramsay, Chief Market Policy Officer, IEX, dated February 13, 2020, at 2-3 ("IEX First Response to Comments").

⁸ See Notice, *supra* note 3, at 71998.

⁹ See *id.* The IEX speed bump applies to all incoming and outgoing messages except for inbound market data from other trading centers and outbound transaction and quote information sent to the applicable securities information processor. In addition, updates to resting pegged orders on IEX are processed within the IEX trading system and do not require separate messages to be transmitted from outside the system.

¹⁰ See, e.g., Securities Exchange Act Release No. 78101 (June 17, 2016), 81 FR 41142, 41157 (June 23, 2016) (File No. 10-222) (granting the application of IEX for registration as a national securities exchange).

¹¹ See *id.*

¹² See *id.* at 41155.

¹ 15 U.S.C. 78s(b)(1).

² 17 CFR 240.19b-4.

³ See Securities Exchange Act Release No. 87814 (December 20, 2019), 84 FR 71997 ("Notice"). Comments on the proposed rule change can be found at <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915.htm>.

⁴ See Securities Exchange Act Release No. 88186 (February 12, 2020), 85 FR 9513 (February 19, 2020).

⁵ See Securities Exchange Act Release No. 88501 (March 27, 2020), 85 FR 18612 (April 2, 2020).

⁶ Notice, *supra* note 3, at 71998.

price.¹³ These order types include the Discretionary Peg (“DPeg”) and the Primary Peg (“PPeg”).¹⁴ DPeg and PPeg orders can “exercise discretion” to trade at prices more aggressive (i.e., higher in the case of a peg order to buy, or lower in the case of a peg order to sell) than their default prices.¹⁵ Specifically, IEX uses a proprietary mathematical calculation, called the crumbling quote indicator (“CQI”), to determine when these pegged order types are eligible to “exercise discretion.”¹⁶ As described in the Notice, the CQI is designed to predict whether a particular quote is unstable or “crumbling,” meaning that the NBB likely is about to decline or the NBO likely is about to increase.¹⁷

C. Crumbling Quote Indicator

The Exchange utilizes real time relative quoting activity of certain Protected Quotations and a proprietary mathematical calculation (the “quote instability calculation”) to assess the probability of an imminent change to the current Protected NBB to a lower price or Protected NBO to a higher price for a particular security (“quote instability factor”).¹⁸ When the quoting activity meets predefined criteria and the quote instability factor calculated is greater than the Exchange’s defined quote instability threshold, IEX treats the quote as “unstable,” and the CQI is on at that price level for up to two milliseconds (hereafter referred to as the “quote instability determination price level” or the “CQI Price”).¹⁹ During all other times, the quote is considered stable, and the CQI is off. When IEX determines, pursuant to the CQI methodology, that the current market for a specific security is unstable—meaning there is a heightened probability of an imminent quote change at the NBB or NBO—IEX’s system will prevent DPeg and PPeg orders on that side of the market from exercising discretion and

trading at a price that is more aggressive than the DPeg or PPeg order’s resting price.²⁰

D. D-Limit Order Type

In this proposal, IEX seeks to adopt the D-Limit order type, which would work in conjunction with the CQI by adjusting its limit price when the CQI is on.²¹ A D-Limit order could be a displayed or non-displayed limit order that, upon entry and when posting to the IEX order book, is priced to be equal to and ranked at the order’s limit price.²²

Most notably, a D-Limit order (including a displayed D-Limit order) would be adjusted to a less aggressive price during periods of quote instability when the CQI is on. Specifically, if, upon entry of a D-Limit buy (sell) order, the CQI is on and the order has a limit price equal to or higher (lower) than the quote instability determination price level (i.e., the CQI Price), IEX will automatically adjust the price of the D-Limit order to one minimum price variant (“MPV”) ²³ lower (higher) than the CQI price. Similarly, when unexecuted shares of a D-Limit buy (sell) order are posted to the order book, if a quote instability determination is made and such shares are ranked and displayed (in the case of a displayed order) by IEX at a price equal to or higher (lower) than the CQI Price, IEX will automatically adjust the price of the resting D-Limit order to one MPV lower (higher) than the CQI Price.²⁴

When the price of a D-Limit order is adjusted, the order will receive a new time priority. If multiple D-Limit orders are adjusted at the same time, IEX will maintain their relative time priority. Further, when the price of a D-Limit order is adjusted, the member that entered the order will receive an order message from the Exchange notifying the member of the price adjustment.

A D-Limit order whose price is adjusted by IEX will not revert back to the price at which it was previously ranked and, in the case of a displayed order, displayed. Rather, the order will

continue to be ranked and, in the case of a displayed order, displayed at the new price, unless the order becomes subject to another automatic adjustment or if the order is subject to the price sliding provisions of IEX Rule 11.190(h).²⁵

III. Discussion and Commission Findings

After careful review, the Commission finds that the Exchange’s proposal is consistent with the requirements of the Exchange Act and the rules and regulations thereunder applicable to a national securities exchange.²⁶ In particular, the Commission finds that the proposed rule change is consistent with Sections 6(b)(5) of the Exchange Act,²⁷ which requires, among other things, that the rules of a national securities exchange be designed to promote just and equitable principles of trade, to remove impediments to and perfect the mechanism of a free and open market and a national market system and, in general, to protect investors and the public interest, and not be designed to permit unfair discrimination between customers, issuers, brokers, or dealers, and 6(b)(8) of the Exchange Act,²⁸ which requires that the rules of a national securities exchange not impose any burden on competition that is not necessary or appropriate in furtherance of the purposes of the Exchange Act.

As outlined below, the Commission has carefully reviewed the proposed rule change, comments received, and IEX’s response to comments to arrive at these findings. Below the Commission first discusses the existence of latency arbitrage on IEX and the effectiveness of CQI in detecting it. Second, the Commission discusses whether the D-Limit order type will lead to “quote fading” or affect IEX’s ability to maintain a protected quotation under

²⁵ IEX Rule 11.190(h) provides for price sliding in the event of a locked or crossed market, to enforce the Midpoint Price Constraint, to comply with the display or execution requirements for a short sale order not marked short exempt during a Short Sale Period, or to comply with the Limit Up-Limit Down Price Constraint. As set forth in IEX Rule 11.190(h), an order that has been subject to price sliding will be repriced back to its more aggressive limit price when the market condition changes such that the condition necessitating the price sliding is no longer applicable. This is in contrast to the normal operation of a D-Limit order when it adjusts due to the CQI being triggered, at which point the D-Limit order’s adjusted price will not reprice.

²⁶ In approving this proposed rule change, the Commission has considered the proposed rule’s impact on efficiency, competition, and capital formation. See 15 U.S.C. 78c(f). The Commission addresses comments about competition below in Section III.E.

²⁷ 15 U.S.C. 78f(b)(5).

²⁸ 15 U.S.C. 78f(b)(8).

¹³ See Notice, *supra* note 3, at 71998 (citing IEX Rule 1.160(f)).

¹⁴ See IEX Rule 11.190(b)(10) and 11.190(b)(8), respectively. DPegs are pegged to one minimum price variation, or “tick,” below the national best bid (“NBB”), in the case of buy orders, or one tick above the national best offer (“NBO”), in the case of sell orders, unless the submitter of the order has specified a limit price that is less aggressive than this default resting price. PPegs are pegged to one tick below the NBB, for a buy order, and one tick above the NBO, for a sell order, but is also available to trade at a price up to the NBB or down to the NBO, unless further restricted by the order’s limit price.

¹⁵ See Notice, *supra* note 3, at 71998.

¹⁶ See *id.*

¹⁷ See *id.*

¹⁸ See *id.* The CQI utilizes a fixed formula that is codified in IEX’s rulebook and thus is publicly known. See IEX Rule 11.190(g).

¹⁹ See *id.* IEX assesses the stability of the Protected NBB and Protected NBO for each security.

²⁰ See *id.* The CQI is thus side specific.

²¹ IEX proposes to amend IEX Rule 11.190(b)(7), which is currently reserved, to add the D-Limit order type.

²² A non-displayed D-Limit order with a limit price more aggressive than the midpoint of the NBBO (“Midpoint Price”) will be subject to the Midpoint Price Constraint and be booked and ranked on the order book at a price equal to the Midpoint Price pursuant to IEX Rule 11.190(h)(2).

²³ See IEX Rule 11.210 (specifying a MPV of \$0.01 for bids and offers priced equal to or greater than \$1.00 per share).

²⁴ In the Notice, the Exchange provides examples of how D-Limit would operate. See Notice, *supra* note 3, at 72000–01.

Regulation NMS.²⁹ Third, the Commission discusses whether the D-Limit order type will permit unfair discrimination between customers, issuers, brokers, or dealers. Finally, the Commission discusses whether the D-Limit order type will impose any burden on competition that is not necessary or appropriate in furtherance of the purposes of the Exchange Act.

A. Protection From Latency Arbitrage

As discussed above, IEX's stated purpose for the D-Limit order type is to protect liquidity providers on IEX from potential adverse selection resulting from latency arbitrage trading strategies, and thereby encourage its members to submit more displayed limit orders to IEX.³⁰

Some commenters opposed to the proposal argue that the data and analysis provided by IEX are insufficient for the Commission to determine that IEX's proposal is consistent with the Exchange Act. They question IEX's characterization of latency arbitrage and the data IEX uses to show that it exists, and ask about the performance of the CQI during periods of higher market volatility and for thinly traded stocks.

One commenter asserts that IEX's data "does not demonstrate the existence, or associated impact, of purported 'latency arbitrage' on IEX, or that the [p]roposal is appropriately tailored to address any such problem."³¹ The commenter further argues that IEX fails to conduct a thorough analysis of liquidity taking orders executed when the CQI is "on" including whether such orders are from retail or institutional investors, are sweep orders taking out a price level across all exchanges, or are hedging activities.³² The commenter states that IEX focuses on the length of time that the CQI is on rather than the trading volume that is affected.³³ In response, IEX states that resting limit orders on IEX "are systematically subjected to adverse impacts of latency arbitrage

strategies."³⁴ In support of its statement, IEX shows that the CQI was on for 1.64 seconds per symbol per day on average, which is 0.007% of the time during regular market hours.³⁵ In that very short period of time, however, the Exchange received 33.7% of marketable orders.³⁶ In other words, the CQI is almost always "off," but during the very short periods of time when it is "on," IEX observes that "certain types of trading strategies are seeking to aggressively target liquidity providers during periods of quote instability."³⁷

As for which market participants are seeking to remove resting liquidity when the CQI is on, IEX estimates, based on how it classifies its members' logical order entry ports,³⁸ that

²⁹ See Notice, *supra* note 3, at 72002.

³⁰ See *id.* at 72001–02 (based on data from September 2019). On a volumeweighted basis, IEX reports that, during the same period, the CQI was on for 5.9 seconds per day per symbol, or 0.025% of the time during regular market hours. See *id.*

³¹ See *id.* at 72001. Further, 24% of displayed volume on IEX is executed when the CQI is on. See *id.* at 71999.

³² See *id.*

³³ See *id.* at 72002. One commenter objects to IEX's classification of it as a "proprietary trading firm" because "over 50% of our trading activity on IEX is on behalf of retail investors." Citadel First Letter, *supra* note 31, at 4. The commenter clarified in a second comment letter that it "typically enter[s] into back-to-back transactions (one on the external venue and one with the retail broker-dealer)." Letter from Stephen John Berger, Managing Director, Citadel Securities, dated July 2, 2020, at 2 ("Citadel Second Letter"). In other words, the Commission understands that the commenter is not directly routing the customer's order to exchanges, but rather is, for example, buying shares for its own account and selling shares to the customer. An anonymous commenter, describing himself or herself as "someone with very intimate knowledge about the retail and wholesaling process" states that "what Citadel is not clarifying is that retail orders are likely sent to Citadel at random times (initiated by actual retail investors living in different parts of the world), but Citadel likely chooses to route these orders to IEX during a CQI condition" (emphasis in original). Letter from Anonymous, undated, at 2. The anonymous commenter asserts that "[d]uring that time, Citadel has a free option to hold the order and decide what to do with it" and believes that "clearly they aren't holding this order and waiting for a better price for retail to show up—they are waiting to make as much money for Citadel as possible." *Id.* The anonymous commenter further asserts that "[n]one of [Citadel's] example has anything to do with retail being harmed." *Id.* at 3. Further, IEX provides updated data from January to April 2020 for firms that are publicly listed as being members of the FIA Principal Traders Group (which IEX notes is a self-described "association of firms that trade their own capital on exchanges . . ."), but excluded that commenter from the analysis even though it is a member of the Principal Traders Group. See Letter from John Ramsay, Chief Market Policy Officer, IEX, dated May 10, 2020, at 13 ("IEX Second Response to Comments"). IEX states that its new data, despite excluding that commenter's trading activity, still "precisely matches patterns seen for all firms classified as proprietary . . ." *Id.* See also Letter from Anonymous, undated (challenging Citadel's characterization of the potential for harm to retail investors) and Letter from John Ramsay, Chief

"proprietary trading firms are more likely to trade against IEX resting orders when the CQI is on," while "sessions classified as full-service and agency are more likely to seek to trade against IEX resting orders during the remainder of the day."³⁹

IEX states that its data evidences that members that enter liquidity-taking orders when the CQI is on "appear to be able to engage in a form of latency arbitrage by leveraging fast proprietary market data feeds and connectivity along with predictive strategies to chase short-term price momentum and successfully target resting orders at unstable prices."⁴⁰

One commenter states that the CQI is overbroad, and thus the impact of D-Limit orders will be "much broader, affecting all types of liquidity takers" including retail.⁴¹ The commenter reports that 15% of the retail-related liquidity-removing orders it routed to IEX in May 2020 arrived when the CQI was on.⁴² The commenter questions whether the CQI may have turned on when it was routing portions of a large retail order to multiple exchanges to fill.⁴³ If those away executions appeared to IEX to sequentially remove liquidity in a "crumbling" manner, then the CQI could have turned on. In that case, had IEX been displaying D-Limit orders, those orders would get repriced before the commenter could remove that interest from IEX at the previously displayed less-aggressive price point unless the commenter accounts for the IEX access delay when routing so that its routing does not cause the CQI to turn on.⁴⁴ Thus, the commenter asserts that the CQI detects more than just latency arbitrage and may broadly discriminate against all types of liquidity takers, in particular orders that are routed *simultaneously* to multiple

Market Policy Officer, IEX, dated August 3, 2020, at 5–6 ("IEX Third Response to Comments").

³⁹ Notice, *supra* note 3, at 72002.

⁴⁰ *Id.*

⁴¹ Citadel Second Letter, *supra* note 38, at 1.

⁴² See *id.* at 3–4. The commenter says that "[t]o provide a sense of scale, we executed over 2.5 million retail orders during the month of May 2020 that required more size than was available at the NBBO across all exchanges at the time of routing." *Id.* at 3. The commenter also states that it is the "leading destination for retail orders, executing approximately 40% of all U.S.-listed retail volume." Citadel Second Letter, *supra* note 38, at note 6.

⁴³ See *id.*

⁴⁴ See *id.* The commenter provided an example of an "actual retail order that removed displayed liquidity on IEX during the month of May 2020." See *id.* at 4. While it is unclear if this is a typical example, and the commenter did not provide timestamps or indicate the execution prices received, the scenario shows that the commenter did not receive a full and immediate execution on two venues (ChoeEDGX and Nasdaq), as it ended up posting 100 shares on each venue. See *id.*

²⁹ See 17 CFR 242.611.

³⁰ See also IEX First Response to Comments, *supra* note 7, at 1 (stating that "certain microsecond-level latency arbitrage strategies . . . act as a powerful disincentive to the posting of displayed quotes by market participants, which has led to a long-term trend of declining displayed liquidity and less transparency in equities trading").

³¹ See also Letter from Stephen John Berger, Managing Director, Citadel Securities, dated April 23, 2020, at 5 ("Citadel First Letter"). See also Letter from Stephen John Berger, Managing Director, Citadel Securities, dated August 14, 2020, at 2 ("Citadel Third Letter").

³² See Citadel First Letter, *supra* note 31, at 5. See also Citadel Third Letter, *supra* note 31, at 2.

³³ See Citadel First Letter, *supra* note 31, at 5.

exchanges to cross the spread and remove liquidity at the bid or offer price.⁴⁵

In light of this, the commenter questions whether D-Limit orders on IEX will require broker-dealers to make changes to their routing strategies, and if so, whether those changes would increase complexity, introduce risks, and be consistent with regulatory requirements applicable to the routing of marketable orders.⁴⁶ Specifically, the commenter asserts that “[a]ccounting” for the IEX speed bump means routing to IEX first and intentionally delaying routing to other exchanges when accessing displayed liquidity” (emphasis omitted).⁴⁷ In response, IEX states that how D-Limit orders will interact with intermarket sweep orders depends on how broker-dealers route, which is “precisely the same choice that brokers face in routing to every other type of displayed order that exists in the market today,” and the ability to account for geographic and technological differences (like the speed bump, which “is the equivalent of physical distance”) when routing is not uniquely affected by the presence of D-Limit orders.⁴⁸ As discussed directly below, the Commission has addressed this concern previously and reaffirms its views on the matter now, based on the understanding that intermarket routing can be accomplished in a manner to avoid such an outcome.

D-Limit orders should not necessitate material changes to routing strategies either for single orders or intermarket sweeps. The Commission previously addressed the commenter’s concern about routing an order to IEX and accounting for its access delay when the Commission approved IEX’s exchange registration. Specifically, the Commission explained that IEX’s speed bump is “well within the range of geographic and technological latencies that market participants experience today” such that “latency to and from IEX will be comparable to—and even less than—delays attributable to other markets that currently are included in the NBBO,” and the Commission found the delay to be de minimis, *i.e.*, so short as to not frustrate the purposes of Rule 611 by impairing fair and efficient access to IEX’s quotation.⁴⁹

In considering the current proposal and concerns raised by the commenter, the same explanation provided to support approval of IEX’s exchange registration pertains here. In effect, IEX’s physical access delay (from the POP to the matching engine) is understood by market participants as the practical equivalent of treating the location of IEX’s matching engine for routing purposes as more than 38 miles further away than its actual geographic location. In the Commission’s view, market participants can, and generally do, account for this fact when routing to IEX. Thus, these routing adjustments do not constitute “preferencing” of IEX because the adjustments do not mean a market participant has to arrive and trade first on IEX.⁵⁰ Rather, to prevent the CQI from observing away executions and turning “on” before an order sent to IEX can execute on IEX, a broker-dealer need only continue to apply current routing techniques prevalent today. Such techniques, such as smart order routing strategies, are commonplace today and already take into account geographic and technological latencies, as well as exchange access delays, to capture liquidity across multiple venues simultaneously without signaling those executions to the market in a way that would impact prices or available liquidity.⁵¹ If utilized, such routing

Secaucus, New Jersey, a user’s electronic message sent to the IEX trading system must physically traverse the IEX “coil,” which is a box of compactly coiled optical fiber cable equivalent to a prescribed physical distance of 61,625 meters (approximately 38 miles) (which is similar to the distance between the Nasdaq and NYSE data centers in Carteret and Mahwah, respectively). After exiting the coil, the message travels an additional physical distance to the IEX trading system, located in Weehawken, New Jersey. The entirety of that trip takes approximately 350 microseconds.

⁵⁰ By analogy, routing parts of a large order to multiple exchanges is similar to a group of friends who live in separate locations meeting at a restaurant for a 7:00 p.m. dinner reservation. Each friend leaves his or her house at a different time, depending on how far away each lives from the restaurant, and all plan to arrive at the same time to make their reservation. While the friend that lives furthest away needs to start the journey first, all arrive together. Routing a large stock order to multiple exchanges is the inverse of that hypothetical, in which the goal is to take liquidity on each exchange as close to simultaneously as possible before other market participants can see and react to those executions. Because of IEX’s speed bump, it often will be the “furthest away” venue and so the journey to reach it starts first, but the execution does not need to occur first on IEX and thus preferencing IEX is not required.

⁵¹ See, e.g., Securities Exchange Act Release No. 84528 (November 2, 2018), 83 FR 58338 (November 19, 2018) (Disclosure of Order Handling Information; Final Rule) and Letter from Daniel Aisen, CEO, Proof Services LLC, dated December 24, 2019, at 5 (“Proof Letter”). See also Securities Exchange Act Release No. 78101, (June 17, 2016), 81 FR 41142, 41153 (June 23, 2016) (File No. 10–222), at note 265 (discussing accounting for the non-variable IEX access delay when routing to IEX).

strategies could avoid triggering the CQI because quotes would not “crumble” in sequence as the various orders are routed to assure coordinated simultaneous executions across venues. The commenter asserts that routing first to IEX would be necessary to avoid triggering the CQI on market sweeps, but that does not mean routing to and arriving at IEX first. Rather, smart order routing that seeks to have orders arrive and execute simultaneously across multiple venues focuses on order arrival and the order transmission time is only a means to an end to achieve that outcome.⁵² Accordingly, the commenter has not presented persuasive evidence that all market sweeps necessarily trigger the CQI or that its liquidity-taking activities on IEX will be materially impacted to the detriment of retail investors. While the commenter presented evidence to show some correlation between its trading and the CQI being on, it did not present evidence that its trading caused the CQI to turn on or that such result is inevitable with current best practices in routing among broker-dealers. For instance, in the commenter’s example it routed to six exchanges but did so in a way that appears to have obtained an execution on IEX last. If the commenter routed in a way that resulted in its order executing on IEX near-simultaneously with its executions on other markets, then the CQI could not have been triggered by the commenter’s routed orders because the various exchange quotes would not have “crumbled” prior to the execution of the order on IEX. The means to route in this manner are common and do not require a broker-dealer to preference any particular exchange over any other exchange, as the point is to achieve simultaneous arrival and executions across multiple exchanges in a coordinated manner.

Rather, “accounting” for IEX’s de minimis speed bump when routing orders is just like accounting for any other technological or geographic latency, and doing so is consistent with applicable rules and regulations and does not require inappropriately preferencing IEX. IEX’s D-Limit proposal, because it does not introduce any new access delays, does not present any new issues in this respect. While the commenter focuses on the fact that IEX will be able to reprice D-Limit orders when the CQI triggers, market participants that post liquidity on IEX also monitor executions on away markets and could change or cancel

⁵² See, e.g., Proof Letter, *supra* note 51, at 5. See also *supra* note 50.

⁴⁵ See Citadel Third Letter, *supra* note 31, at 1–2.

⁴⁶ See *id.* at 5–6. See also Citadel Third Letter, *supra* note 31, at 4.

⁴⁷ Citadel Third Letter, *supra* note 31, at 4.

⁴⁸ Letter from John Ramsay, Chief Market Policy Officer, IEX, dated August 20, 2020, at 2–3.

⁴⁹ See Securities Exchange Act Release No. 78101 (June 17, 2016), 81 FR 41142, 41161 (June 23, 2016). Specifically, after entering through the POP in

their quotes on IEX in response to that same information. In other words, with or without D-Limit orders, if a broker-dealer does not seek to maximize its fill rates while minimizing information leakage by accounting for latencies (e.g., technological, geographic, or access delay) when it routes portions of large orders to multiple venues near-simultaneously, the broker-dealer runs the risk of missing out on executions at displayed prices. Accounting for those latencies is possible, using affordable and readily-available technology, and addresses the commenter's concern, which is unrelated to whether IEX has or does not have D-Limit orders.

IEX further asserts that investors will not be negatively impacted when trying to access a quote on IEX that contains a displayed D-Limit order because brokers representing investor orders or trading on their behalf generally are not able to time their orders to arrive with the level of sophistication required to engage in latency arbitrage because they lack the "low-latency tools to aggregate data from all the markets and react to price changes in microseconds."⁵³ Further, referencing the commenters that support the proposal, IEX explains that "market participants who rely on the ability to access liquidity through 'intermarket sweep' orders have clearly said they do not believe D-Limit would limit their ability to access liquidity at displayed prices."⁵⁴ With respect to hedging activities, IEX argues that market makers hedge throughout the day and "[i]t is not credible to suppose that orders from market makers sent to hedge risks on various markets happen to converge at IEX in the tiny time increments when the CQI signal is on."⁵⁵ IEX further states that market makers that rely on the equities markets to hedge have supported the proposal.⁵⁶ Finally, IEX states that marketable orders to take liquidity when the CQI is on account for over 20% of displayed volume executions on IEX, but less than 2% of total trading volume of non-displayed interest.⁵⁷

IEX's characterization of latency arbitrage is supported by commenters asserting that latency arbitrage negatively impacts liquidity and price discovery. One commenter believes that "speed advantages . . . have tilted the

playing field in favor of firms specializing in 'latency arbitrage,' reducing the willingness of both long-term investors and market makers to display quotes, to the detriment of price discovery and market efficiency."⁵⁸ Another commenter states that "[t]he disincentive to all market participants to provide displayed quotes in fear of getting 'picked off' when the price of a security is in transition to a new price level continues to plague displayed markets."⁵⁹ Similarly, another commenter opines that "[a] growing body of academic research suggests that latency arbitrage strategies are equivalent to zero-sum 'races' between high-frequency traders (HFT) and may actually discourage liquidity provision by both HFT and non-HFT."⁶⁰ Other commenters similarly assert that latency arbitrage causes some institutions to

⁵³ See Letter from Kevin Duggan, Managing Director, Capital Markets, Ontario Teachers' Pension Plan, Benoit Gauvin, Vice-President, CDPQ, and Alex Done, Deputy Comptroller, Office of New York City Comptroller, et al., at 1 ("Joint Letter from 27 Asset Managers"). See also Letter from Lev Bagramian, Senior Securities Policy Advisor, Better Markets, Inc., dated May 15, 2020.

⁵⁴ See, e.g., Letter from Mehmet Kinak, VP & Global Head of Systematic Trading & Market Structure and Jonathan D. Siegel, VP & Senior Legal Counsel—Legislative & Regulatory Affairs, T. Rowe Price, dated February 5, 2020, at 1 ("T. Rowe Letter"); and Letter from Brian Urey, Senior Trader, Allianz Global Investors, dated May 12, 2020 ("Allianz Letter").

⁵⁵ See and Letter from Marius-Andrei Zoican, Assistant Professor of Finance, University of Toronto-Mississauga, dated January 20, 2020, at 1 ("Zoican Letter"). See also Letter from John Christofilos, Senior VP, Chief Trading Officer, AGF, dated February 11, 2020, at 2 ("AGF Letter") ("It is clear that latency arbitrage affects investors, brokers, and market makers, and as a result, the market is less liquid and more susceptible to volatile price swing."); Letter from Sean Paylor, Trader, AJO, dated February 10, 2020, at 4 ("AJO Letter") ("Trading from the back of the queue leads to lower fill rates during periods when the quote is stable and greater adverse selection when it is not ('last man standing' before a price change)."); Letter from Kenneth A. Bertsch, Executive Director, Jeffrey P. Mahoney, General Counsel, Council of Institutional Investors, dated February 11, 2020, at 2 ("CII Letter") ("Long-term investors are at real and substantial risk from speed advantages of a small number of trading firms that specialize in 'latency arbitrage,' which imposes a multibillion-dollar tax on institutional investors. A recent study sponsored by the Financial Conduct Authority suggested that this activity is endemic, and results in substantial losses to all liquidity providers."); Letter from Philip Berlinski, Co-Chief Operating Officer, Equities, Global Markets, Goldman Sachs & Co. LLC, dated February 26, 2020, at 3 ("Goldman Sachs Letter") ("Adverse selection stemming from latency arbitrage can have a negative effect on the national market system because liquidity providers may be more inclined to provide less liquidity at wider spreads."); and Letter from Sanjana Kapur, Compliance Officer, Equity Compliance, Jefferies LLC, dated February 5, 2020, at 3 (explaining that one of its agency algorithms seeks "the ability to participate in price discovery by displaying your interest without having to bear the cost of adverse selection [getting 'picked off' by arbitrage-based strategies relying primarily on speed]").

avoid posting displayed liquidity on exchanges, which they say can impact liquidity and price discovery.⁶¹

The comment file on this proposal reflects a dichotomy of views on the issue of latency arbitrage. On the one hand, some commenters question IEX's characterization of latency arbitrage and the performance of IEX's CQI, which is intended to detect it. On the other hand, other commenters, including investment firms with longer-term investment horizons and agency broker-dealers, state that they are adversely impacted by latency arbitrage.

Even though the CQI is mostly off and comes on only when certain market-moving conditions are present, those small increments of time are meaningful on IEX because, as discussed above, a material amount of activity occurs during those moments.⁶² In those rare moments when market prices are in transition, a race condition exists between liquidity providers who want to reprice their on-exchange displayed liquidity to reflect the changing market prices and the liquidity takers who want to take before those updates can occur.⁶³ This creates information asymmetries and can lead to other externalities, which can affect the willingness of many market participants to post displayed liquidity because it subjects their orders to adverse selection when prices move and they are not able to see or react as fast to those changing conditions.⁶⁴ In turn, this race can have a meaningful effect on all market participants because it can incentivize investors to trade in the dark, either off exchange or through non-displayed

⁶¹ See, e.g., AGF Letter, *supra* note 60, at 1; Letter from Joseph Scafidi, Global Head of Trading, Carlos Oliveira, Head of Trading Analytics and Market Structure, Brandes Investment Partners, LP, dated February 20, 2020, at 1 ("Brandes Letter"); Letter from Ray Ross, Chief Technology Officer, Clearpool Group, dated January 21, 2020, at 2 ("Clearpool Letter"); Goldman Sachs Letter, *supra* note 60, at 4; Letter from Gary Thompson, Executive Director, Head of Global Trading, Vontobel Asset Management, Inc., dated February 14, 2020, at 1.

⁶² See *supra* note 36 and accompanying text.

⁶³ See, e.g., Letter from Joseph Saluzzi, Themis Trading, dated February 6, 2020, at 2 ("[t]hose who have paid top-dollar for high speed data related products, race each other to pick off stale orders from investors who haven't necessarily paid for the same low latency technology.") ("Themis Letter"); Zoican Letter, *supra* note 60, at 1 (stating that there is a growing body of academic research suggests that latency arbitrage strategies are equivalent to zero-sum 'races' between high-frequency traders).

⁶⁴ See *supra* notes 58–61 and accompanying text. See also, e.g., AJO Letter, *supra* note 60, at 2 ("Instead of offering all market participants equal access at the same speed, the exchanges have created a multi-tiered system, effectively tilting the odds in favor of a small subset of firms that possess the resources to invest in the lowest latency infrastructure") and Allianz Letter, *supra* note 59, at 2.

exchange order types.⁶⁵ The result is that a valuable source of liquidity may instead seek out dark non-exchange trading venues where the speed traders' advantages are moot, but in doing so this liquidity is no longer displayed to and accessible by the market as a whole. Such an outcome does not advance the Exchange Act's goal of promoting fair and orderly securities markets. IEX's D-Limit order type seeks to compete with those other trading venues by incentivizing more displayed liquidity through improved execution quality for liquidity providers.

Furthermore, exchange functionality that protects resting displayed orders against adverse selection resulting from latency arbitrage will improve the execution quality experienced by market participants that post displayed liquidity and are affected by such adverse selection. This improved execution quality could encourage more displayed liquidity, which in turn, would contribute to fair and orderly markets and support the public price discovery process. Specifically, if sufficiently protected against being "picked off" when the conditions for latency arbitrage are present, long term investors will no longer experience those relatively poor executions and thus will have less incentive to avoid posting displayed orders on exchanges. Accordingly, as suggested by some commenters, long term investors could shift a portion of their order flow back onto the exchange as displayed orders. The result would be an increase in displayed liquidity as a more diverse group of long term investors participates in the exchange market, which would in turn facilitate fair competition, economically efficient executions, and an opportunity for long term investors' orders to be executed without the participation of a dealer.⁶⁶

A substantial amount of IEX's overall execution volume is attributable to non-displayed interest. Thus, when the CQI is on, the marketable interest IEX receives is not seeking merely to access liquidity on IEX in the normal course, but rather is seeking specifically to remove a displayed quote on IEX in a manner consistent with what IEX identifies as latency arbitrage.⁶⁷ In other words, IEX has demonstrated that the market conditions that trigger activation of the CQI are the same short-term

market conditions that the most highly sophisticated latency-sensitive traders detect and seek to trade on. The Commission also finds persuasive comments from asset managers supporting the proposal who argue that the prevalence of such trading strategies impacts their willingness to participate in the on-exchange displayed market where the public price discovery process takes place. IEX responds to those long-standing concerns by now offering a narrowly tailored tool that balances the ability of long-term investors to access displayed liquidity in the ordinary course against the current structural advantages enjoyed by short-term latency arbitrage trading strategies that rely on superior access to the fastest data and connectivity, while also encouraging liquidity providers to post more displayed liquidity.⁶⁸ Many commenters state that the proposal would be useful in addressing such concerns without being overbroad.⁶⁹ The Commission concludes that this attempt by IEX to promote the ability of long-term investors to post liquidity on its exchange by counterbalancing the existing advantages used in short-term trading strategies is consistent with the Exchange Act.⁷⁰ As discussed below, exchanges should be able to innovate to address this concern.

Therefore, because IEX's proposal promotes the interest of long term investors in a narrowly tailored manner that will inure to the benefit of displayed markets, leading to increased displayed liquidity from which all market participants ultimately will benefit, IEX's proposal is consistent with the Exchange Act, including the protection of investors and the public interest.

One commenter questions whether the statistics IEX presented in its filing, which were based on data collected during September 2019, "are representative of other time periods or of different market environments."⁷¹

⁶⁵ IEX notes in its filing that other exchanges use transaction rebates and volume tiers "to essentially compensate market makers and other liquidity providers for posting aggressive limit orders" that are subjected to the effects of latency arbitrage, but IEX believes "that these pricing schemes can contribute to a number of conflicts of interest and market distortions including, among others, conflicts of interests, excess intermediation and potential adverse selection, market fragmentation, complexity, the proliferation of new order types to enable avoidance of fees, and elevated fees to subsidize rebates" and does not use them. See Notice, *supra* note 3, at 72002.

⁶⁶ See, e.g., Allianz Letter, *supra* note 59; Brandes Letters, *supra* note 61; and T. Rowe Letter, *supra* note 59.

⁶⁷ See *infra* Sections III.C–E.

⁷¹ Letter from Joan C. Conley, Senior Vice President & Corporate Secretary, Nasdaq, Inc., dated

The commenter considers that period to be normal market conditions and believes that IEX should provide more "representative data" to address how the CQI operates during periods of market volatility.⁷² The commenter states that IEX's failure to do so renders IEX's proposal incapable of showing whether the proposed D-Limit order is appropriately tailored to address the issue that IEX seeks to address.⁷³ The commenter also states that IEX provided aggregated data across all stocks but does not provide data in its filing for different types of stocks, like thinly traded symbols, which may or may not experience the CQI in the same way as more liquid stocks, and thus questions whether IEX's "figures hold true for all categories of symbols, including those which are subject to frequent or routinely-high levels of volatility."⁷⁴

In response to these concerns, IEX provided additional data in its two responses to comments and asserts that the results remain consistent.⁷⁵ With respect to longer periods and periods of higher market-wide volatility, IEX looked at data from the fourth quarter of 2019 and found that, "based on IEX average-weighted volume, the CQI was on for 0.021% of the time during regular market hours, virtually the same as the 0.025% of the time it was on for September 2019."⁷⁶ IEX also looked at the period of January to April 2020, which IEX states involved an "unprecedented 'stress test' as a result of market volatility connected to the COVID-19 pandemic," and IEX reports that the CQI was on between 0.026% (for January 2020) and 0.125% (for March 2020) of the time during regular market hours.⁷⁷ IEX asserts that, "during

January 21, 2020, at 6 ("First Nasdaq Letter"). See also, e.g., Letter from Jeffrey Bacidore, Ph.D., President, The Bacidore Group, LLC, dated February 13, 2020, at 2–3 ("Bacidore Letter").

⁷² See First Nasdaq Letter, *supra* note 71, at 4–7. See also Bacidore Letter, *supra* note 71, at 2–3.

⁷³ See First Nasdaq Letter, *supra* note 71, at 4–7. See also Bacidore Letter, *supra* note 71, at 2–3.

⁷⁴ See First Nasdaq Letter, *supra* note 71, at 4–7. See also Bacidore Letter, *supra* note 71, at 2–3.

⁷⁵ See IEX First Response to Comments, *supra* note 7; IEX Second Response to Comments, *supra* note 38.

⁷⁶ See IEX First Response to Comments, *supra* note 7, at 15. IEX also examined aggregate CQI data during a December 2018, which IEX asserts was a period of relatively higher market-wide volatility. While IEX found that the CQI was "on" for relatively more time (an average of 0.060% per symbol per trading day during December 2018 compared to 0.025% during September 2019), IEX characterized that relatively higher level as "nonetheless extremely low, corresponding to approximately 15 seconds per day per symbol on average during regular market hours." *Id.* at 16.

⁷⁷ IEX Second Response to Comments, *supra* note 38, at 16. IEX states that the CQI was "on" longer in March 2020 "because of the extraordinary

Continued

one of the most volatile and unprecedented months in the history of the stock market," the CQI was only on for 0.125% of the trading day on average.⁷⁸ Further, IEX calculated the percent of displayed volume that traded when the CQI was "on" during that period, and found that it ranged between 22% (for March 2020) to 24.4% (for January 2020).⁷⁹ IEX concludes that its updated data "confirms that the CQI signal performs extremely well and consistently in extreme, and in 'normal' market conditions."⁸⁰

IEX also examined data from the fourth quarter of 2019 on the amount of time that the CQI was on for certain individual stocks that IEX classified as subject to greater volatility, and found that for those stocks, "which tend to also be stocks that are more thinly-traded, the CQI was on less often than for stocks with lower volatility."⁸¹

With respect to the CQI and thinly traded stocks, IEX examined data from the fourth quarter of 2019 and found that "the CQI was actually significantly less active than for stocks with higher [average daily volume]."⁸² IEX explained that, for thinly traded stocks, "the CQI necessarily will fire less often . . . because there are fewer data points from which to draw in making predictions" and thus "there will be less repricing of the orders, which directly counters the arguments that CQI would have a larger impact in those symbols."⁸³ IEX concludes that its data shows that D-Limit would not impede access to thinly-traded or other stocks.⁸⁴

A few commenters also request that IEX provide more information on the accuracy of the CQI. One commenter states that IEX's filing provides no data of when CQI "activates mistakenly in circumstances where it should not do so" where it would cause "needless missed executions for liquidity takers."⁸⁵ Similarly, another commenter states that IEX should provide more data on the accuracy of the CQI, such as the percentage of time that the CQI accurately predicts NBBO changes, the

increase in the number of quote changes per stock in that month." *Id.* at 17.

⁷⁸ *Id.* at 16–17. IEX notes that during that period, the market-wide circuit breakers were triggered 4 times in 8 trading days and the VIX volatility index recorded its highest-ever value on March 16. *See id.*

⁷⁹ *Id.* at 17. IEX observes that this proportion remained "remarkably consistent throughout the period" even though the absolute number of trades was much higher in March and April. *See id.*

⁸⁰ *Id.*

⁸¹ IEX First Response to Comments, *supra* note 7, at 16.

⁸² *Id.* at 15.

⁸³ *Id.* at 16.

⁸⁴ *Id.*

⁸⁵ *See* First Nasdaq Letter, *supra* note 71, at 7.

number of times the CQI inaccurately predicted a change, and how the accuracy rate has evolved with each update.⁸⁶

IEX responds to those comments by providing data on CQI performance during the January to April 2020 period, where it found that the CQI "accurately predicted the direction of the next price change (not time bound) in 75.5% of the cases, on a volume-weighted basis."⁸⁷ A 75% accuracy rate for the CQI shows that it succeeds, far more often than not, in detecting the conditions for latency arbitrage, and thus it is successful in informing order types designed to mitigate the impact of that latency arbitrage.⁸⁸ The CQI is on so infrequently and detects events that are such idiosyncratic anomalies within the current market structure that only a small number of market participants are capable of detecting and acting upon them. As such, the CQI does not result in needless missed executions for most traders, though it will make it more difficult for a few latency arbitrage traders to profit by taking advantage of idiosyncrasies in market structure to trade with stale-priced displayed quotes on IEX, as further discussed below.

Thus, in response to all of the commenters' concerns discussed in this section about the CQI, and the representativeness of the data that IEX provided in support of its proposal, IEX provided additional data and analysis and concludes that "(i) the fraction of the trading day that the CQI is on is consistent in different time periods; (ii) the CQI is on for less of the trading day for thinly-traded securities compared to all securities; (iii) the CQI is on for less of the trading day for securities that experienced higher volatility than for lower volatility securities; and (iv) during a high volatility period, the period of time the CQI was on

⁸⁶ *See* Letter from Joanna Mallers, Secretary, FIA Principal Traders Group, dated January 21, 2020, at 6 ("FirstFIA PTC Letter").

⁸⁷ IEX Second Response to Comments, *supra* note 38, at 21. The CQI's accuracy is substantial.

⁸⁸ Further, the D-Peg order type, which is designed to protect non-displayed midpoint orders from latency arbitrage and is informed by the CQI, has been popular with the agency brokers and full-service brokers that it is intended to protect. Those market participants carefully monitor their execution quality and thus must regard the CQI as accurate or else they would not be expected to use D-Peg orders. *See id.* at 15 (noting that "D-Peg alone has executed over 111 billion shares of volume since IEX was approved as an exchange with over 90% of that volume from agency and full-service brokers"). If the CQI were not accurate, then users of D-Peg orders would needlessly miss out on executions and, as a result, would receive poor execution quality and be expected to avoid using the D-Peg order type. Because the D-Peg order type is popular on IEX, its ability—by function of the CQI—to protect the liquidity provider is apparent.

continued to be extremely low (about 15 seconds during the trading day)."⁸⁹

IEX has provided a sufficient amount of data and analysis to allow the Commission to consider the full range of issues raised by IEX's proposal. Based upon the overall record, including the data and analysis submitted by IEX described immediately above and the comments received, the Commission finds that the proposal is consistent with the Exchange Act. As explained above, the Commission finds that the proposal promotes the interest of long term investors in a narrowly tailored manner that will inure to the benefit of displayed markets, leading to increased displayed liquidity from which all market participants ultimately will benefit.⁹⁰ The Commission finds that the proposed D-Limit order type is narrowly and appropriately tailored to achieve those benefits. As discussed above, some commenters characterize the data IEX provided in its filing as insufficient, and question how the CQI performs during periods of higher market volatility and for thinly traded stocks.⁹¹ However, the Commission is persuaded by the data and analysis IEX provided in its filing and its responses to comments.

Market events over the past several months have provided an environment for IEX to test the consistency of its CQI in periods of significant volatility. IEX did so and shows that the period of market volatility observed in early 2020 did not cause the CQI's behavior to differ markedly from the data IEX provided in its filing. Further, IEX shows that more volatile stocks and thinly traded stocks were not unduly impacted by the CQI. Thus, the Commission concludes that the data IEX provided in its filing on latency arbitrage and the CQI was "representative of other time periods or of different market environments" and IEX's "figures hold true" for different types of stocks that were subject to higher levels of volatility.

B. Prior Commission Consideration of the CQI

One commenter expressed concern that the use of CQI by IEX "shifts the exchange's role from a platform designed to facilitate price discovery into an active participant in the price discovery function."⁹²

⁸⁹ *Id.*

⁹⁰ *See supra* notes 62–70 and accompanying text.

⁹¹ *See supra* notes 32–34 and accompanying text.

⁹² Letter from Adam Nunes, Head of Business Development, Hudson River Trading LLC, dated January 21, 2020, at 2 ("HRT Letter"). *See also* Citadel First Letter, *supra* note at 31, at 11. One commenter argues that IEX will be exercising actual

The Commission disagrees with the commenter's characterization of the CQI functionality. The CQI does not place IEX into the role of an active participant in the price discovery function, nor is IEX making investment pricing decisions through the CQI. As the Commission previously addressed in connection with IEX's exchange registration, IEX's discretionary orders are "unique in the way that the discretion functionality will be turned 'on' or 'off' depending on IEX's quote stability determination."⁹³ Among other things, IEX has "encoded in its rule the totality of the discretionary feature" and therefore the "hardcoded conditionality" of the discretionary feature does not provide IEX "with actual discretion or the ability to exercise individualized judgment when executing an order."⁹⁴ Though the Commission was discussing the DPeg order type in the IEX exchange registration context, the Commission recognizes that D-Limit will use the exact same CQI functionality that underlies the DPeg. And, as the Commission has previously stated, the CQI, which uses a "pre-determined, objective set of conditions that are detailed in IEX's [rules]" and which any market participant can thus recreate on its own, will allow users to submit D-Limit orders and have them operate as designed and as reflected in IEX's rules. In so doing, users of D-Limit orders can better achieve their goals when their orders operate efficiently. Further, as discussed below, D-Limit orders and any associated liquidity-provision and

discretion over the execution of a D-Limit order and thus will be engaging in traditional broker-dealer activities in offering the D-Limit order type informed by the CQI, because "its predictive nature and potential for error makes it difficult to distinguish from typical broker-dealer order routing and execution algorithms (which are also codified)." See First FIA PTC Letter, *supra* note 86, at 5-6. The Commission previously addressed the CQI and IEX's discretionary order type functionality and determined that it does not result in IEX engaging in traditional broker-dealer activities. See, e.g., Securities Exchange Act Release No. 78101 (June 17, 2016), 81 FR 41142, 41153 (June 23, 2016) (File No. 10-222) (granting the application of IEX for registration as a national securities exchange). Such also is the case with IEX's D-Limit proposal, as D-Limit orders will not allow IEX to exercise any discretion on any particular order by deviating from the CQI and D-Limit functionality, which is hardcoded in the IEX rulebook. While broker-dealer algorithms also are codified, broker-dealers do not have to align their algorithms with Section 6 of the Exchange Act nor do they need to file all applicable rules with the Commission as IEX does for the CQI and its order types.

⁹³ Securities Exchange Act Release No. 78101, (June 17, 2016), 81 FR 41142, 41153 (June 23, 2016) (File No. 10-222).

⁹⁴ *Id.*

price discovery benefits will be available to all users.

C. Quote Accessibility and Impact on Displayed Markets

Discretionary order types informed by the CQI, discussed above, are not new for IEX. What is novel in this proposal is that (1) the proposed D-Limit order could be *displayed* and (2) the CQI functionality would be used to "exercise discretion" to move the order to a *less* aggressive price.⁹⁵

Several commenters assert that the D-Limit Order type, if displayed, would result in what the commenters refer to as "quote fading" or "phantom liquidity" on IEX if market participants have difficulty accessing the D-Limit order liquidity that is displayed on IEX, and some question whether the D-Limit order type would be consistent with the requirements of Regulation NMS.

1. Quote Fading and Phantom Liquidity

Quote fading and phantom liquidity both refer, from the perspective of the liquidity taker, to the ability to execute against a displayed quote. The ability of any market participant to successfully execute against any particular displayed quote is subject to a number of factors and is not guaranteed on any market, as at any time any market participant can be seeking to execute against an order that is being repriced, changed, cancelled, or executed by a different market participant.

Some commenters express concern that displayed D-Limit orders might result in more "quote fading" or "phantom liquidity" and thus negatively impact market participants seeking to take liquidity. One commenter describes its concern "about quote accessibility as a result of a displayed limit order being repriced based on IEX's crumbling quote indicator" and recommends that consideration be given "to the impact on an order attempting to seek liquidity from a posted D-Limit Order" when the CQI causes the D-Limit order to reprice because "the IEX protected quote that broker-dealers are attempting to access

⁹⁵ Nondisplayed D-Limit orders would not raise any new or novel issues not previously considered in connection with IEX's nondisplayed discretionary order types. See, e.g., Securities Exchange Act Release No. 78101 (June 17, 2016), 81 FR 41142 (June 23, 2016) (File No. 10-222) (In the Matter of the Application of: Investors' Exchange, LLC for Registration as a National Securities Exchange; Findings, Opinion, and Order of the Commission). See also Letter from R. T. Leuchtkafer, dated August 3, 2020, at 2-3 ("Leuchtkafer Letter") (pointing out that IEX currently reprices displayed orders in certain scenarios, including to comply with other regulatory requirements, in situations that do not involve the CQI).

would no longer be at the price that they are trying to execute against" ⁹⁶ That commenter further asserts that even if the extent of quote fading caused by the use of D-Limit Orders is not "meaningful," there is no "*de minimis* threshold to the Firm Quote Rule" and therefore D-Limit orders "could further erode the integrity of a displayed quotation." ⁹⁷ Another commenter raises the same concern, stating that D-Limit quotes would be "nothing more than a 'maybe' quote or indication of interest." ⁹⁸ Likewise, another commenter articulates this concern by asserting that firms "would be blind as to how to make informed trading decisions" because there will be no indicator in the market data feeds that will reveal when IEX's quote includes a D-Limit order, which the commenter characterizes as an order type that is subject to fading. ⁹⁹ More specifically, a different commenter opines that specific types of market participants— institutional and retail traders—will be harmed by the quote fading concerns presented by D-Limit orders when those traders experience declining fill rates as they sweep the market by sending a portion of a larger order to IEX but are unable to access D-Limit orders that are being repriced by IEX. ¹⁰⁰

In contrast, other commenters, including institutional investors and asset managers that trade equities, do not believe D-Limit orders will less accessible. For example, one commenter says that it is "confident that D-Limit

⁹⁶ Letter from Ellen Greens, Managing Director, Equity and Options Market Structure, SIFMA, dated February 5, 2020, at 1-2 ("SIFMA Letter"). See also Letter from Andrew Stevens, General Counsel, IMC, dated January 22, 2020, at 2 ("IMC Letter") (characterizing D-Limit as a "perilous gimmick that creates a safe zone for illusory orders" that would allow "systematic quote fading").

⁹⁷ See SIFMA Letter, *supra* note 96, at 3. Cf. Goldman Sachs Letter, *supra* note 60, at 2-3 (arguing that, with respect to firm quote requirements, "D-Limit Orders are no different from the operation of peg order types" and thus comply with the requirements of Rule 602 of Regulation NMS). The possibility that a displayed quotation can change before someone can access it does not, by itself, mean that the quote is not firm for purposes of Rule 602. For example, the liquidity provider that posted the order might have changed or cancelled it.

⁹⁸ See Letter from Mary M. Dunbar, D.I.A Piper LLP (US), dated March 10, 2020, at 2-3 ("D.I.A Piper Letter"). D-Limit orders are not equivalent to an indication of interest, because indications of interest normally require a subsequent and manual "firm up" instruction by the submitting member in order to execute against incoming trading interest. See, e.g., Securities Exchange Act Release No. 83663 (July 18, 2018), 83 FR 38768, 38847-48 (August 7, 2018) (Regulation of NMS Stock Alternative Trading Systems; Final Rule) (discussing IOIs).

⁹⁹ See First Nasdaq Letter, *supra* note 71, at 2-3.

¹⁰⁰ See First FIA PTC Letter, *supra* note 86, at 3-4.

orders will be as accessible to our orders and those representing other institutional asset managers as any other liquidity is available today from other venues.”¹⁰¹ Other commenters opine that D-Limit orders could be more accessible than liquidity on other venues.¹⁰² Another commenter asserts that quote fading “is not a new phenomenon” and says that they “experience quote fading every day on every other exchange” as latency arbitrage commonly trade ahead of their liquidity seeking orders.¹⁰³ The commenter explains that, in its trading experience, “latency arbitrage are front-running our liquidity-seeking child orders and taking the posted liquidity we seek before our orders arrive.” The commenter regards this as “unnecessary intermediation” and argues it “is not an example of price discovery, it is a form of predatory trading that increases investor costs” and is “also the very behavior that D-Limit seeks to combat.”¹⁰⁴ Further, one commenter believes that institutional investors would not be harmed by displayed D-Limit orders “since institutional order ‘taking’ strategies are driven by a fundamental demand for liquidity and are not intentionally seeking to trade while the CQI is ‘on.’”¹⁰⁵ Another commenter agreed, noting that institutional traders “almost always initiate orders during stable markets, so they should have little trouble accessing displayed D-Limit quotes” and when they do trade using “reactive” strategies “there is such a dramatic gap in speed between the elite proprietary trading firms and even the relatively fast agency brokers, that by the time these reactive agency tactics are able to act, any favorable opportunities have already been exploited by their

faster counterparts” and “they would have likely been too late either way.”¹⁰⁶

Responding to the concerns about quote fading and phantom liquidity, IEX asserts that “an exchange quote is accessible only to the degree that the participant is able to send a message that can execute against the quote before someone else accesses it or the quote is cancelled before the taker’s order arrives.”¹⁰⁷ On that point, IEX argues that the commenters raising concerns over quote fading are implying that the D-Limit repricing “will deprive investors or their agents of prices they otherwise would be able to access.”¹⁰⁸ IEX does not believe that to be the case, and explains that “exchange quotes are not equally accessible to all participants” but rather “are only reliably accessible to participants using the fastest trading strategies and the fastest market information.”¹⁰⁹

For example, with respect to situations in which the NBB or NBO is in transition, IEX believes that “quotes on exchanges that remain at the soon-to-be stale price will either be accessed first by a fast market participant, or they will be canceled before they can be executed by anyone” and thus concludes that “[i]n either event, quotes on other exchanges will not be accessible in these moments to institutional investors, which are not seeking to trade in these moments.”¹¹⁰ IEX instead focuses on the prospect that if the D-Limit functionality “gives liquidity providers more incentive to provide displayed liquidity, then any investor seeking to trade in the 99.96% of the day when the CQI is off will have more opportunities to access liquidity on IEX, without the need to buy new low-latency tools.”¹¹¹

Some commenters question whether similar order types, if adopted by other exchanges, could collectively result in quote fading or have other negative market-wide effects. One commenter believes that the market-wide impact “may be profound for symbols that are subject to routinely-high levels of price volatility” or “during times of market duress,” and that such impact would be “amplified” if other exchanges adopt something similar.¹¹² Another

commenter states that approval of this proposal would “open a Pandora’s box, as other exchanges introduce similar order types, leading to more and more liquidity ‘fading’ in a correlated manner, an effect which could be most pronounced in times of market stress.”¹¹³ Likewise, another commenter states that “it is important to consider that most of the liquidity on IEX is ‘dark’/non-displayed . . . if one or more exchanges with significant lit liquidity were to offer the same D-Limit Order or similar order types, this functionality would exacerbate the number of inaccessible quotes in the marketplace.”¹¹⁴

IEX’s D-Limit order type, if displayed, would not impair access to IEX’s quotation because IEX is not introducing any additional access delay. Further, IEX would only rarely reprice the order in response to a very targeted and specific pre-defined signal that suggests a high potential for latency arbitrage. When the CQI is off, which, as discussed above, is virtually the entire regular trading hours session, a D-Limit order is simply a regular limit order and thus will be as equally accessible as any other limit order on IEX. For the small part of the day when the CQI is on, market participants that are not engaging in latency arbitrage trading strategies are unlikely to be seeking to trade with a D-Limit order precisely when it is in the process of being repriced by IEX because IEX’s data shows that latency arbitrage trading (as signaled by the CQI) is very highly concentrated and reactive in nature.¹¹⁵ Conversely, dozens of commenters that represent institutional traders and investors say that they do not trade in this manner and are unable to compete with the small number of firms that purchase the necessary systems, connectivity, and exchange proprietary market data to target their trading to those precise periods when crumbling quotes cause the CQI to turn “on.”¹¹⁶ Exchanges should be able to innovate to

¹⁰¹ Letter from David Brooks, Director of Trading, The London Company of Virginia LLC, dated February 20, 2020, at 2 (“The London Company Letter”).

¹⁰² See Joint Letter from 27 Asset Managers, *supra* note 58 (noting that “other exchanges with protected quotations sell multiple speeds of technology and data, which may make their quotations less accessible to those who do not purchase the same tier of access from the exchange. So, in practical terms, we believe that IEX D-Limit meets the current standards of a protected quote and these quotes will likely be more accessible to traditional investors than quotes on other exchanges.”).

¹⁰³ AJO Letter, *supra* note 60, at 4.

¹⁰⁴ *Id.*

¹⁰⁵ T. Rowe Letter, *supra* note 59, at 2. The commenter believes that D-Limit orders would help liquidity providers defend themselves against “reactive strategies used by a small subset of proprietary trading firms that invest in high speed infrastructure to predict price changes, leverage small latency advantages, and opportunistically trade against stale quotes.” *Id.*

¹⁰⁶ Proof Letter, *supra* note 51, at 5.

¹⁰⁷ See IEX First Response to Comments, *supra* note 7, at 7.

¹⁰⁸ *See id.*

¹⁰⁹ *See id.*

¹¹⁰ *See id.* at 8.

¹¹¹ *See id.* at 7–8.

¹¹² See First Nasdaq Letter, *supra* note 71, at 3. But see IEX Second Response to Comments, *supra* note 38 (discussing data from January to April 2020 to show that the CQI does not behave profoundly differently during periods of exceptional price volatility and market duress).

¹¹³ See Bacidore Letter, *supra* note 71, at 3.

¹¹⁴ See SIFMA Letter, *supra* note 96, at 4. See also DLA Piper Letter, *supra* note 98, at 5; Citadel First Letter, *supra* note at 31, at 10. To the extent that another exchange seeks to adopt its own speed bump, crumbling quote indicator, and D-Limit order type, the Commission would carefully analyze it and the comments received thereon, and consider whether the new proposal is narrowly tailored to achieve its stated objectives and consistent with the Exchange Act and the rules and regulations thereunder.

¹¹⁵ See, e.g., IEX Second Response to Comments, *supra* note 38, at 13–14.

¹¹⁶ See, e.g., T. Rowe Letter, *supra* note 59, at 2 (noting that “institutional order ‘taking’ strategies are driven by a fundamental demand for liquidity and are not intentionally seeking to trade while the CQI is ‘on.’”).

address this competitive imbalance in a manner that is consistent with the Exchange Act.

Given how narrowly tailored the CQI is and how infrequently it activates, IEX's D-Limit order type will not result in the average market participant experiencing significant quote fading when trying to take liquidity on IEX, though, as discussed above, it will by design effect speed traders engaging in latency arbitrage. By protecting liquidity providers in a narrowly tailored way, IEX may attract additional liquidity through D-Limit orders, including from new types of market participants, which will promote more displayed liquidity that will be available to all market participants.¹¹⁷ Therefore, the Commission finds that the D-Limit order type is consistent with Section 6(b)(5) of the Exchange Act in that it is designed to promote just and equitable principles of trade, to remove impediments to and perfect the mechanism of a free and open market and a national market system and, in general, to protect investors and the public interest.

2. Automated Quotes and Rule 611

In general, Rule 611 under Regulation NMS protects the best "automated" quotations of exchanges by obligating other trading centers to honor those "protected" quotations by not executing trades at inferior prices, or "trading through" such best automated quotations.¹¹⁸ Only an exchange that is an "automated trading center" displaying an "automated quotation" is entitled to this protection.¹¹⁹ Among other things, an "automated quotation" must be immediately and automatically executable.¹²⁰

Several commenters argue that D-Limit orders will not be "automated quotations" under Regulation NMS, and thus they should not be "protected" quotations under Rule 611 of Regulation NMS.¹²¹ They argue that the D-Limit functionality, combined with the IEX speed bump and IEX's ability to bypass it to adjust the price of displayed D-Limit orders when the CQI is on, is inconsistent with the requirements for immediate and automatic execution

required for automated quotations to be protected under Rule 611.¹²²

Other commenters disagree and argue that D-Limit orders would qualify as protected quotations under Regulation NMS. For example, one commenter notes that the Commission, when it approved IEX's exchange registration, already concluded that IEX is an automated trading center with protected quotations.¹²³ The commenter asserts that "[t]he introduction of the D-Limit Order does not alter that analysis" and that "[t]here is no delay embedded within D-Limit Orders."¹²⁴ The commenter concludes that "D-Limit Orders are no different" and "are as accessible as any other quote."¹²⁵

The Commission previously determined that IEX can maintain a protected quotation when it approved IEX's exchange registration.¹²⁶ Because IEX is not introducing any new delay or modifying its speed bump in connection with D-Limit orders, IEX's quote can continue to be an "automated quotation" that is "protected" under Rule 611 even if it contains a D-Limit order.¹²⁷

D. Unfair Discrimination

Section 6(b)(5) of the Exchange Act requires, among other things, that rules of the Exchange may not be designed to

permit unfair discrimination between customers, issuers, brokers, or dealers.¹²⁸ Several commenters argue that the D-Limit order type will unfairly discriminate against liquidity takers in favor of liquidity providers. One commenter asserts that D-Limit orders are "specifically designed to advantage liquidity providers, and to allow them to avoid unfavorable executions," but "displayed quotations on IEX will be more difficult to access for liquidity takers when the market is moving in their favor" to the extent that IEX's quote is composed of D-Limit orders that get repriced.¹²⁹ Another commenter similarly believes that D-Limit orders will unfairly discriminate against liquidity takers, "particularly for orders that are sent to more than one venue for execution, such as intermarket sweep orders," if market participants need to modify their routing practices.¹³⁰

One commenter critiques IEX's argument that the CQI is "on" only for a limited duration by referring to IEX's data that shows "a very significant portion of total trading activity will be affected, as 33.7% of marketable orders are received and 24% of displayed volume is executed during these periods."¹³¹ Similarly, another commenter points to commentary from an IEX officer saying that "[i]n November 2019, just 3 member firms at IEX were responsible for 55% of all the lit taking volume while the [CQI] Signal was 'on,' even though those firms accounted for only 13% of the total volume on IEX."¹³² The commenter asserts that "[b]ased upon such statistics, the Commission should consider whether latency arbitrage on

¹¹⁷ 15 U.S.C. 78f(b)(5).

¹¹⁸ First FIA PTG Letter, *supra* note 86, at 4. See also First Nasdaq Letter, *supra* note 71, at 7; DLA Piper Letter, *supra* note 98, at 6. Another commenter believes that D-Limit orders will discriminate between fast and slow liquidity takers, and says that liquidity takers that do not engage in latency arbitrage will be forced to protect against financial harm from the D-Limit functionality by building technology to mimic or predict the CQI functionality. See DLA Piper Letter, *supra* note 98, at 6. The commenter argues that doing so would be "a prohibitively expensive option for many of them and completely impractical if other markets adopted similar rules." *Id.* As further explained below, the repricing of D-limit orders will be applied only in rare and discrete moments of time when the CQI is triggered, which significantly reduces the possibility of D-Limit being applied to the detriment of liquidity takers not engaged in latency arbitrage strategies. Furthermore, as noted above, several commenters who engage in liquidity-taking trading activity—but do not employ latency arbitrage—state that they have been harmed by latency-arbitrage strategies, and do not believe the CQI will inhibit their ability to access IEX's quote. See *supra* notes 103–104.

¹¹⁹ Citadel First Letter, *supra* note 31, at 7–8.

¹²⁰ *Id.* at 7.

¹²¹ First Nasdaq Letter, *supra* note 71, at 7.

¹¹⁷ See also *supra* note 68–70 and accompanying text. See also *infra* note 138 and accompanying text.

¹¹⁸ See 17 CFR 242.611. "Trading through" refers to the purchase (sale) of NMS stock at a price lower (higher) than the best bid (offer).

¹¹⁹ See 17 CFR 242.600(b)(4) (defining "automated trading center") and 17 CFR 242.600(b)(3) (defining "automated quotation").

¹²⁰ See 17 CFR 242.600(b)(3).

¹²¹ See, e.g., First Nasdaq Letter, *supra* note 71, at 9; HRT Letter, *supra* note 92, at 4; SIFMA Letter, *supra* note 96, at 3; and IMC Letter, *supra* note 96, at 2.

¹²² See, e.g., SIFMA Letter, *supra* note 96, at 3; Citadel First Letter, *supra* note 31, at 7; DLA Piper Letter, *supra* note 98, at 4; Letter from Kristen Malinconico, US Chamber of Commerce, Center for Capital Markets Competitiveness, dated April 23, 2020, at 1; Letter from John L. Thornton, Co-Chair, Hal S. Scott, President, and R. Glenn Hubbard, Co-Chair, Committee on Capital Markets Regulation, dated April 23, 2020, at 1–2 ("Committee on Capital Market Regulation Letter"); Bacidore Letter, *supra* note 71, at 2; First FIA PTG Letter, *supra* note 86, at 7; HRT Letter, *supra* note 92, at 4; First Nasdaq Letter, *supra* note 71, at 10–11. Another commenter argues that "the combination of the 'Discretionary Limit' order type and the IEX speed bump will impair fair and efficient access to IEX displayed quotes, meaning that the intentional access delay can no longer be considered *de minimis* under the Commission's Automated Quotations Interpretive Guidance in the context of this specific order type. Therefore, displayed quotes using the 'Discretionary Limit' order type will not qualify as 'automated quotations' for purposes of Rule 611." See First FIA PTG Letter, *supra* note 86, at 6–7. See also HRT Letter, *supra* note 92, at 2–3; DLA Piper Letter *supra* note 98, at 4–5.

¹²³ See Goldman Sachs Letter, *supra* note 60, at 2.

¹²⁴ *Id.*

¹²⁵ *Id.* See also IEX Second Response to Comments, *supra* note 38, at 10.

¹²⁶ See Securities Exchange Act Release No. 78102, (June 17, 2016), 81 FR 40785, 41162 (June 23, 2016) (File No. S7-03-16). See also Securities Exchange Act Release No. 78102 (June 17, 2016) (File No. S7-03-16) (Commission Interpretation Regarding Automated Quotations Under Regulation NMS). See also 17 CFR 242.600(b)(4) (defining "automated quotations").

¹²⁷ See also *supra* Section III.A (discussing the CQI).

IEX is actually the serious and widespread problem that IEX asserts it to be" and urges the Commission to "consider whether it would be fair for IEX to discriminate against 45% of its lit taking volume to address a perceived problem with only three firms."¹³³

Other commenters believe that D-Limit orders will not be unfairly discriminatory. For example, one commenter believes that D-Limit orders will "equally benefit long-term investors, their brokers, and market makers alike" and notes that "[a]ny market participant can use D-Limit, regardless of their sophistication or technological capability, and any speed or information advantage they may or may not have."¹³⁴ Another commenter explains that D-Limit will "provide any Member with narrowly targeted protection . . . but without the need for the Member to have any geographical or informational advantages or its own predictive analytical capabilities."¹³⁵ Some commenters suggest that the D-Limit order type will benefit all market participants in that it has the potential to increase the depth of displayed liquidity and narrow spreads in some stocks,¹³⁶ contribute to price discovery,¹³⁷ and encourage more market participants, particularly long

term investors, to submit displayed liquidity on IEX.¹³⁸

In response to these comments, IEX agrees that the D-Limit order type will benefit liquidity providers "by protecting their orders during discrete moments when latency arbitrage strategies are most aggressive" but argues that such benefit is not unfair because, in turn, liquidity takers will benefit from "an increased supply of liquidity from a more diverse group of participants" and will attract "more stable liquidity that is not driven by sub-millisecond price moves, whether they are 'making' or 'taking' liquidity."¹³⁹ IEX emphasizes that the D-Limit order type "is not intended to ensure that trades are profitable" but rather is designed to promote displayed liquidity.¹⁴⁰

IEX's proposal identifies a legitimate disadvantage in latency arbitrage, which many market participants say they face when posting displayed liquidity on exchanges. As further explained above, the proposed D-Limit order type (operating in conjunction with the CQI) is designed to operate in a manner that protects investors and the public interest because it is narrowly tailored to address this concern.¹⁴¹ Additionally, a large number of market participants (including a diverse group of agency brokers, institutional traders, asset managers, and pension funds that collectively manage trillions of dollars' worth of investor assets) commented on this proposal to confirm that aggressive liquidity taking activity during CQI events is of such concern and impact on them, and the investors on whose behalf they invest, that it affects their trading

and can dissuade them from posting liquidity.

The Commission has critically reviewed and considered the data and analysis that IEX provides in its submissions to show that non-pegged limit orders on IEX are systematically subjected to adverse impacts of latency arbitrage strategies. IEX has provided extensive information in its filing and the letters it submitted in response to comments. Further, because the CQI formula is codified in IEX's rulebook, it is fully transparent and commenters had the opportunity not only to review IEX's material and critique it, but to submit their own trading data and analysis to the Commission on the existence of latency arbitrage, the effectiveness of the CQI in detecting it, and the efficacy of IEX's discretionary order types in combating it, though no commenter did so.

IEX's receipt of a significant percentage of marketable orders in a short period of time during crumbling quote events negatively affects market participants that post displayed liquidity on IEX. As discussed above, IEX states that though the CQI was, on average, on for only 0.007% of the trading day for each security, IEX received 33.7% of marketable orders and executed 24% of displayed volume during this short time period.¹⁴² Subsequently, during the very volatile period of January to April 2020, IEX reports that the CQI was on between 0.026% (for January 2020) and 0.125% (for March 2020) of the time during regular market hours¹⁴³ and the percent of displayed volume that traded when the CQI was "on" during that period ranged between 22% (for March 2020) to 24.4% (for January 2020).¹⁴⁴ The displayed volume figures reflect the fact that relatively little of IEX's overall transaction volume currently involves the execution of displayed orders.¹⁴⁵ The effects of latency arbitrage therefore appear more pronounced for liquidity providers that display interest on IEX. Further, IEX has shown that the CQI performs consistently over calm markets and periods of more volatile trading, so

¹³³ *Id.* IEX responds by suggesting that the commenter "misread[s]" that data and asserts that "aggressive taking orders while the CQI is on are strongly correlated to latency arbitrage strategies, not just those from the top three takers." IEX First Response to Comments, *supra* note 7, at 14. As discussed above, the Commission concludes that latency arbitrage is occurring on IEX and it is a problem for liquidity providers on IEX. As further discussed above, the Commission concludes that IEX's D-Limit proposal appropriately balances competition among orders and the interests of long term and short term traders in a manner that meets the needs of long term investors through a narrowly tailored order type that will promote liquidity for all market participants.

¹³⁴ T. Rowe Price Letter, *supra* note 59, at 2. See also CII Letter, *supra* note 60, at 2; The London Company Letter, *supra* note 101, at 2; Joint Letter from 27 Asset Managers, *supra* note 58, at 2; Vanguard Letter, *supra* note 65, at 3.

¹³⁵ Letter from Tyler Gellach, Executive Director, Healthy Markets Transparency and Trust, dated February 14, 2020, at 7-8 ("Healthy Markets First Letter"). See also Leuchtkafer Letter, *supra* note 95, at 2-3.

¹³⁶ See, e.g., Letter from Thomas M. Merritt, Deputy General Counsel, Virtu Financial, LLC, dated January 16, 2020, at 2; and Letter from Eric Swanson, CEO, XTX Markets LLC (Americas), dated January 17, 2020, at 5.

¹³⁷ See, e.g., Letter from Peter D. Shutsman, Global Head of Equity Trading, Capital Group, dated March 16, 2020, at 2; Letter from Curtis F. Bradbury, Chief Operating Officer, Stephens Inc., dated February 28, 2020, at 2; and Letter from David Cannizzo, Managing Director, Head of Electronic Trading & Market Structure, and Rich Delayo, Director, Electronic Trading & Market Structure, Raymond James & Associates, Inc., dated February 24, 2020 ("Raymond James Letter"), at 2; Allianz Letter, *supra* note 59.

¹³⁸ See, e.g., Joint Letter from 27 Asset Managers, *supra* note 58; Clearpool Letter, *supra* note 61, at 2; Themis Letter, *supra* note 63, at 3; CII Letter, *supra* note 60, at 4; Zoican Letter, *supra* note 60, at 2; and Vanguard Letter, *supra* note 65, at 2.

¹³⁹ IEX Second Response to Comments, *supra* note 38, at 20. See also Allianz Letter, *supra* note 59 (asserting that latency arbitrage "hurts the interests of our clients and the performance of their investments" not just when they post liquidity but also when they take liquidity because they "have to pay more (when buying) or sell for less as there is less displayed liquidity at the best bid and offer prices").

¹⁴⁰ IEX Second Response to Comments, *supra* note 38, at 20. IEX also notes that "there are trade-offs between using D-Limit, as opposed to a pegged or standard limit order, including in cases where a firm's priority is obtaining a faster execution rather than avoiding adverse selection." *Id.* at 21. Further, IEX asserts that "unlike the 'asymmetric advantages' that other exchanges routinely sell today (different connectivity, market data, trading protocols) for millions of dollars, IEX is offering D-Limit to all of its members equally and at no additional cost compared to any other limit order." *Id.* at 5.

¹⁴¹ See *supra* Section III.A.

¹⁴² See *id.* at 72001-02.

¹⁴³ See IEX Second Response to Comments, *supra* note 38, at 16. IEX states that the CQI was "on" longer in March 2020 "because of the extraordinary increase in the number of quote changes per stock in that month." *Id.* at 17.

¹⁴⁴ See *id.* IEX observes that this proportion remained "remarkably consistent throughout the period" even though the absolute number of trades was much higher in March and April. See *id.*

¹⁴⁵ IEX states in its filing that only 13% of traded volume was from displayed limit orders. See Notice, *supra* note 3, at 72002.

its application to D-Limit orders is well understood.

Based on the Commission's understanding of broker-dealers, as also reflected in the comment letters from institutional traders, most broker-dealers have not purchased the fastest connectivity and market data from multiple individual exchanges that are necessary to be able to trade at the precise moments in time identified by the CQI. In the race to access a "stale" quote, speed is paramount, and the systems, connectivity, and data needed to achieve the necessary speed to take advantage of the information asymmetries that underlie latency arbitrage are expensive and uncommon among broker-dealers.

The CQI formula, by design, identifies only successively crumbling markets. As shown by the data above, it is not overbroad and does not, for example, turn on in response to intermarket sweeps from large orders that execute simultaneously across multiple markets. Thus, D-Limit orders will not reprice in response to normal market conditions and regular liquidity sweeps, and thus will not harm long-term investors who take liquidity. Rather, the unique crumbling market conditions that the CQI identifies are rare, and can only be recognized and acted on by the most sophisticated broker-dealers whose ability to profit from these moments comes at the expense of the institutional investors who do not or cannot reasonably compete. IEX has proposed an order type to offset the speed advantage that some traders have in a manner that is not overbroad in its application.¹⁴⁶

Accordingly, the Commission concludes that IEX has identified and quantified a latency arbitrage concern that adversely impacts a diverse set of participants on its exchange, many of which are sophisticated about market structure and have commented on how they have seen first-hand the impact as they trade in the markets on behalf of others, including public investors. In addressing the adverse impacts of latency arbitrage, the Commission acknowledges that D-Limit orders will provide a benefit to liquidity providers but not liquidity takers, and will negatively impact liquidity takers that employ latency arbitrage strategies. For the reasons discussed below, the Commission finds that neither the benefit provided to liquidity providers nor the negative effects on liquidity takers employing certain strategies is unfairly discriminatory under the Exchange Act.

IEX has narrowly tailored the functionality of D-Limit orders to address a very specific trading dynamic that takes place during an exceptionally small fraction of the trading day (for any one CQI event and collectively for all CQI events across different types of stocks and under different market conditions). Disparate capabilities with respect to systems, connectivity, and market data between liquidity providers and liquidity takers using latency arbitrage strategies disadvantage the ability of liquidity providers to update potentially stale quotes. While displayed D-Limit orders may impact a large number of marketable orders that seek to access a stale D-Limit quote precisely when IEX is in the process of repricing it, that is the specific harm against which IEX is seeking to protect liquidity providers—the combination of a large number of marketable orders all collectively targeting those infrequently occurring precise moments when disparate access to low-latency systems, connectivity, and data sources favors a few short term traders at the expense of long term traders.¹⁴⁷

IEX's D-Limit order attempts to address that advantage through a narrowly tailored order type that carries out the liquidity provider's instructions by exercising discretion infrequently to update the D-Limit order's limit price using predetermined, transparent, and rule-based automated standards.¹⁴⁸ Further, IEX will allow all traders to use D-Limit orders on the same terms and without additional charge to protect their limit orders from targeted latency arbitrage. D-Limit orders consequently have the potential to encourage more types of market participants to post more displayed liquidity on an exchange, and may contribute to price

discovery and displayed depth to the benefit of all market participants.

Thus, the proposal is not unfairly discriminatory because it makes available a benefit that any liquidity provider can readily access and provides a narrowly focused protection that is calibrated to impact only the small number of liquidity takers that engage in latency arbitrage in order to incentivize liquidity providers to post orders for the benefit of all market participants. While protecting against latency arbitrage with this order type will affect a large number of marketable orders received in small increments of time, those orders dissuade many liquidity providers from posting limit orders on exchanges. Consequently, D-Limit orders should benefit all market participants by incentivizing more firms to post limit orders and thereby contribute to liquidity that all market participants can access. Finally, as discussed above, D-Limit orders will encourage long term investors to participate in the displayed exchange market by protecting them against one particular strategy employed by short term traders. It is not unfairly discriminatory for an exchange to address that advantage in a narrowly tailored manner that promotes investor protection and the public interest. Accordingly, the Commission concludes that IEX's proposal is not designed to permit unfair discrimination between customers, issuers, brokers, or dealers.

The Commission is mindful that, in considering the IEX proposal, some commenters compare it to a recent proposal from CboeEDGA to adopt an access delay of up to 4 milliseconds for all liquidity taking orders during which liquidity providers could continue to access their orders without delay.¹⁴⁹ One commenter states that IEX's D-Limit proposal is similar to the CboeEDGA proposal in that liquidity takers would be disadvantaged in favor of liquidity providers, and liquidity providers on the host exchange could be advantaged vis-à-vis liquidity providers on other exchanges that do not offer similar protections.¹⁵⁰ One commenter questions the data provided by both exchanges as to the existence of a problem, its purported impact, or the

¹⁴⁷ In addition, non-displayed D-Limit orders will not affect liquidity takers that cannot see non-displayed orders and thus would not purposefully and knowingly route to IEX to trade with them.

¹⁴⁸ As discussed above, IEX's rules set forth the precise predetermined mathematical formula that IEX uses to determine whether a "crumbling quote" situation exists and the D-Limit order abides by those rules based on system logic in an entirely automated manner. Neither IEX nor the member submitting the order has any actual discretion or ability to exercise individualized judgment when using a D-Limit order. When the CQI is off (on average approximately 99% of the regular trading day), a D-Limit order will behave like any other limit order. When the CQI is on, IEX will only reprice the specific side (bid or offer) at issue and will only move the price to a less aggressive price that is only one MPV away (lower for a bid or higher for an offer) from the CQI price and IEX will not provide the user with any optionality to do otherwise. Further, when a D-Limit order reprices, it will receive a new timestamp, and thus will not receive any priority advantage over other orders.

¹⁴⁹ See Securities Exchange Act Release No. 88261 (Feb. 21, 2020), 85 FR 11426 (Feb. 27, 2020) (CboeEDGA-2019-012) (order disapproving proposed rule change to introduce a liquidity provider delay mechanism) ("CboeEDGA Order").

¹⁵⁰ See First FIA PTG Letter, *supra* note 86, at 4. See also IMC Letter, *supra* note 96, at 2 and Letter from Joanna Mallers, Secretary, FIA Principal Traders Group, dated April 23, 2020, at 1 ("Second FIA PTG Letter"). The Commission addresses concerns about competitive disadvantages in the next section.

¹⁴⁶ See *supra* Section III.A.

benefits of the proposed solution.¹⁵¹ The commenter also questions why IEX is providing a benefit “to sophisticated proprietary trading firms acting as liquidity providers without a corresponding obligation.” As noted above, any user will be able to submit D-Limit orders.¹⁵² Two other commenters assert that the concerns raised about the CboeEDGA proposal are similar to the concerns that these commenters raise for this proposal, notably the quote fading, unfair discrimination, burden on competition, and narrowly tailored concerns discussed above.¹⁵³

Another commenter notes that, unlike CboeEDGA’s proposal, the D-Limit order will not provide market participants with an “option” to change their order but rather will reprice 100% of the time when the CQI triggers, which process is both “transparent and certain.”¹⁵⁴

Another commenter that opposed CboeEDGA’s proposal and supports IEX’s proposal distinguishes the two proposals by explaining that D-Limit is “non-discriminatory” in that any market participant can use it without “technological investment that is generally outside the reach for most institutional investors and their brokers.”¹⁵⁵ The commenter also explains that IEX’s proposal is “deterministic and transparent” and presents less of a quote fading concern for institutional investors since the CQI is narrowly tailored.¹⁵⁶

The two proposals differ substantially.¹⁵⁷ Specifically, IEX, unlike CboeEDGA, presented substantial evidence of latency arbitrage occurring on its market and has narrowly tailored D-Limit orders to specifically protect against it. In the CboeEDGA disapproval order, the Commission stated that CboeEDGA did not “provide specific analysis as to why it is appropriate to apply the 4 millisecond delay to all incoming executable orders that would remove liquidity from the EDGA Book from all market participants as opposed to tailoring a response to target the trading of a relatively small number of market participants who engage in latency arbitrage.”¹⁵⁸ Second, CboeEDGA did not address the impact on relatively slower liquidity providers, who might be unable to cancel or modify their quotes during the 4 millisecond delay and thus “would continue to face the risk of adverse selection” and would be unable to benefit from the CboeEDGA delay.¹⁵⁹ Finally, CboeEDGA did not “provide[] specific analysis or demonstrate[] that the proposed rule change would not permit unfair discrimination against liquidity taking orders that are not related to latency arbitrage as they would be treated in the same manner as orders engaged in latency arbitrage that the Exchange seeks to target in its effort to protect EDGA liquidity providers.”¹⁶⁰

In contrast, as further explained above, IEX provides data and analysis that demonstrate a harm caused by latency arbitrage strategies employed by liquidity takers with significant technological advantages over liquidity providers and those liquidity takers that do not engage in latency arbitrage trading strategies.¹⁶¹ Because IEX will reprice all D-Limit orders without further action from the user, all users will benefit equally regardless of their technological capabilities and ability to take action within a prescribed period. Likewise, D-Limit orders will be repriced only in rare and discrete moments of time when the CQI is triggered, which would significantly reduce the possibility of D-Limit being applied to the detriment of liquidity takers not engaged in latency arbitrage

strategies. Thus, IEX’s D-Limit proposal does not raise the same issues as those raised by the CboeEDGA proposal because D-Limit is narrowly tailored to be triggered only at precise and specific moments in time during which resting orders may be exposed specifically to latency arbitrage.¹⁶²

E. Burden on Competition

Finally, the Exchange’s proposal is consistent with Section 6(b)(8) of the Exchange Act,¹⁶³ which requires that the rules of a national securities exchange not impose any burden on competition that is not necessary or appropriate in furtherance of the purposes of the Exchange Act. Several commenters question whether D-Limit orders will impose a burden on competition that is inconsistent with the Exchange Act.¹⁶⁴ One commenter argues that D-Limit orders “are designed to take advantage of the fact that other market participants are subject to the IEX speed bump when updating prices, whereas D-Limit orders are not.”¹⁶⁵ Another commenter claims that the D-Limit order type “advantages liquidity providers on IEX over liquidity providers on other venues, as IEX liquidity providers can free-ride on the pricing heuristics and risk taking capabilities of others by posting prices equal to the NBBO and relying on IEX to observe away executions and reprice D-Limit orders to frequently avoid unfavorable executions.”¹⁶⁶

Other commenters that support the proposal say it is a narrowly tailored competitive response that facilitates the ability of natural liquidity providers to protect themselves from microsecond liquidity arbitrage, and therefore it furthers competition.¹⁶⁷ For example,

¹⁶² See also IEX First Response to Comments, *supra* note 7, at 7-9; Clearpool Letter, *supra* note 61, at 2-3; and Healthy Markets First Letter, *supra* note 135 (each contrasting D-Limit with the CboeEDGA proposal).

¹⁶³ 15 U.S.C. 78f(b)(8).

¹⁶⁴ See 15 U.S.C. 78f(b)(8) (requiring that the rules of a national securities exchange not impose any burden on competition that is not necessary or appropriate in furtherance of the purposes of the Exchange Act).

¹⁶⁵ See HRT Letter, *supra* note 92, at 3. The commenter stated that “[e]xchanges should not have the ability to make investment pricing decisions such as pricing orders using price predictions” and argues that the resulting “competition will not be on fair terms as exchanges have inherently better access to the matching engine . . .” *Id.* at 2.

¹⁶⁶ Citadel First Letter, *supra* note 31, at 10. See also First FIA PTG Letter, *supra* note 86, at 4.

¹⁶⁷ See, e.g., Allianz Letter, *supra* note 59. But see Letter from Joan C. Conley, Senior Vice President & Corporate Secretary, Nasdaq, Inc., dated March 26, 2020, at 2 (“Some broker-dealers choose to compete with proprietary trading firms, and purchase data and connectivity products that allow them to do so, while others choose not to do so.”).

¹⁵¹ See Citadel First Letter, *supra* note at 31, at 5-6. The commenter further states that a D-Limit order will “lose much of its value if IEX is alone at the NBBO and therefore routed to first, as the CQI signal will not provide added protection in this situation” and therefore may “not generally be expected to narrow prevailing market-wide spreads.” *Id.* at 6. As the Commission discusses above, D-Limit orders are intended to incentivize investors to display limit orders in general and at any price. Even if D-Limit orders are not used to narrow the best displayed quotes, to the extent they add displayed liquidity at the best displayed quotes liquidity takers would still benefit as they would have access to more liquidity at the best prices.

¹⁵² See Citadel First Letter, *supra* note at 31, at 9. The consideration of benefits provided to registered market makers in return for obligations to the market recognizes that market makers are typically afforded special privileges by exchanges, including preferential priority and margin treatment, in return for their undertaking quoting and other obligations. D-Limit orders will be available for use by any market participant and will not entitle the user to any additional benefits.

¹⁵³ See First Nasdaq Letter, *supra* note 71, at 11; Citadel First Letter, *supra* note at 31, at 2. See also HRT Letter, *supra* note 92, at 3; Committee on Capital Market Regulation Letter, *supra* note 122, at 2. Cf. SIFMA Letter, *supra* note 96, at 3.

¹⁵⁴ Healthy Markets First Letter, *supra* note 135, at 2. See also T. Rowe Letter, *supra* note 59, at 2.

¹⁵⁵ T. Rowe Letter, *supra* note 59, at 2.

¹⁵⁶ See *id.*

¹⁵⁷ The CboeEDGA proposal would have broadly imposed a non-tailored access delay constantly and consistently during trading hours to all liquidity taking messages, but liquidity providers would have been able to access their displayed orders (e.g., to change or cancel them) without being subject to the delay.

¹⁵⁸ CboeEDGA Order, *supra* note 149, at 11436.

¹⁵⁹ *Id.* at 11436.

¹⁶⁰ *Id.* at 11435.

¹⁶¹ See *supra* note 40 and accompanying text and Section II.A.

one commenter believes that “the D-Limit order type is pro-competitive” because it offers market participants that do not buy the fastest market data “a potential way to mitigate the risk of posting liquidity without participating in a costly high-speed race to minimize latency.”¹⁶⁸

In response to the comments, IEX asserts that “[t]he asymmetry involved in the latency arbitrage strategies that are the focus of D-Limit favors the few participants that can take liquidity using the most sophisticated tools, in contrast to both market makers and brokers acting for investors that provide liquidity by posting displayed quotes.”¹⁶⁹ In particular, IEX argues that brokers representing investors “must cope with the latency caused by geographic dispersion of exchanges, the additional latency caused by systems configurations required to comply with regulatory and risk parameters in their capacity as agent, and the need to route orders in different ways to meet the needs of their various clients” and, as a result, they are “destined to lose out to firms that can prioritize speed over all other factors.”¹⁷⁰ IEX concludes that the resulting “imbalance in market competition between those who provide liquidity, versus those who take it, necessarily reduces the incentives to provide displayed quotes and therefore reduces liquidity available to investors.”¹⁷¹ Further, IEX argues that because every D-Limit order will “be required to specify a limit price, which may or may not be equal to the NBBO,” these orders should “contribute meaningfully to price discovery, as commenters have stated.”¹⁷²

As discussed at length above, the D-Limit order type is narrowly tailored to accomplish its objectives by mitigating the effects of latency arbitrage for long-term investors while incentivizing more displayed liquidity on the Exchange. Presently, as noted by several commenters with institutional trading experience, many market participants are reluctant to post displayed liquidity because of their prior experience with having that interest be adversely selected by latency arbitrage traders with whom they cannot reasonably

compete.¹⁷³ To take advantage of their low-latency systems and technology, latency arbitrage traders purchase connectivity and proprietary market data from exchanges, which they utilize to react faster to changing market prices than other market participants. Those other market participants might not be able to afford those same low-latency systems, or purchase high-end connectivity and market data from multiple individual exchanges to protect themselves. The resulting competitive imbalance between latency arbitrage traders and others can make those other market participants reluctant to post displayed limit orders on exchanges. The lack of displayed liquidity can, in turn, harm price discovery and lead to greater off-exchange trading, which can negatively impact markets and market participants. Exchanges should be able to innovate to address this competitive imbalance in a manner that is consistent with the Exchange Act.

IEX’s proposal seeks to better balance the interests of liquidity providers and long-term investors seeking liquidity with those of short-term investors utilizing latency arbitrage strategies. The D-Limit functionality will help mitigate the effects of latency arbitrage on liquidity providers and, as explained above, will likely lead to more displayed liquidity on the Exchange, which benefits all market participants through additional liquidity and enhanced public price discovery.¹⁷⁴ Further, because it is so narrowly tailored, liquidity takers who are not employing latency arbitrage strategies are unlikely to be seeking to remove a D-Limit order when it is being repriced, and thus D-Limit orders will not impose a burden on liquidity.

Accordingly, the Commission finds that D-Limit orders will not impose any burden on competition that is not necessary or appropriate in furtherance of the purposes of the Exchange Act. The D-Limit order type is IEX’s competitive response to mitigate current competitive imbalances between liquidity providers and latency arbitrage liquidity takers. It is designed to encourage market participants to post more priced limit orders, including displayed orders, on IEX, and thereby promotes just and equitable principles of trade, removes impediments to and perfects the mechanism of a free and open market and a national market, and, in general, protects investors and the public interest.

IV. Conclusion

It is therefore ordered, pursuant to Section 19(b)(2) of the Exchange Act,¹⁷⁵ that the proposed rule change (SR-IEX-2019-15) be, and it hereby is, approved.

By the Commission.
Vanessa A. Countryman,
Secretary.

[FR Doc. 2020-19204 Filed 8-31-20; 8:45 am]
BILLING CODE 8011-01-P

SECURITIES AND EXCHANGE COMMISSION

[Release No. 34-89693; File No. 265-33]

Asset Management Advisory Committee

AGENCY: Securities and Exchange Commission.

ACTION: Notice of meeting.

SUMMARY: Notice is being provided that the Securities and Exchange Commission Asset Management Advisory Committee (“AMAC”) will hold a public meeting on September 16, 2020, by remote means. The meeting will begin at 9:00 a.m. (ET) and will be open to the public via webcast on the Commission’s website at www.sec.gov. Persons needing special accommodations to take part because of a disability should notify the contact person listed below. The public is invited to submit written statements to the Committee. The meeting will include a discussion of matters in the asset management industry relating to the ESG and Private Investments Subcommittees; and improving diversity and inclusion. It will also include a follow-up discussion on COVID-19 matters relating to AMAC’s meeting of May 27, 2020.

DATES: The public meeting will be held on September 16, 2020. Written statements should be received on or before September 11, 2020.

ADDRESSES: The meeting will be held by remote means and webcast on www.sec.gov. Written statements may be submitted by any of the following methods. To help us process and review your statement more efficiently, please use only one method. At this time, electronic statements are preferred.

Electronic Statements

- Use the Commission’s internet submission form (<http://www.sec.gov/rules/other.shtml>); or
- Send an email message to rule-comments@sec.gov. Please include File Number 265-33 on the subject line; or

¹⁷⁵ 15 U.S.C. 78s(b)(2).

¹⁶⁸ Vanguard Letter, *supra* note 65, at 3 (further noting that “[o]rganizations that do not pay for data products that provide unparalleled speed advantages are discouraged from posting liquidity on exchanges because they may receive unfavorable executions”). See also Allianz Letter, *supra* note 59; Raymond James Letter, *supra* note 137.

¹⁶⁹ IEX First Response to Comments, *supra* note 7, at 3.

¹⁷⁰ *Id.*

¹⁷¹ *Id.*

¹⁷² IEX Second Response to Comments, *supra* note 38, at 21.

¹⁷³ See *supra* note 116 and accompanying text.

¹⁷⁴ See *supra* Section III.A.

Admin Record No.

18

June 24, 2025

VIA Email

Vanessa Countryman
Secretary
U.S. Securities and Exchange Commission
100 F Street, N.E.
Washington, D.C., 20549-1090

Re: Securities Exchange Act Release No. 34-102895 (SR-IEX-2025-02), IEX Options Application

Dear Ms. Countryman:

I appreciate the opportunity to comment in connection with IEX's proposal for an Options Facility.

I presently serve as an Associate Professor of Law with tenure at the George Mason University School of Law.

I recently served on the Investor Advisory Committee of the Securities and Exchange Commission and was the chairman of the Market Structure Subcommittee of that Advisory Committee. I previously served on the FASB's FASAC advisory Committee.

I currently serve on FINRA's Investor Issues Committee.

I am writing in my individual capacity, and my views are my own.

My views are however informed by my work as a professor of securities law. My views are also informed by my experience as Republican Senior Counsel and Chief Economist to the House Committee on Financial Services, where I took academic leave from my teaching position to serve from May 2013 until April 2015 as an advisor to Chairman Hensarling on a variety of financial regulatory issues.

I write in support of IEX Group's application for an Options Facility.

This proposal represents the kind of market-driven innovation that Congress envisioned when it directed the Commission to facilitate competition and technological advancement in our national market system. For nearly a decade, IEX and other new exchanges have successfully demonstrated that exchanges can simultaneously protect liquidity providers from predatory trading while improving execution quality for all market participants.

The data regarding this applicant's analogous equity market innovations is unambiguous: IEX's speed bump and D-Limit order type have increased displayed liquidity, reduced adverse selection, and enhanced price discovery—all while maintaining fair access for legitimate trading strategies.

The opposition to IEX's options facility employs many of the same arguments that incumbent exchanges and high-frequency trading firms advanced against IEX's original exchange application in 2016 and later against IEX's D-Limit order type.

In both instances, critics claimed IEX's innovations would harm retail investors, create "quote fading" or other concerning market effects, and undermine market efficiency. In both instances, these concerns proved unfounded. The Securities and Exchange Commission unanimously approved D-Limit, and when Citadel challenged that decision in federal court, the D.C. Circuit unanimously upheld the Commission's approval, finding that IEX's protections were properly designed to combat latency arbitrage while preserving fair access to displayed quotes.

Now, this comment file sees deployment of similar arguments against IEX's options proposal, despite offering limited new evidence and notwithstanding the successful track record of IEX's existing protections on the equity side. Rather than building upon empirical evidence from prior innovations, critics continue to advance theoretical concerns about this options facility.

As I observed in my 2015 comment letter supporting IEX's original application, regulatory decisions that unduly favor incumbent interests would represent precisely the kind of market distortion that undermines the public interest. This letter demonstrates that IEX's options facility deserves approval for five fundamental reasons:

First, the criticisms leveled by opponents fundamentally mischaracterize how IEX's protections operate and ignore the substantial body of evidence supporting their effectiveness.

Second, IEX's innovations represent exactly the kind of competition-enhancing market structure improvements that the 2005 Atkins-Glassman NMS dissent championed as superior to rigid regulatory mandates.

Third, the historical record—from IEX's initial approval through the D-Limit litigation—establishes a pattern of opposition from certain firms that has not been substantiated by subsequent market evidence.

Fourth, a detailed legal and empirical analysis of IEX's proposal confirms that it will benefit market participants by reducing the adverse effects of latency arbitrage while preserving fair access to displayed liquidity.

Fifth, the unique characteristics of options markets—including their reliance on exchange-based trading and their vulnerability to cross-asset latency arbitrage—make IEX's protections even more valuable in the options context than they have proven to be in equities.

The Commission should approve IEX's application and continue its important work of fostering innovation and competition in our national market system.

Section I: Responding to Criticisms and Correcting Mischaracterizations

The opposition letters filed against IEX's options application employ many of the same arguments that have been previously examined by both the Commission and the federal judiciary. These critiques rest on fundamental misunderstandings of how IEX's protections operate and fail to account for the substantial empirical evidence demonstrating their benefits.

The "Quote Fading" Characterization

Central to the opposition is the claim that IEX creates improper "quote fading"—a term that obscures the distinction between what IEX prevents (predatory latency arbitrage) and what it enables (legitimate risk management). The Citadel/Schwab letter's use of this terminology appears inconsistent with Citadel's own research acknowledging that "a wide array of market participants seek to lower their risk of inopportune executions by constantly updating their orders to reflect changing market conditions" and that this "can lead to higher quote cancellation rates and frequent quote updates to reflect accurate prices."

This is precisely what IEX's Options Risk Parameter (ORP) facilitates—it allows market makers to maintain accurate pricing rather than being forced to either widen spreads dramatically or face systematic adverse selection.

More fundamentally, the concept of absolute quote firmness that opponents invoke simply does not reflect current practice in modern options markets. Options markets are characterized by ubiquitous repricing and canceling, with market makers employing sophisticated risk controls precisely because the alternative—maintaining stale quotes in rapidly moving markets—would be economically ruinous. The Commission itself has recognized that the options market structure inherently involves significant quote updating, with legitimate risk management tools already widely employed across exchanges.

Retail Investor Impact Analysis

The opponents' claim that IEX's protections harm retail investors warrants careful examination. Retail investors simply do not trade within the microsecond timeframes during which IEX's protections operate.

As the Commission found in approving D-Limit, institutional investors and retail customers "generally transact during stable market conditions" and if they do happen to send an order during a period of market instability, "they would have likely been too late either way" to obtain execution at the stale price. The only parties systematically disadvantaged by IEX's protections are those engaged in trading strategies that seek to exploit temporary pricing dislocations.

Moreover, the opponents do not fully account for the substantial benefits that IEX's protections provide to retail investors. By reducing the risk of adverse selection, IEX's protections encourage market makers to post tighter spreads and deeper liquidity.

Risk Controls and Market Structure

The opposition attempts to distinguish IEX's protections from the risk controls offered by other exchanges, suggesting that IEX provides special advantages. This characterization misunderstands both IEX's protections and existing market practice.

Options exchanges currently offer market makers sophisticated risk controls. These tools allow market makers to automatically cancel or modify quotes when certain thresholds are exceeded. IEX's ORP operates on similar principles but with greater transparency and precision—it uses objective, publicly disclosed mathematical formulas to identify when quotes are likely to become stale, rather than relying on opaque, proprietary risk management systems.

The key distinction is not that IEX provides protections while other exchanges do not, but that IEX's protections are transparent, predictable, and designed to address the specific problem of latency arbitrage. Unlike less transparent risk controls employed elsewhere, market participants can understand exactly when and why IEX's protections will activate. This transparency benefits the entire market by making price discovery more reliable and reducing the uncertainty associated with hidden liquidity that may disappear without warning.

The Innovation and Competition Framework

The opposition's arguments ultimately reflect a preference for the status quo over innovation. But as the D.C. Circuit recognized in upholding the Commission's D-Limit decision, the Securities Exchange Act "encourages competition and innovation" in market structure. The Commission has repeatedly emphasized that exchanges should have "structural flexibility in order to enhance competition between market centers, while promoting market fairness, efficiency, and transparency."

IEX's options proposal represents exactly this kind of beneficial innovation. It offers market participants a choice—they can continue to trade on exchanges with traditional market structures, or they can utilize an exchange that provides additional protections against latency arbitrage. This choice enhances competition by requiring all exchanges to compete not just on fees and rebates, but on the quality of their market structures and the protection they provide to liquidity providers.

The opponents' preference to prevent this competition is noteworthy. Rather than improving their own offerings to compete with IEX's innovations, they seek regulatory intervention to block IEX's application entirely. This approach, while understandable from a business perspective, does not align with regulatory policy that encourages market-driven solutions and competitive innovation.

The Commission should evaluate these criticisms in this context and recognize them as reflections of commercial interests rather than compelling evidence of market harm.

Section II: Vindicating the Atkins-Glassman Vision Through Market-Driven Innovation

Twenty years ago, Commissioners Paul Atkins and Cynthia Glassman issued a prescient dissent to the adoption of Regulation NMS that reads today like a roadmap for the very problems IEX's innovations have been designed to solve. Their warnings about "massive regulatory intrusion" stifling competition

and innovation have proven remarkably accurate, making IEX's market-driven solutions particularly valuable as correctives to the unintended consequences of rigid regulatory mandates.

The Atkins-Glassman Dissent: A Framework for Analysis

Commissioner Atkins and Commissioner Glassman warned that Regulation NMS represented "a massive regulatory intrusion...unwarranted" given the lack of evidence of market failure, and they expressed concern that the rules would "do significant harm... by unduly interfering with the operation of competitive forces" that had benefited investors. They specifically criticized the majority's decision to abandon "competitive market forces to determine the appropriate market structure" in favor of "government-controlled competition."

Subsequent market developments have validated many of these concerns. Rather than enhancing market quality, Regulation NMS's rigid trade-through protections created new opportunities for exploitation by high-frequency traders who could use their speed advantages to game the system. The rules that were designed to protect investors instead enabled new forms of predatory trading, as speed-based competition intensified rather than price-based competition for genuine value creation.

IEX as the Market-Driven Alternative

IEX's innovations represent precisely the kind of market-driven solution that Commissioners Atkins and Glassman envisioned as superior to rigid regulatory mandates. Rather than imposing uniform rules across all market participants, IEX offers optional protections that market participants can choose based on their individual needs and preferences.

The speed bump and D-Limit order type exemplify this philosophy. They do not mandate that all exchanges adopt similar protections or require all market participants to trade under identical conditions. Instead, they provide an alternative venue where liquidity providers can access additional protections against latency arbitrage while maintaining fair access for all legitimate trading strategies.

This competitive dynamic has produced exactly the benefits that the Atkins-Glassman dissent predicted would flow from market-driven innovation. IEX's entry encouraged incumbent exchanges to improve their own offerings and compete on market quality rather than just fees and rebates. Several exchanges developed their own latency protection mechanisms in response to IEX's success, demonstrating how competition drives innovation far more effectively than regulatory mandates.

Section III: A Pattern of Concerns and Market Experience

The current opposition to IEX's options application follows a recognizable pattern that has characterized responses to IEX innovations since 2016. Understanding this pattern provides useful context for evaluating the current concerns and their relationship to actual market experience.

The 2016 Opposition: Concerns and Outcomes

When IEX first applied for exchange status in 2016, the response from certain incumbent exchanges and high-frequency trading firms was immediate and emphatic. They argued that IEX's speed bump would harm retail investors, create inaccessible quotes, and undermine market efficiency. The arguments were presented with considerable certainty about the damage IEX would supposedly inflict on market quality.

Nasdaq and other incumbent exchanges argued that the speed bump would create "phantom liquidity" and prevent retail investors from accessing the best available prices. High-frequency trading firms warned that any delay in order processing would fragment markets and reduce price discovery. They predicted that IEX's innovations would fail in the marketplace and harm the very investors they purported to protect.

The Commission unanimously rejected these arguments, recognizing them as commercially motivated rather than reflecting legitimate policy concerns. The SEC found that IEX's speed bump was designed to prevent latency arbitrage while maintaining fair access for all market participants. Far from harming market quality, the Commission determined that IEX's innovations would enhance competition and provide valuable alternatives for market participants seeking protection from predatory trading.

The subsequent years demonstrated the accuracy of the Commission's judgment. IEX successfully launched and grew based on the quality of its execution services. Market participants voluntarily chose to trade on IEX because it provided superior protection against latency arbitrage without the negative consequences opponents had predicted. Rather than harming retail investors, IEX's protections enhanced execution quality for all market participants.

The D-Limit Deja-vu Experience: Repeated Concerns and Empirical Validation

When IEX proposed its D-Limit order type, many of the same firms deployed similar arguments against the innovation. Once again, they claimed that the innovation would harm retail investors, create quote fading, and undermine market efficiency. The arguments against this Options Facility bore strong resemblance to the 2020 objections to D-Limit.

Citadel and other high-frequency trading firms argued that D-Limit would create "inaccessible quotes" that would frustrate legitimate trading interest. They predicted that the order type would reduce market quality and harm the very price discovery process that markets depend upon.

Once again, the Commission thoroughly evaluated these arguments and unanimously rejected them. The SEC's analysis demonstrated that D-Limit was carefully designed to address latency arbitrage while preserving fair access for legitimate trading strategies. The Commission found that the order type would enhance market quality by encouraging more aggressive quoting and deeper displayed liquidity.

Independent Judicial Review

The concerns raised by opponents received independent evaluation when Citadel and other firms challenged the Commission's D-Limit decision in federal court.

The D.C. Circuit Court of Appeals unanimously rejected Citadel's challenge, providing a comprehensive analysis that addressed the opponents' core arguments. The court found that substantial evidence supported the SEC's determination that D-Limit accurately identifies latency arbitrage and provides appropriate protections without creating unfair discrimination.

Particularly significant for evaluating current opposition arguments was the court's finding that claims about retail investor harm lacked evidentiary support. The court noted that retail investors "generally transact during stable market conditions" and would have been "too late either way" to access quotes that D-Limit protections modify. Rather than harming retail investors, the court found that D-Limit's protections enhance market quality by encouraging more competitive quoting.

The D.C. Circuit's decision provides more than legal validation for IEX; it offers authoritative judicial confirmation that the arguments repeatedly deployed against IEX's innovations often lack factual foundation. When subjected to rigorous scrutiny by independent federal judges, many opposition claims about investor harm and market disruption cannot withstand analysis.

The Current Application: Familiar Arguments and Market Context

Now, faced with IEX's options application, many of the same firms are presenting similar arguments to those they have used previously. The opposition submissions read like applications of the same analytical framework, with "options" substituted for "equities" and "ORP" replacing "D-Limit" but otherwise maintaining similar theoretical concerns.

They claim once again that IEX's protections will harm retail investors, despite the D.C. Circuit's finding that retail investors do not trade during the microsecond periods when these protections activate. They raise concerns once again about "quote fading," despite their own research acknowledging that quote updating is ubiquitous in options markets and necessary for accurate pricing. They predict once again that IEX's innovations will undermine market quality, despite the demonstrated success of similar protections in the equities market.

The consistent nature of these arguments across multiple proceedings suggests a systematic approach rather than innovation-specific analysis. If the opponents had developed new insights based on IEX's actual track record or identified options-specific concerns, their arguments might merit additional consideration. Instead, their consistency in opposing IEX innovations suggests a fundamental resistance to this type of market structure change, regardless of empirical evidence supporting its benefits.

Section IV: The Legal and Regulatory Record Supporting IEX's Innovations

The extensive legal and regulatory history surrounding IEX's exchange registration and subsequent innovations provides compelling support for the Commission's approval of the options application. This record demonstrates not only the thorough vetting that IEX's protections have received, but also the consistent validation of the principles now being applied to options markets.

The 2016 Exchange Approval: Unanimous Commission Support

In 2016, the Commission approved IEX's application to become a national securities exchange, finding that IEX's innovative features would "help to prevent latency arbitrageurs from executing trades at 'stale' prices that have not been updated to reflect current market conditions." The Commission's analysis, detailed in Securities Exchange Act Release No. 78101, recognized that IEX's 350-microsecond speed bump constituted a de minimis delay that would not frustrate the purposes of the Exchange Act while providing meaningful protection against predatory trading practices.

The 2016 approval established several important precedents directly applicable to the current options proposal. First, the Commission determined that IEX's speed bump does not impair fair access to quotations because the delay is so brief as to be imperceptible in normal trading operations. Second, the Commission found that IEX's protections enhance rather than undermine market quality by encouraging more aggressive quoting and reducing adverse selection. Third, the Commission recognized that protecting displayed liquidity from latency arbitrage serves the broader goal of promoting fair and efficient markets.

The D-Limit Approval: Thorough Analysis and Empirical Validation

The Commission's 2020 unanimous approval of IEX's D-Limit order type, documented in Securities Exchange Act Release No. 89686, provided even more detailed analysis of the principles underlying IEX's protections. The Commission found that D-Limit orders "protect liquidity providers...from potential adverse selection by latency arbitrage" in a manner consistent with the Exchange Act's goals of fair and orderly markets.

The D-Limit approval process involved extensive public comment and regulatory analysis. The Commission solicited comments, considered opposition arguments, and ultimately determined that D-Limit would enhance market quality without creating unfair discrimination. The SEC's analysis specifically addressed concerns about quote accessibility, finding that D-Limit orders remain accessible to legitimate trading strategies while providing protection against predatory practices.

Importantly, the Commission's D-Limit analysis relied on careful analysis and reasonable determinations rather than theoretical speculation. IEX provided data demonstrating how the crumbling quote indicator narrowly targets periods of market instability when quotes are likely to become stale. This data pattern demonstrates precisely what IEX's protections are designed to address: concentrated periods of latency arbitrage activity that systematically disadvantage displayed liquidity providers.

The D.C. Circuit Decision: Independent Judicial Validation

The most authoritative validation of IEX's approach came from the D.C. Circuit Court of Appeals in Citadel Securities LLC v. Securities and Exchange Commission, which unanimously upheld the Commission's D-Limit approval against Citadel's challenge. The court's analysis in this 2022 decision directly addresses and evaluates many of the arguments now being raised against the options proposal.

The D.C. Circuit found that "substantial evidence supports the SEC's finding that the crumbling quote indicator accurately identifies latency arbitrage." The court noted that the SEC had reasonably

determined "that short-term investors engage in latency arbitrage during the same 'small increments of time' that the crumbling quote indicator turns on," directly addressing claims that IEX's protections target legitimate trading activity.

Particularly relevant to current opposition arguments, the court addressed Citadel's claim that D-Limit orders harm retail investors. The court found this argument unpersuasive, noting that the SEC had reasonably concluded that retail investors "generally transact during stable market conditions" and if they do encounter periods when D-Limit protections activate, "they would have likely been too late either way" to obtain execution at stale prices.

The court also rejected arguments that D-Limit orders create improperly discriminatory treatment. Instead, the court found that D-Limit "narrowly targets latency arbitrage because IEX changes a price only when the crumbling quote indicator turns on," and that this targeted approach serves legitimate regulatory objectives without creating improper advantages.

Perhaps most significantly for the current options proposal, the D.C. Circuit specifically endorsed the SEC's finding that D-Limit orders qualify as protected quotations under Regulation NMS because they execute "immediately and automatically." The court rejected arguments that the intentional nature of IEX's delay disqualifies D-Limit orders from protected status, finding that orders can execute "immediately" even when subject to a de minimis delay.

The Precedential Framework for Options Markets

The extensive legal and regulatory record supporting IEX's protections creates strong precedent for approving the options application. The core principles underlying IEX's approach—protecting displayed liquidity from latency arbitrage while maintaining fair access for legitimate strategies—apply with equal or greater force in options markets.

The Commission has already determined that IEX's speed bump constitutes a de minimis delay that does not impair fair access. The D.C. Circuit has confirmed that this analysis satisfies legal requirements under both the Exchange Act and Regulation NMS. These determinations establish that IEX's approach is legally permissible and practically beneficial.

Section V: The Compelling Case for Extending IEX's Protections to Options Markets

The extensive historical record and legal precedent supporting IEX's innovations leads logically to the conclusion that the Commission should approve IEX's options application. When the proven benefits of IEX's protections are considered alongside the unique characteristics of options markets, the case for approval becomes compelling. Far from representing an untested experiment, IEX's options proposal offers a carefully designed application of validated protections to a market structure that will benefit significantly from latency arbitrage safeguards.

Options Markets: Enhanced Vulnerability to Latency Arbitrage

The characteristics that distinguish options markets from equities markets make them particularly vulnerable to the latency arbitrage that IEX's protections address. As the SEC's own Staff Report on Equity and Options Market Structure Conditions in Early 2021 noted, options market structure is "broadly similar to equities market structure" but with the critical difference that "displayed liquidity is primarily derived from market maker quotes" and all options trading occurs on exchanges.

This complete reliance on exchange-based trading means that options markets cannot benefit from the off-exchange mechanisms that provide some latency arbitrage protection in equities markets. While approximately 50% of equity trading occurs off-exchange through mechanisms that are not subject to rigid order protection rules, 100% of options trading must comply with the Options Order Protection and Locked/Crossed Markets Plan. This creates a more rigid environment where latency arbitrageurs can systematically exploit temporary pricing dislocations without the competitive pressure that off-exchange venues provide in equities.

The relationship between options prices and underlying securities creates additional vulnerabilities that do not exist in single-asset markets. Options Market Makers must continuously update quotes across hundreds or thousands of series as underlying stock prices change, creating numerous opportunities for latency arbitrageurs to exploit temporary pricing misalignments. A single movement in an underlying stock can trigger arbitrage opportunities across multiple strikes and expirations, multiplying the potential for systematic adverse selection.

Addressing Concerns About Price Improvement Auctions

The Citadel/Schwab letter raises concerns that IEX's protections might interfere with price improvement auctions on other exchanges, but this concern reflects a misunderstanding of how these mechanisms operate. Price improvement auctions are designed to provide better prices than the prevailing NBBO, not to guarantee access to quotes that are in the process of becoming stale.

If IEX's quotes are genuinely superior and accessible, other exchanges' auctions will need to provide even better prices to compete—exactly the kind of competitive dynamic that benefits investors. If IEX's quotes become unavailable due to ORP activation, it indicates that they were likely to become stale anyway, meaning that auction participants were never realistically competing against accessible liquidity.

Conclusion: The Path Forward

The case for approving IEX's options application rests on multiple reinforcing foundations: the proven success of IEX's protections in equity markets, the greater vulnerability of options markets to latency arbitrage, the sophisticated and transparent design of the Options Risk Parameter, and the competitive benefits that approval would generate for the entire industry.

Opposition arguments have not identified any compelling basis for distinguishing options markets in a way that would make IEX's protections harmful rather than beneficial. Their concerns rely on many of the same theoretical considerations that have been examined through both regulatory analysis and

practical market experience in the equities context, and have been addressed by both regulatory and judicial authorities.

The Commission should approve IEX's proposed rules and allow market participants to benefit from the innovation and competition that approval will generate. This decision would validate the market-driven approach to addressing market structure problems while demonstrating the Commission's continued commitment to fostering beneficial innovation in our national market system.

The Commission faces a clear choice. IEX's protections have survived rigorous regulatory scrutiny, unanimous judicial review, and practical market testing. They represent exactly the kind of competition-enhancing innovation that Commissioner Atkins and Commissioner Glassman envisioned as superior to rigid regulatory mandates. The historical record demonstrates that opposition to IEX's innovations stems from understandable business concerns about preserving existing competitive advantages rather than compelling evidence of market harm.

The Commission should approve IEX's proposed rules and allow market participants to benefit from the enhanced competition and improved market quality that approval will generate.

Respectfully submitted,

J.W. Verret

Admin Record No. 19



June 28, 2025

Vanessa Countryman
Secretary
Securities and Exchange Commission
100 F Street, N.E.
Washington, DC 20549-1090
rule-comments@sec.gov

Re: **Proposed Rule Change to Adopt Rules to Govern the Trading of Options on the Exchange for a New Facility Called IEX Options; Release No. 34-102190; File No. SR-IEX-2025-02**

Dear Ms. Countryman:

CTC, LLC¹ (“CTC” or “we”) appreciates the opportunity to comment on the recent Investors Exchange LLC (“IEX”) filing (“Proposal”) to launch an options market (“IEX Options”). The Proposal includes a hardware-based latency mechanism adding a *de minimis* delay of 0.000350 seconds (“speedbump”) to each incoming order and quote message designed to enable IEX to obtain the most accurate view of the national market prior to processing quotes and orders, accompanied by an Options Risk Parameter (“ORP”) designed to protect IEX Market Makers from excessive execution of quotes at stale prices (e.g., before the market maker can update them in response to rapidly-changing market conditions). The ORP would reprice or cancel market maker quotes on IEX within that window in certain narrowly-prescribed circumstances which IEX estimates would occur significantly less than 0.001% of the time.² (It should be noted that this mechanism in no way offers a “last look” or opportunity for market makers to “change their mind” about a quote by examining inbound orders in some nefarious way; rather, any quote adjustments will be performed entirely by the exchange.)

This mechanism—including the speedbump of duration about one one-thousandth that of the blink of an eye³—would help foster competition by ensuring all market participants have at least some minimum amount of time to react to price changes in related markets, reducing the advantage that would otherwise be held only by the small number of players who use extreme low-latency

¹ CTC is a global proprietary trading firm that provides liquidity in regulated futures and securities markets in the U.S. and internationally. CTC, LLC is a registered broker-dealer and market maker in equities, ETFs, equity options, and index options and a member of Cboe, C2, BZX, EDGX, NYSE Arca, NYSE American, Nasdaq ISE, Nasdaq Phlx, and FINRA.

² See Proposal, Amendment No. 3.

³ <https://bionumbers.hms.harvard.edu/bionumber.aspx?id=100706&ver=4>

technology to “pick off”⁴ participants who take slightly longer to reprice resting orders in response to new information. The Proposal offers an important opportunity to improve the fairness of the market and, in so doing, directly encourage additional displayed liquidity and efficient price discovery to the benefit of all investors and end users. **IEX’s proposal is thoughtful, well-argued, and pro-competitive, and CTC recommends its approval.**

Latency Arbitrage Costs Investors Money

Market makers take unique risks to display liquidity

Options market makers are liquidity providers who contribute directly to transparency and price discovery by publicly disseminating the prices at which they would buy and sell a wide range of options on multiple exchanges. The role of market makers in facilitating risk transfer is particularly important in less-liquid instruments, including many series of options, where the likelihood of natural buyers and sellers posting simultaneous offsetting orders would otherwise be very low.

Unlike liquidity takers, market makers stand at continuous risk to the market, and—due to regulatory quoting obligations that typically require they post a very large number of bids and offers simultaneously—risk significant losses due to instantaneous adverse selection. Professional “high-frequency” trading firms with price feeds that are faster than a market maker’s by tiny, otherwise-insignificant fractions of a second can be a significant source of such adverse selection (“latency arbitrage”). Consequently, market participants who provide liquidity must invest progressively greater sums in direct exchange feeds, ticker plant infrastructure, and related systems to avoid being “picked off.” These technology expenditures by all parties serve only to protect professional market participants from one another (and, perhaps, to generate a profit for purveyors of market data and trading infrastructure), and do not fill any expressed need for additional speed on the part of end users.

The costs associated with these risks lead to inferior executions for investors

These costs could be justified if they provided meaningful benefit to investors. On the contrary, however, in order to earn a return commensurate with their level of risk, market makers must account for all the costs of running their business when determining the amount of quoted size and the tightest possible bid-ask spread they are able to disseminate to the marketplace. Increased risk of instantaneous adverse selection, and the increased infrastructure cost necessary to mitigate that risk, is therefore a direct cause of market makers’ quoting wider spreads and/or smaller size in order to generate sufficient risk-adjusted returns—thereby increasing costs for investors. As a result, the endless furtherance of this technology “arms race” acts counter to investor protection and the public interest. This phenomenon has been confirmed by peer-reviewed academic research.⁵

⁴ “Pickoffs” are trades immediately regretted by one party (for example, because a cancellation request had already been transmitted but not yet processed by an exchange matching engine). Trade executions best serve to advance a fair, orderly, and efficient market when, on average, they represent mutually-beneficial transfer of risk.

⁵ For example, Eric Budish (University of Chicago Booth School of Business), Peter Cramton (University of Maryland), and John Shim (University of Chicago Booth School of Business) have identified this situation as “a never-ending arms race for speed” and characterize the result as “a classic prisoner’s dilemma: snipers invest in speed to try to win the race to snipe stale quotes; liquidity providers invest in speed to try to get out of the way of the snipers; and all trading firms would be better off if they could collectively commit not to invest in speed, but it is in each firm’s private interest to invest.” They conclude, “Our results say that sniping is negative for liquidity and that the speed race is socially wasteful.” See Eric Budish,

The Proposal Would Reduce Latency Arbitrage, and Save Investors Money

The Proposal would reduce latency arbitrage costs for investor orders on IEX Options in a thoughtful and deliberate way. By providing a mechanism to intelligently revise or cancel market maker quotes in limited circumstances, the Proposal would reduce the disadvantage incurred by those liquidity providers *who are slower by the smallest margin* than aggressive liquidity takers. By helping to establish a market structure where massive technology expenditures are no longer as critical in preventing trading losses, the Proposal will help foster a fairer marketplace with superior liquidity provision and tighter bid-ask spreads.

Indeed, the Proposal is explicitly pro-competitive. Using an exchange-controlled mechanism to mitigate “pick-off” risk that could otherwise only be addressed through massive technology expenditures will lower technological barriers to entry and allow *more firms*—including those unwilling or unable to spend giant sums to build a microsecond-latency trading platform—to become competitive at providing liquidity, and empower many existing liquidity providers to quote more aggressively. Anecdotally supporting the pro-competitive nature of the Proposal, as of this writing, the two *largest* U.S. equities options market making wholesalers have commented opposing the filing, while three comparatively *smaller* ones (including CTC) have all commented in support (including one international firm which we understand to be a new entrant to US options market making)—implying the Proposal may support/encourage more upstart competitors in the market making space.

Notably, while the Proposal would promote competition among liquidity providers, it still allows for benefits to those who choose to invest in higher-performance trading systems since, under the Proposal, the first liquidity-taker to pass through the speedbump will uniquely be able to execute against the entire remaining posted size, if desired—even if he or she beats out other would-be takers by only a single nanosecond. At the same time, the Proposal represents a new and innovative approach that will encourage competition among exchanges in accordance with the Exchange Act. The ultimate benefit of all this, of course, will go to the investing public.

Negative Commenters’ Claims are Misleading

Market Maker Risk Controls (MMRCs) provide irrefutable precedent

IEX’s proposal is far from novel or unprecedented despite commenters’ claims to the contrary. One comment letter states:

[W]e are extremely concerned about introducing a novel mechanism into the US options market that is explicitly designed to facilitate the fading of Market Maker displayed quotations, while nonetheless continuing to treat those displayed quotations as “protected.” This means that every market participant, including retail investors, will be compelled (by regulatory fiat) to route orders (or have their orders routed) to Market Maker displayed quotations on IEX, despite individual assessments of whether the novel IEX quote fading functionality is harmful to their execution quality.⁶

Peter Cramton, and John Shim (2015). “The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response.” *Quarterly Journal of Economics*, 130.4. <https://academic.oup.com/qje/article/130/4/1547/1916146>

⁶ IMC, Schwab, Citadel letter of June 4.

This sounds ominous until we realize that *this precisely describes the market as it already exists* if we simply replace the phrase “quote fading functionality” with “Market Maker Risk Controls and/or whatever matching engine latency any given exchange, broker, or price feed happens to have”.

Market Maker Risk Controls (“MMRCs”) are the various mechanisms implemented by options exchanges that allow exchanges, on their own initiative, to cancel a market maker’s displayed quotes in response to certain events with no explicit, contemporaneous instruction to do so. Triggers for these cancels may include trading a prescribed number of contracts, executions, or complete quote fills; the relevant parameters are set by market makers within ranges allowed by the exchanges pursuant to rules approved by the SEC.

Importantly:

1. Canceling a quote in one options series *need not involve ever executing any order in that particular option series* – typically, MMRCs are configured to operate across all instruments in a given options class.
2. These cancellations are typically initiated *atomically and in sequence directly by the exchange matching engine*, meaning they “jump the queue” and result in cancellations that prevent execution of otherwise-marketable orders that may already be in flight to the exchange, or even already queued up for execution in the exchange matching engine.

These are desirable and necessary features of MMRCs and have been in place for about 20 years, yet are substantively identical to aspects of the Proposal that some commenters claim are “unprecedented” or uniquely “self-serving”.

For example, an exchange competitor of IEX objects to “allowing IEX to degrade the quality of published quotations by displaying liquidity that may or may not be accessible”⁷ despite also claiming, in the same letter, that they have themselves “introduced innovative new active risk controls” which precisely create (again, in a very much not-unprecedented way) the possibility that displayed liquidity may or may not always be accessible!

Another exchange operator concedes the comparison to MMRCs, but then mistakenly claims that MMRCs (in a way presumably different from the Proposal) “honor displayed quotes”.⁸ This is incorrect: consider the following example of a Market Maker (“MM”) in AAPL options who has configured MMRCs to pull quotes on Nasdaq ISE after trading a total of 50 contracts:

Time	Event
T+0	Nasdaq ISE receives MM bid of 6.00 for 50 July 200 calls
T+1	Nasdaq ISE receives MM bid of 4.00 for 50 July 205 calls
T+2	Retail Customer A sends an order to sell 50 AAPL July 200 calls at 6.00

⁷ MEMX letter of May 14.

⁸ Nasdaq letter of May 20.

T+3	Retail Customer B sends an order to sell 50 AAPL July 205 calls at 4.00
T+4	Nasdaq ISE receives and executes Retail Customer A's order, MMRP is tripped, all of MM's AAPL quotes are canceled
T+5	Nasdaq ISE receives Retail Customer B's order

At time T+5, when Nasdaq ISE receives Retail Customer B's order, **the quote he or she was trying to hit has already been canceled by Nasdaq's MMRC**. It is unclear how these existing MMRCs on a Nasdaq exchange, which the SEC has approved and which are to our knowledge currently offered in various flavors on all Nasdaq markets, "honor displayed quotes" in some way that the IEX Proposal does not.

Similarly, another commenter argues, "If the Options NMS Plan [governing order protection] did permit IEX to treat its displayed quotations as 'protected' (it does not), the Plan itself would be unlawful."⁹ But if this critique were sound (it is not), it would imply that the Options NMS Plan is *already* unlawful because it *already* allows displayed quotes from exchanges supporting MMRCs to be protected. Presuming we do not accept the conclusion that the well-established Options NMS Plan is unlawful, we must therefore also reject the premise, and set this argument aside.

The chance of missing a fill by the blink of an eye has always – and will always – exist

The same commenter states elsewhere

... rather than innovating or competing on merit, IEX asks the Commission to effectively *force* all investors to chase after fleeting quotes that can be canceled and repriced *after* an investor order has been sent to IEX¹⁰

But *this has always been true*. Orders sent to every exchange are always at risk of missing the market due to a cancel that may already be in flight (but not yet reflected in the published quote), or an order update from another participant with a faster connection (for example, microwave vs. fiber) or a shorter geographical distance to the relevant exchange data center. **IEX quotes would not be "stale" in any new or unique way.**

Similarly, an exchange competitor of IEX argues

IEX's Proposal creates a 350 microsecond Delay before processing incoming orders which results in a delay in transmitting quotation updates to the Options Price Reporting Authority ("OPRA"). ... If every options market were to adopt a delay similar to IEX then no options market would display timely quotes. The NBBO would no longer be relied upon...¹¹

The problem with this argument is the OPRA NBBO is *already* stale by more than 350 microseconds. The OPRA NBBO is aggregated and computed from all constituent exchange NBBOs which must then be transmitted to NYSE's data center in Mahwah, NJ and then transmitted nationwide. The summed aggregation, calculation, and round-trip transmission times to (for example) Nasdaq's data center in

⁹ Citadel letter of June 23.

¹⁰ Citadel letter of June 23 (*italics in original*).

¹¹ Nasdaq letter of May 10.

Carteret, NJ are estimated by Nasdaq to be about 500 microseconds,¹² which already exceeds IEX's proposed speedbump.

Given the above, all of these arguments claiming that the risk of an order missing a displayed quote is unprecedented are simply wrong and should be set aside. A good-faith objection one could raise is that the Proposal could substantially increase the *extent* to which this happens, but again, in response to requests from multiple commenters, IEX has provided data which effectively rebuts this concern.

Market structure has always differed across markets

One letter objects:

To the extent Market Makers seek to match the IEX price on other options exchanges, they will be at a competitive disadvantage as there is no similar mechanism to reprice their quotations.¹³

But this is in fact true of *any* distinguishing market structure feature on *every* options exchange today. For example, as a market maker on various options exchanges, we may already be “at a competitive disadvantage” if we match prices:

- from NYSE Arca on our Cboe quotes, since Arca pays maker rebates and Cboe doesn't
- from Cboe on our BZX quotes, since BZX's high take fees are often worth incurring only by the most highly informed (and, therefore, adverse to market makers) order flow
- from BZX on our Nasdaq ISE quotes, since ISE permits auctions that could allow other market makers to lean on our lit quotes without displaying their interest, perhaps leaving us to be filled only by more toxic orders
- from MEMX on our Nasdaq Phlx quotes, since the Phlx specialist model may provide a preferred allocation to another market maker even if we were on the quote first or for larger size

and so on. It is, and has always been, true that sometimes posting the same quote is more likely to result in a favorable execution on one market than another. But, unless we intend to force all exchanges to operate under identical rules, this is a red herring as pertains to approving the Proposal.

Negative commenters' hypothetical examples ignore the near-certainty of better fills for investors

An exchange competitor of IEX states:

¹² Phil Mackintosh, Chief Economist and Senior Vice President at Nasdaq, “Time is Relativity: What Physics Has to Say About Market Infrastructure”. <https://www.nasdaq.com/articles/time-is-relativity-what-physics-has-to-say-about-market-infrastructure-2020-04-09>

¹³ IMC, Schwab, Citadel letter of June 4.

Nasdaq disagrees with IEX's assertion in its Proposal that without the ORP, market makers may be forced to widen their spreads, show less liquidity, or simply exit the market.¹⁴

What IEX actually said is that “Without *robust liquidity protection mechanisms* [generally] ... market makers may be forced to widen their spreads” or show less liquidity. If Nasdaq disagrees with this, they are mistaken (and we respectfully note that we are a market maker and Nasdaq is not). It is indisputably true – as we are certain that Citadel and IMC would agree – that market makers widen spreads *today and every day* to partially offset the effects of toxic order flow, including that driven by latency arbitrage, relative to what they would do if such toxic flow did not exist. Competitors may disagree about the magnitude of this effect and whether IEX's Proposal addresses it, but claiming that market makers don't widen spreads due to delta pickoffs is patently incorrect. Because market makers bake pickoffs into their required edge to quote, any mechanism that reduces assessed pickoffs will inevitably narrow quoted spreads to some degree.

It is of course possible that the Proposal will be *ineffective* in materially addressing delta pickoffs, and that therefore no market makers would quote tighter on IEX than elsewhere. **But that is a problem that solves itself**—if in fact the ORP turns out to be badly designed such that IEX Market Makers *won't* actually quote tighter spreads, then all commenters' arguments that a tighter quote on IEX would improperly force participants to route orders to IEX are irrelevant, since in fact no tighter quote on IEX would actually exist. It is therefore inconsistent for commenters to argue, as some have, that the Proposal would both (1) be ineffective in improving market quality but also (2) somehow damage the validity of the NBBO. It is literally impossible for both of those things to be true.

IEX Proposes Tweaks, Not a Revolution

Use of underlying equities data in the options markets is standard and normal

Another commenter claims that the Proposal “introduces the novel concept of a displayed options quote being repriced by the exchange based on a signal driven from equities trading activity. For the first time, an exchange will adjust prices of a quote in one asset class using inputs from another asset class.”¹⁵

To our knowledge, this is correct as a technicality. But it is not a material point. It is an undisputed fact that equities market activity is the underlying driver of the entire options market, and multiple exchanges use SEC-approved mechanisms that reference equity trades or prices, including

1. Protections against selling options under parity
2. Obvious Error rules that use the underlying price to come up with a theoretical price for a given option for purposes of making rulings to bust or adjust trades
3. Automatically halting options markets (i.e., making previously displayed quotes suddenly inaccessible) when the underlying equity price has entered a Limit Up/Limit Down Volatility Trading Pause

¹⁴ Nasdaq letter of May 10.

¹⁵ FIA PTG letter of February 26.

Given these and many other aspects of options trading that rely on equities market data, we cannot take seriously the argument that allowing contemporaneous equities pricing to impact options market mechanics is somehow objectionably novel.

The SEC has already approved de minimis intentional delays on the IEX SRO (among others)

IEX's proposal for a 350-microsecond speed bump appropriately recognizes the realities of U.S. market structure, where highly correlated instruments including equities, futures, and ETFs are variously traded in data centers across the New York-New Jersey metro area as well as in and around Chicago.¹⁶ As speed-of-light transmission times between the most distant relevant venues takes multiple milliseconds, and the latency differential between various transmission modalities can introduce swings of several more milliseconds among market participants, IEX's proposed speedbump remains *de minimis* given the technological realities of cross-market securities and derivatives trading and hedging strategies.

The speedbump's duration also exactly matches that of other, SEC-approved speedbumps in equities markets (including IEX and formerly NYSE American). Though some commenters have claimed that an "intentional" speedbump is novel and problematic in an options market, this is an artificial distinction: "jitter" or random noise in exchange matching engines and market data technology routinely exceeds the duration of the proposed speedbump.¹⁷ We understand that exchanges often choose not to immediately address this "jitter" to save money on infrastructure costs or due to other technical priorities—so it could more accurately be considered *intentional latency through inaction*. The idea that this latency, quietly left in place for years in some cases, is perfectly acceptable – whereas IEX's fully-disclosed speedbump, executed in hardware, and described in a detailed regulatory filing, is somehow unacceptably new and risky – doesn't pass the smell test.

Another of IEX's competitors¹⁸ claims that the ORP would force firms to route flow to IEX first even when multiple exchanges are at the *same* price, since otherwise the ORP might trip as (presumably) traders who receive fills on other options markets hedge with stock, pushing its price. Again, though, the speedbump is far too brief for this to be realistic: the total latency for a market maker to receive an options fill, transmit a hedging order in a stock, have that hedging order processed and executed by a stock exchange, and have that stock exchange disseminate a new BBO to the SIP which then transmits it to IEX for processing and application to the ORP, will vastly exceed 350 microseconds in practice.

¹⁶ A market maker, for example, might trade (1) E-Mini S&P 500 futures on the Chicago Mercantile Exchange at their primary data center in Aurora, Illinois; (2) S&P 500 ("SPX") index options on Cboe at their data center in Secaucus, New Jersey; and (3) S&P 500 ETF ("SPY") options on various options exchanges with data centers elsewhere in New Jersey. The market maker's ability to provide good liquidity in *all* of these instruments is impaired by its risk of being "picked off" in *any* of them, which is in part a function of competitive transmission times between the Chicago and New York/New Jersey metros.

¹⁷ As an indicative example of "jitter" in only a small part of the entire tick-to-trade-to-print chain, the difference between reported OPRA 10th and 99th percentile latencies has ranged from hundreds of microseconds to over 2,870 microseconds since 2021 (99th percentile or greater latencies are expected to be in effect about 4 minutes out of every 6.5 hour trading day). <https://cdn.opraplan.com/documents/OPRA-SIP-Metrics-February-2024.pdf>

¹⁸ NYSE letter of June 17.

In fact, in 2017, NYSE defended their own speedbump on an equities exchange (later approved by the SEC) in part by stating that 350 microseconds was “well within the geographic and technological latencies experienced by market participants when routing orders ... and would provide Exchange systems sufficient time to update prices based on market data disseminated by other markets.”¹⁹ We agree – and geographical latencies have not changed since 2017. And as Nasdaq argued in 2020 when defending SIP infrastructure, even a delay longer than IEX’s proposed 350 microseconds would have “no impact on human traders or professionals watching a screen.”²⁰

Finally, one letter makes the odd argument that the problem with the Proposal is that it will be *too successful* at tightening the NBBO:

Retail investor orders typically receive price improvement above and beyond the marketwide best price through “price improvement auctions” that are initiated on US options exchanges. However, to initiate such an auction, a liquidity provider must be willing to trade at the existing market-wide best price or better... Hence the problem – to the extent the market-wide best price is set by a Market Maker on IEX (and is a better price because of the quote fading mechanism available to the Market Maker on IEX), liquidity providers on other exchanges will be less likely to be willing (or able) to provide retail investors with additional price improvement (as the market-wide best price is already artificially aggressive).²¹

Setting aside the suggestion that the IEX NBBO is “artificial” (which IEX has decisively rebutted with data showing that it would in fact be accessible over 99.99% of the time), the meat of this argument seems to be that *a tighter NBBO on one exchange would be problematic because it means market makers will have to display better prices elsewhere too*. This argument is similar to claiming that a company should refrain from introducing innovative features to its products because doing so would increase competitive pressure on its rivals. We are confident the SEC will not find this argument compelling.

The Commission should not stifle competition through administrative inaction or regulatory fiat

We note recommendations from objecting commenters include the usual delaying tactic of seeking further review, requiring answers to unspecified new questions, or just explicitly asking for a slower regulatory process. For example, one commenter insists:

Procedurally, given that this latest amended filing contains material changes and was made over 150 days after IEX’s initial proposal, the Commission must consider it as a new filing, subject to a full review timeline...²²

which is of course spurious: SROs routinely amend filings to address SEC questions or commenters’ concerns without restarting the clock.

¹⁹ NYSE letter to the SEC, <https://www.sec.gov/comments/sr-nysemkt-2017-05/nysemkt201705-1748553-151659.pdf>

²⁰ Phil Mackintosh, Chief Economist and Senior Vice President at Nasdaq, “Time is Relativity: What Physics Has to Say About Market Infrastructure”. <https://www.nasdaq.com/articles/time-is-relativity-what-physics-has-to-say-about-market-infrastructure-2020-04-09>

²¹ IMC, Schwab, Citadel letter of June 4; Nasdaq makes an identical argument in their comment letter of May 20.

²² Citadel letter of June 23.

Another commenter suggests that

IEX must assess the impact its novel quote fading mechanism could have on the execution of retail investor orders generally, including the estimated reduction in “price improvement auctions” and associated price improvement more broadly,²³

notwithstanding the fact that, to our knowledge, *no such supporting data has ever been required for the approval of MMRCs*, despite their being universally available, and, as we have explained above, offering a materially equivalent “quote fading mechanism”.

We respectfully submit that blocking the Proposal by regulatory mandate would be irreconcilable with the statement of then-Commissioner, now-Chair Atkins that, according to Congress, “competitive forces, rather than unnecessary regulation” should guide market development.²⁴

* * * * *

For all the reasons cited above, we encourage the Commission to approve the Proposal. We believe it will: (1) protect investors and the public interest by enhancing price discovery and transparent liquidity provision through improved disseminated quotes, (2) encourage new entrants and competition among liquidity providers by lowering technological barriers to entry, (3) promote competition among exchanges on the basis of market structure innovation, and (4) continue to remove impediments to the operation of fair and orderly markets that arise from the trading technology arms race—all directly in line with the explicit goals of the Exchange Act. The risk introduced by microsecond-level pick-offs, and the corresponding cost of the never-ending arms race for speed, has resulted in less-attractive pricing and inferior displayed liquidity on many markets and in many asset classes. The Proposal offers a reasonable attempt to mitigate this risk and further encourage competition to display liquidity.

Should you have any questions with respect to this letter, please contact our General Counsel, James Ongena at james.ongena@chicagotrading.com or (312) 863-4713. We would welcome the opportunity to discuss it further and appreciate the opportunity to respond.

Sincerely,



Steve Crutchfield
Head of Business Development

²³ IMC, Schwab, Citadel letter of June 4.

²⁴ Dissent of Commissioners Cynthia A. Glassman and Paul S. Atkins to the Adoption of Regulation NMS, <https://www.sec.gov/files/rules/final/34-51808-dissent.pdf>

Admin Record No.

72



BETTER MARKETS

July 18, 2025

Vanessa A. Countryman
Secretary
U.S. Securities and Exchange Commission
100 F Street, NE
Washington, DC 20549-1090

Re: SR-IEX-2025-02

Dear Ms. Countryman:

Better Markets¹ appreciates the opportunity to comment on a proposed rule change filed by the Investors Exchange LLC (IEX) to adopt rules to govern the trading of options on IEX Options LLC, a new facility that IEX proposes to operate for the trading of options.² IEX's proposal is designed to counteract the deleterious effects of latency arbitrage. The Commission should approve the proposal as it will benefit investors and the public interest.

I. Latency arbitrage harms retail investors

Latency arbitrage involves exploiting the unavoidable delay between the moment an order is transmitted to a public exchange and the moment the order is reflected in the national best bid/offer. This delay presents an arbitrage opportunity for anyone who can access pricing data faster than the data is reflected in the national best bid/offer, but only high-frequency traders are able to obtain this pricing data so quickly. This informational advantage allows them to essentially guarantee a profit during the period of latency. So high-frequency traders trade “in the sub-second time windows between when market prices moves and when market makers update their quoted prices.”³ By doing so, they can trade at one price essentially knowing what the stock's next price will be. In other words, since high-frequency traders process information much more quickly than slower, traditional traders, they “profit at the slower traders' expense.”⁴

These slower traders include retail investors. Retail investors submit the orders that high-frequency traders use to determine where the market is going to go, and they use their greater speed and access to information to trade against these orders. So the high-frequency traders who

¹ Better Markets is a non-profit, non-partisan, and independent organization founded in the wake of the 2008 financial crisis to promote the public interest in the financial markets, support the financial reform of Wall Street, and make our financial system work for all Americans again. Better Markets works with allies—including many in finance—to promote pro-market, pro-business, and pro-growth policies that help build a stronger, safer financial system that protects and promotes Americans' jobs, savings, retirements, and more.

² SR-IEX-2025-02.

³ Kristin N. Johnson, *Regulating Innovation: High Frequency Trading in Dark Pools*, 42 J. Corp. L. 833, 860 (2017) (quoting Matt Prewitt, *High Frequency Trading: Should Regulators Do More?*, 19 Mich. Telecomm. Tech. L. Rev. 131, 136 (2012)).

⁴ *Id.*

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interact with slower retail traders “essentially free ride off of the information in the retail traders’ orders.”⁵ They “act on available information signals faster than retail investors are able to.”⁶ This allows high-frequency traders to “transfer wealth from retail investors to themselves.”⁷

Estimates vary as to the extent of this wealth transfer. An early study found the total profit potential from latency arbitrage opportunities in S&P 500 ticker symbols was over \$3 billion in 2014.⁸ A more recent study concluded that latency arbitrage practices cost investors an average of \$5 billion per year.⁹ One study even found the potential profits from latency arbitrage to be as high as \$21 billion.¹⁰ Regardless of the precise estimates, there should be little question that latency arbitrage is a highly profitable scheme that comes at the expense of retail investors.

II. IEX’s speed bump would mitigate the harmful effects of latency arbitrage

IEX’s proposal is designed to prevent this wealth transfer from retail investors to high-frequency traders by reducing latency arbitrage. It would do so by introducing a “speed bump,” as it has on its equities exchange. Some recent studies demonstrate why innovations like IEX’s speed bump are so important to protect investors and markets from high-frequency traders.

For example, one recent study sought to “explore the multifaceted impacts of [high-frequency trading] on financial stability and small investor welfare.”¹¹ By analyzing how [high-frequency trading] affects market access and returns for retail participants, the study aim[ed] to assess whether the technological arms race in trading disadvantages the very investors financial markets are designed to serve.”¹² The results “underscore[d] a growing concern about fairness.”¹³

The findings further reveal that small investors face considerable disadvantages due to latency arbitrage and information asymmetry introduced by [high-frequency trading]. Retail investors typically lack access to co-location services and ultra-low latency networks, placing them at a structural disadvantage. Empirical evidence from trading logs showed that in fast-moving markets, small investors were more likely to experience adverse price execution and increased slippage compared to institutional or [high-frequency trading]-driven accounts.¹⁴

⁵ Jonathan R. Macey, *Securities Regulation and Class Warfare*, 2021 Colum. Bus. L. Rev. 796, 823 (2022).

⁶ *Id.*

⁷ *Id.*

⁸ Elaine Wah, *How Prevalent and Profitable are Latency Arbitrage Opportunities on U.S. Stock Exchanges*, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2729109 (Feb. 8, 2016).

⁹ J.W. Verret, *Efforts to Sue the SEC over Broker-Inducement Regulation Unlikely to Succeed*, 17 Ohio St. Bus. L.J. 180, 202 (2023).

¹⁰ Alexander Abedine, *The Symbiosis of High Frequency Traders and Stock Exchanges: A Macro Prospective*, 14 N.Y.U. J.L. & Bus. 595, 617 (2018).

¹¹ Anuj Kumar Shah, *How high-frequency trading affects market stability and small-investor welfare*, International Journal of Social Impact, at 76 (2025), <https://ijsi.in/wp-content/uploads/2025/07/18.02.013.20251003.pdf>.

¹² *Id.*

¹³ *Id.* at 80.

¹⁴ *Id.* at 80.

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This is why IEX’s speed bump is needed—so retail investors are not left behind as high-frequency traders prosper. The study’s conclusion makes the benefits explicit: “Mechanisms such as **speed bumps**, order-to-trade ratios, and improved surveillance technologies can help ensure that [high-frequency trading] serves the broader goals of market stability and fairness.”¹⁵

Similarly, another recent study found that “restricting latency arbitrage improves liquidity by reducing toxic arbitrage.”¹⁶ Although the study tested the effects of a ban on proprietary traders taking liquidity, its conclusions have implications for the benefits of a speed bump. That’s because such a ban “grants liquidity providers sufficient time to revise stale quotes” and is thus “analogous to an infinitely long delay in market orders from fast traders.”¹⁷

This study found that a ban on aggressive proprietary trading caused bid-ask spreads to decline by 11% and adverse selection costs to decline by 21%.¹⁸ These findings, according to the study, were “consistent with the theory of speed bumps.”¹⁹ As a result, the study provided “additional evidence for the efficiency and complexity of passive liquidity protection mechanisms which extend to membership bans, trading restrictions, and speed bumps.”²⁰

The study’s conclusion was that “restricting fast arbitrageurs from taking liquidity protects liquidity suppliers from being sniped and mitigates adverse selection costs, hence improving market liquidity.”²¹ Yet the study recognized that “banning aggressive proprietary traders is a unique and harsh solution to latency arbitrage.”²² A speed bump along the lines of IEX’s proposal offers a less harsh solution to latency arbitrage that would still benefit investors.

III. IEX’s proposal would increase competition

The rise of high-frequency traders also impacts more than retail investors. Other market participants must contend with increased participation costs as they seek to compete in the “arms race” prompted by high-frequency trading strategies.²³ These costs have led major market participants to “take their business to trading venues free from competition from HFT traders.”²⁴

This means that, as a result of high-frequency traders, other professional traders have less “incentives to participate meaningfully in the market.”²⁵ “Over time, this disengagement might lead informed traders to either leave the market entirely or to transact on less transparent

¹⁵ *Id.* at 81 (emphasis added).

¹⁶ Chengcheng Qu, *Latency Arbitrage and Market Liquidity*, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3799550 (Dec. 14, 2022).

¹⁷ *Id.* at 1.

¹⁸ *Id.* at 1-2.

¹⁹ *Id.* at 19.

²⁰ *Id.* at 3.

²¹ *Id.* at 2.

²² *Id.* at 19.

²³ Yesha Yadav, *How Algorithmic Trading Undermines Efficiency in Capital Markets*, 68 Vand. L. Rev. 1607, 1661 (2015).

²⁴ *Id.* at 1661-62.

²⁵ *Id.* at 1162.

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venues.”²⁶ The withdrawal of informed traders from the market will make prices less reflective of fundamental information, and thus less able to help capital get to its most productive uses, which means high-frequency trading “imposes serious costs on the major function that securities markets perform: allocating capital efficiently and productively across the real economy.”²⁷

The Commission should be concerned about the inability of other market participants to compete with high-frequency traders. The “technological and real estate advantages needed to successfully compete in the HFT space” mean that a “consolidation of trading amongst a small number of market players” is not surprising.²⁸ IEX’s proposal would help level the playing field.

The race for speed and co-location calls into question the notion of a level playing field because the small advantages that accrue to HFT can never be made up by other traders. In the zero-sum game of trading, traders with speed advantages will always win the competition to come in first, as others can never make up for any initial delay. Since trading in the current markets often reduces to a mere competition for speed, slower traders feel that they cannot hope to compete with co-located traders for the best trades, leading to the belief that there is a two-tiered market. . . . That said, the concept of fairness in the financial markets requires that regulators strive to keep access to information and the inherent ability to profit from it open to all. While in theory all may have the opportunity to use this information, in practice HFT calls into question the notion of a level playing field by restricting the ability to profit from it to only the very most sophisticated actors.²⁹

IEX’s proposal would alleviate this state of affairs by allowing other market participants to better compete with high-frequency traders. Not only should the Commission seek to facilitate this greater competition, but it has already approved similarly pro-competitive innovations. In a similar context, the Commission expressed the view that market innovations to increase competition with high-frequency traders are in the public interest. Five years ago, the Commission found that IEX’s D-Limit order type protected investors and the public interest as a “competitive response to mitigate current competitive imbalances between liquidity providers and latency arbitrage liquidity takers.”³⁰ The Commission relied on evidence that institutional traders and investors were “unable to compete with the small number of firms that purchase the necessary systems, connectivity, and exchange proprietary market data to” engage in latency arbitrage.³¹ In approving the D-Limit order, the Commission found that “[e]xchanges should be able to innovate to address this competitive imbalance.”³² There is no reason to reach a different result here.

²⁶ *Id.*

²⁷ *Id.* at 1670.

²⁸ Hilary J. Allen, *The SEC as Financial Stability Regulator*, 43 J. Corp. L. 715, 748 (2018).

²⁹ Steven McNamara, *The Law and Ethics of High-Frequency Trading*, 17 Minn. J.L. Sci. & Tech. 71, 145 (2016).

³⁰ *Order Approving a Proposed Rule Change to Add a New Discretionary Limit Order Type Called D-Limit*, 85 Fed. Reg. 54,438, 54,451 (Sept. 1, 2020).

³¹ *Id.* at 54,446.

³² *Id.* at 54,446-47.

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Conclusion

We hope these comments are helpful as the Commission considers this matter.

Sincerely,

Benjamin Schiffrin

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(202) 618-6464

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Admin Record No. 119

United States Senate

August 1, 2025

The Honorable Paul S. Atkins
U.S. Securities and Exchange Commission, Chairman
100 F Street, NE
Washington, D.C. 20549

Re: *IEX Exchange LLC's Proposal to Launch a National Options Exchange, File No. SR-IEX-2025-02*

Dear Chairman Atkins,

I write to express my support for innovation and competition in U.S. capital markets. These two defining attributes have underpinned the enduring global leadership of those markets for decades. In that spirit, I encourage the Commission to give full and fair consideration to IEX Exchange LLC's proposal to launch a new national exchange for the trading of equity options.

The options market plays a vital role in modern financial markets and offers investors a critical tool for price discovery, risk management, and portfolio strategy. Despite rapid growth in options trading volumes and increased retail participation in recent years, competition among liquidity providers has not kept pace. Instead, liquidity has become more concentrated, raising questions about long-term resiliency and investor access.

It is my understanding that one of the structural challenges contributing to this trend is the presence of latency arbitrage, an activity by which sophisticated firms use speed advantages on a timescale of microseconds to exploit market makers' quotes during price transitions. These practices act as a hidden tax on liquidity providers, leading to wider spreads and higher barriers to entry, particularly for firms that lack the extraordinary resources for systems needed to limit the impact of this tax. The recent exit of a major market participant such as Morgan Stanley from options market making underscores the real-world impact of this structural inefficiency.

IEX's proposal offers a market-based response to this issue. As I understand it, the design of IEX's proposal includes an optional quote protection mechanism that would automatically adjust market maker quotes during microsecond-level windows of heightened risk, mitigating the effects of latency arbitrage without disrupting price discovery. This innovation builds on IEX's success in equities, where its D-Limit order type has already demonstrated the ability to attract liquidity and foster fairer trading conditions by mitigating the costs of latency arbitrage imposed on liquidity providers.

Rather than relying on regulation to solve this structural challenge, IEX's proposal leverages technology and design to encourage competition, empower market participants, and

enhance investor outcomes. I believe this approach reflects the best tradition of American markets, where private-sector ingenuity drives public benefit.

I encourage the Commission to evaluate IEX's proposal with the seriousness it merits and to support an environment in which innovative, pro-investor market designs can flourish.

Sincerely,

A handwritten signature in black ink, reading "Roger W. Marshall". The signature is written in a cursive, flowing style with a large, prominent "R" and "M".

Roger Marshall, M.D.
United States Senator

Admin Record No. 121



One World Trade Center • Suite 47M • NEW YORK • NY 10007 • PHONE: +1 (646) 804 7900

Via E-mail
Ms. Vanessa Countryman
Secretary
U.S. Securities and Exchange Commission
100 F Street N.E.
Washington, D.C. 20549-1090

Re: Investors Exchange LLC: Notice of Filing of a Proposed Rule Change to Adopt Rules to Govern the Trading of Options on the Exchange for a New Facility Called IEX Options; Release No. 34-102190; File No. SR-IEX-2025-02

Dear Ms. Countryman:

Volant Trading appreciates the opportunity to comment on the above-referenced proposed rule change submitted by Investors Exchange LLC ("IEX"). We write in strong support of IEX's proposal to launch an options exchange and, in particular, to implement a novel quoting protection mechanism designed to mitigate the harmful effects of latency arbitrage in the U.S. equity options markets.

Volant Trading was formerly a registered market maker on multiple U.S. equity options exchanges. Until our exit from US options market making in 2023, we were an active liquidity provider in hundreds of underlying names. Despite our deep experience and strong quantitative capabilities, Volant was ultimately forced to exit the business due to the escalating costs and arms race associated with maintaining latency competitiveness. The playing field has increasingly tilted toward a small number of firms with the financial and technological resources and economies of scale to operate at the bleeding edge of latency-sensitive infrastructure. These developments have created unsustainable barriers to entry and reduced the diversity of liquidity providers, ultimately harming market quality for end investors.

Latency arbitrage in the options markets arises from a structural mismatch while option prices are highly sensitive to movements in their underlying equities, there are unavoidable delays in how exchanges and market participants receive and process those underlying price updates. Sophisticated firms exploit this latency gap by executing against stale option quotes before slower market makers can adjust their prices. The result is a one-sided dynamic where passive liquidity providers are routinely "picked off" at prices that no longer reflect fair value. This leads to defensive quoting behavior, wider spreads, and thinner markets. Firms with deep expertise in option pricing and risk management – those best positioned to provide tight spreads and meaningful size – are often pushed out due to the steep technical demands required merely to remain competitive.

The costs of maintaining latency competitiveness are staggering. To begin, liquidity providers must subscribe to the fastest available proprietary equity market data feeds, many of which cost tens of thousands of dollars per month. To transmit this data among the three major East Coast data centers – Mahwah, Carteret, and Secaucus – vendors offer ultra-low latency millimeter wave networks at a cost of approximately \$200,000 per month. A similar dynamic exists in futures markets, where microwave networks transmit S&P 500 E-mini futures prices from the CME's Aurora, Illinois data center. These futures are among the most liquid and price-influential instruments in the world and are highly correlated with U.S. equity prices. As a result, access to this feed has become an additional latency requirement, often costing another \$200,000 per month.

And yet, access to premium data is only the starting point. A latency-competitive quoting operation also requires massive ongoing infrastructure investments: maintaining multiple local feed instances for redundancy and speed; deploying customized FPGA hardware to reduce internal processing delays; keeping exchange gateways continuously warmed; implementing arbitrage logic to shift between faster exchange sessions; and ensuring near-instant access to quote cancellation facilities. While technically impressive, these measures serve solely to defend against predatory strategies. They do not enhance price discovery or benefit end investors. Instead, they concentrate market making power in the hands of the firms best equipped to invest in speed, not necessarily those best suited to provide reliable and competitive liquidity.

IEX's proposed quote protection mechanism addresses this inefficiency directly. By deterministically canceling stale quotes in response to rapid moves in the underlying, IEX reduces the exploitable window that latency arbitrageurs depend on. Importantly, this protection is symmetric and nondiscretionary—it does not provide any special access or "last look" privileges. Rather, it ensures that market participants are not executed at stale prices due solely to physical or processing delays outside their control.

Some critics argue that the proposed speed bump and quote protection mechanism will reduce quote availability and impair liquidity. In our view, this criticism misrepresents the actual functioning of today's markets. Quote availability is already limited by frequent and necessary bulk cancellations as a means of risk control. Market makers regularly cancel entire swaths of quotes in response to volatility or systemic risk, often acting conservatively due to the lack of more targeted tools. In contrast, the IEX mechanism offers a precise, event-driven tool that reduces the need for broad cancels and encourages market makers to quote more continuously and competitively.

It is also important to dispel the notion that quotes are always fully available across exchanges. In reality, quote fragmentation, latency asymmetries, and differing exchange protocols contribute to periods of unavailability. These issues are well known to practitioners and are exacerbated during volatile market conditions. IEX's proposal would mitigate these issues by making quotes more durable and defensible against fleeting but aggressive latency arbitrage strategies.

As a firm with firsthand experience in options market making and a deep interest in improving the fairness and resilience of U.S. markets, Volant believes that IEX's proposal is an important and necessary step forward. The mechanism does not inhibit access or confer unfair advantages – it levels the playing field. By helping reduce the dependency on expensive infrastructure and allowing market makers to quote more competitively without being exposed to asymmetric risks, the proposal will enhance liquidity, improve price discovery, and ultimately benefit all investors.

We strongly urge the Commission to approve this proposal and consider its merits as a model for broader market structure reform. The long-term health of the U.S. options markets depends on maintaining a competitive, transparent, and fair trading environment. IEX's proposal supports these goals through thoughtful innovation and a measured response to a real market failure.

Thank you for the opportunity to share our perspective.

Sincerely,



Brian P Donnelly
Founder
Volant Trading

Admin Record No. 125

August 20, 2025

Ms. Vanessa Countryman
Secretary, U.S. Securities and Exchange Commission
100 F Street N.E.
Washington, D.C. 20549-1090

Re: **Investors Exchange LLC; Notice of Filing of a Proposed Rule Change to Adopt Rules to Govern the Trading of Options on the Exchange for a New Facility Called IEX Options; Release No. 34-102190; File No. SR-IEX-2025-02**

Dear Ms. Countryman:

Investors Exchange LLC (“IEX” or “Exchange”) is writing with regard to its proposal to establish a new options exchange (“Options Proposal”)¹, to respond to a third comment letter from Citadel Securities, Inc. (“Citadel”).²

In our previous response to comments,³ IEX explained that a central issue before the Commission is whether IEX will be permitted to offer on its options market a narrowly-tailored tool designed to protect market makers from latency arbitrage — a tool of the same type that the Commission has previously approved, that a federal appellate court has upheld,⁴ and that has been proven effective on IEX’s equities exchange. Rather than engaging with this core issue of market competition and latency arbitrage, Citadel sidesteps it entirely, choosing instead to make unfounded assertions about the method used to address it. Citadel’s assertions amount to nothing more than rearticulations of arguments that have failed in front of the Commission and the federal appellate court before. Citadel offers no evidence to back up its opposition and disregards the substantial additional data and analysis that independently support the Options Proposal. Perhaps recognizing the weak basis for its opposition, Citadel now adds in its third comment letter the remarkable objection that the SEC has not given it enough time to object.

Overall, Citadel’s third letter amounts to another attempt to rely on conjecture and alarmist rhetoric, rather than facts, in an effort to sow doubt. But all of the objections fail to withstand even minimal scrutiny. And they wither even more in the face of the broad

¹ See Amendment No. 3 to a Proposed Rule Change to Adopt Rules to Govern the Trading of Options on the Exchange for a New Facility Called IEX Options (“Rule Filing”), avail. at [34-103290.pdf](#).

² Letter from Stephen Berger, Citadel, to Vanessa Countryman, SEC, dated August 12, 2025.

³ Letter from John Ramsay, IEX, to Vanessa Countryman, SEC., dated June 19, 2025 (“First Response Letter”). Unless otherwise defined, capitalized terms used below have the same meanings as in the First Response Letter.

⁴ Citadel Securities LLC v. Securities and Exchange Commission, No. 20-1424, (D.C. Circuit 2022) (“D.C. Circuit Decision”).

and forceful support expressed by a wide array of market participants in comments that both endorse the proposal and contradict Citadel's assertions against it. These include many market makers wishing to compete and hundreds of investors speaking directly for their own interests.⁵

Using an Approved and Proven Approach to Support Competition in Options Markets

To summarize the points laid out in the Rule Filing and our previous response, the objections from Citadel are focused on a tool that IEX is proposing to offer to market makers on a purely voluntary basis, the Options Risk Parameter, or "ORP". The ORP directly targets a well-documented problem, latency arbitrage,⁶ and would serve to strengthen competition among more market makers and improve price quality for

⁵ There is a substantial compilation of letters from investors – numbering more than six hundred – that have been submitted reflecting broad support for IEX's Options Proposal. While too numerous to cite individually, the following are examples of comments made by such investors:

("I am an individual investor. PFOF Brokers and Wholesalers do not represent my interests"); ("The reality is that laying fiber, co-locating servers, and deploying complex arbitrage strategies are tactics inaccessible to everyday investors"); ("The assertion by large market participants that this harms retail is disingenuous; fairness isn't bad for retail; it's bad for firms that profit from information asymmetries."); ("Please, for the sake of the little guys and individual investors support the proposed IEX Options Exchange."); ("An overtone of many of the critics was a general concern about their business losing market share, which again completely ignores individual investors that are not invested in those businesses" and "IEX's proposal would increase competition"); ("Please do not equate volume with representation. The ability to route trades does not give these firms the right to define what is best for individual investors"); ("Stating that the [critics] are acting in retail's best interest by advocating against transparent solutions such as IEX and attempting to stifle free market competition is a blatant manipulation of facts that are publicly available."); ("These PFOF broker firms and Wholesalers claim that they represent retail investor's best interest, but I do not agree. These firms do not represent me or my interest in the slightest, and in a free market IEX should be able to proceed with their proposed options exchange. This will create additional avenues for both retail and institutional investors, and help foster competition that in the long-run is healthy for free and open markets."); ("If any financial institution actually does represent the interests of retail investors, it's IEX who has historically fought to level the playing field between retail investors and financial institutions. I would like to express support for IEX's proposed options exchange."); (IEX's options "fosters trust and empowers retail traders to participate in options markets with enhanced confidence in price fairness.");

⁶ See, e.g., Letter from American Securities Association (ASA) (July 9, 2025) (commenting that "[l]atency arbitrage basically imposes a tax on firms that provide liquidity in both equity and options markets," and that it "harms American investors because it skews the market by allowing a limited number of firms, who can afford the extremely high costs of competing at a microsecond speed level, the ability to set prices."); Letter from Volant Trading (Aug 8, 2025) (strongly supporting IEX's proposal to implement a quoting protection mechanism designed to mitigate effects of latency arbitrage and describing the harmful effects of latency arbitrage on the markets whereby certain firms exploit a latency gap by executing against stale quotes before slower market makers can adjust their prices); Letter from Open to the Public Investing, Inc. (July 9, 2025) (commenting that latency arbitrage is "a practice in which traders exploit microsecond delays in market data dissemination to trade ahead of price updates, harming liquidity providers, widening spreads, and ultimately increasing costs for market participants including our retail customers."); Letter from Select Vantage (July 8, 2025) (explaining that latency arbitrage is a form of adverse selection and "Citadel knows very much what this concept is.");

investors.⁷ It is supported by a wide range of market makers, public advocates, and investors who would benefit from more competition.⁸

Moreover, the design of the ORP closely tracks the approach IEX has used with great success on its equities market, with its D-Limit order type. As with D-Limit, it is supported by data confirming that it is narrowly-tailored to provide protection in time increments, measured in microseconds, that add up to mere seconds during the trading day. It also draws on the experience of other options markets in providing similar risk tools for market makers that are offered and implemented by every options exchange.

Although options markets have experienced growth from increased demand, most notably from retail investors, this is not matched by an increase in competition among the market makers that are virtually the only suppliers of liquidity. In fact, the number of firms willing to compete has steadily fallen, and no new firms have emerged to replace them. The latest evidence of this ongoing trend is the exit of Morgan Stanley from the options market maker business and the acquisition of its key assets, including exchange specialist appointments, which are highly coveted and important to market maker volume, by Citadel.⁹

A recent industry report illustrates the trend, finding that retail trades rose to a new high of 561 million contracts in April 2025, with a few firms dominating this segment: “Within the level of retail engagement is the dominance of just a few firms, on both sides of the trade...Among wholesale market makers, Citadel Securities and IMC Financial Markets (via Dash Financial) executed roughly 65% of retail option volume for June 2025.”¹⁰ ORP is designed to support competition by protecting market makers from the latency arbitrage tax levied on liquidity providers. This tax depresses competition by adding to

⁷ Letter from HAP Trading LLC (June 16, 2025) (commenting that IEX’s protections protect market makers “from pure underlying security price latency arbitrage which can only be mitigated with the costliest latency solutions,” and noting that, “[i]n the current structure, customers may be absorbing most of the costs of such systems via worse execution outcomes”); Letter from Maven Securities (August 14, 2025) (commenting that their options market-making business has faced “tremendous complexity” and substantial costs “to combat predatory latency arbitrage,” and that the “impact of predatory latency arbitrage is significant and detrimental to the vast majority of liquidity providers who seek to make the markets more efficient for end users”).

⁸ See *supra* n. 5, see also, e.g., Letter from Open to the Public Investing, Inc. (July 9, 2025); Letter from Virtu Financial, Inc. (July 9, 2025); Letter from HAP Trading LLC (June 16, 2025); Letter from CTC, LLC (June 28, 2025); Letter from All Options USA LLC (June 12, 2025); Letter from Select Vantage (July 8, 2025); Letter from Better Markets (July 18, 2025); Letter from J.W. Verret (June 24, 2025); Letter from American Securities Association (ASA) (July 9, 2025); Letter from Volant Trading (Aug 8, 2025); Letter from Maven Securities (August 14, 2025).

⁹ See Bloomberg, “Citadel Securities Buys Morgan Stanley’s Options Market Maker, avail. at <https://www.bloomberg.com/news/articles/2025-07-10/citadel-securities-buys-morgan-stanley-s-options-market-maker>.

¹⁰ See [The Retail Option Landscape – Q2 2025 Update – Alphacution Research Conservatory](#).

market maker costs and erecting barriers to entry.¹¹ By reducing the impact of latency arbitrage, this tool is designed to enable market makers to quote better bid-offer prices in some cases and to encourage new firms to compete as market makers, as many such firms have commented they would seek to do. This will inure to the benefit of investors, who rely exclusively on market makers to supply liquidity in options.¹²

Response to Comments

The Purpose: Reducing the Latency Arbitrage Tax on Liquidity

Citadel does not address the impact of latency arbitrage on competition, or the state of competition in options markets, at all. Nor does it suggest a different way to address the problem. In its latest letter, it simply asserts that IEX has not clearly defined the term latency arbitrage and that the Commission should “demand clarity.” In our earlier response, IEX clearly defined the term by repeating word for word the description the Commission used in its D-Limit approval order. Nothing more is needed.¹³

Order Protection/Firm Quote

Citadel implausibly asserts that approving the ORP would amount to a “sweeping expansion” of the Order Protection Rule (“OPR”) in the options market and would take advantage of the Regulation NMS structure, while referencing Chairman Atkins’ call for a review of the OPR in equities and options.¹⁴

This is a gross and calculated distortion. Like the letters in “OPR” and “ORP”, Citadel’s thinking is inverted. A review of the Order Protection Rule and its unintended effects is important and needed, but that review has nothing to do with IEX’s ability to innovate to

¹¹ As market maker CTC commented, the technology expenses required to defend from latency arbitrage serve only to “protect professional market participants from one another (and, perhaps, to generate a profit for purveyors of market data and trading infrastructure), and do not fill any expressed need for additional speed on the part of end users.” Letter from Steve Crutchfield, CTC, LLC to Vanessa Countryman, SEC, dated June 28, 2025.

¹² Again, from CTC: “Increased risk of instantaneous adverse selection, and the increased infrastructure cost necessary to mitigate that risk, is therefore a direct cause of market makers’ quoting wider spreads and/or smaller size in order to generate sufficient risk-adjusted returns—thereby increasing costs for investors.” *Id.* See also Letter from Maven Securities (August 14, 2025) (“Our strong belief is that IEX’s innovative de minimis delay and ORP technology seems to accomplish all of these goals,” including supporting fair and orderly markets, greatly reducing the “technology tax” imposed by the presence of latency arbitrage, as well as improving the overall quality of markets by allowing market-makers to tighten spreads and increase the size of their quotes.).

¹³ First Response Letter, at 2-3.

¹⁴ See “SEC Announces Roundtable on Trade-Through Prohibitions”, <https://www.sec.gov/newsroom/press-releases/2025-99-sec-announces-roundtable-trade-through-prohibitions>.

address market needs *within the existing rule set* in ways that the Commission has clearly and repeatedly determined is permitted.

Judge Walker of the D.C. Circuit captured the point perfectly. During oral argument on the failed challenge to the D-Limit order type, he responded in this way to Citadel's counsel about its claim that IEX was seeking to interfere with market forces:

Your last statement about just letting the market forces, you know, play out, here that's what it seems IEX wants to do and it's you who is going to a federal agency and saying stop a private entity, IEX, from doing what they want to do. You're the one who's trying to kind of regulate your way into a market victory.¹⁵

As it attempted to do in challenging the D-Limit order type, Citadel misleadingly conflates the ORP with an earlier proposal by the Cboe EDGA Exchange to adopt a "one-sided" delay. But the Commission explained why that earlier proposal was fundamentally distinguishable for many reasons, and the court concurred.¹⁶ Every distinction that was identified in that context applies equally to the ORP.

Citadel argues that more than geographic latency, what matters is "what is permitted to happen during the delay." It suggests that allowing quotes to cancel during a delay would "thwart the pro-competitive transformation that has taken place in the U.S. options market over the last several decades." Leaving aside that richly ironic note, the SEC has been clear that how an exchange protects resting orders matters in assessing a proposal's overall merits, but that regulation does not bar an exchange from using technology in targeted ways, which may include use of a "de minimis" delay, to protect participants from the negative effects of latency arbitrage. Otherwise, the OPR would act as an absolute block to any efforts to offer such protection, which is exactly what Judge Walker meant about Citadel seeking to "regulate [its] way into a market victory."

Options Market Structure and Precedent

Differences in options markets provide even stronger reasons to reach the same conclusion in that context. As we observed in our first letter, because all options trading must be on exchange, there is no way for participants to avoid the latency arbitrage tax, and the vastly larger number of instruments means that market makers are even more

¹⁵ Transcript of Oral Argument at 48:7–12, *Citadel Sec. LLC v. SEC*, No. 20-1424 (D.C. Cir. Oct. 25, 2021) (Walker, J.). See also Letter from Select Vantage (July 8, 2025) (commenting that "Citadel has a right to make any arguments it wants to protect its business, but it takes a special kind of arrogance (and a certain lack of respect for your regulator) to repeatedly use the same playbook and count on the SEC to accept your alarmist rhetoric at face value").

¹⁶ The D.C. Circuit Court stated: "Further, the SEC noted that Cboe's proposed delay (1) was four milliseconds long; (2) applied asymmetrically to liquidity providers and takers; and (3) allowed liquidity providers to manually reprice their bids and offers during the entire trading day....IEX's delay, by contrast, (1) is 350 microseconds long, which is approximately 1/11th the length of Cboe's delay; (2) applies to incoming communications from both liquidity providers and takers; and (3) permits D-Limit orders to automatically reprice, which occurs during just 0.007% of the trading day." D.C. Circuit Decision, at 14.

susceptible to microsecond level price discrepancies across a larger quantity of active quotes.¹⁷

The same result is also buttressed by the extremely high rate of market maker cancellations that already exist and the fact that existing activity-based risk controls and “purge ports” can “atomically” cancel quotes across classes and series without notice to those seeking to access them.¹⁸ If IEX quotes were disqualified because of the ORP, it is hard to see how any market maker quotes on existing markets could be considered “protected” or “firm”.¹⁹

Citadel ties itself in knots trying to distinguish those examples by stating that a market maker loses “queue priority” when using a purge port to cancel but asserting that when triggered ORP would allow a firm to keep the same priority for execution. But Citadel is wrong on the facts and flawed in its logic. If an IEX quote using the ORP is repriced, it will be assigned a new timestamp and will have no advantage over other orders that are already resting at the new price. In addition, because IEX is proposing to operate as a “pro rata” market, a feature that is clearly disclosed in the Rule Filing, quotes for a given series at a single price level do not gain priority solely based on the time they arrive, so

¹⁷ See, e.g., Letter from CTC, LLC (June 28, 2025) (commenting that “market makers stand at continuous risk to the market, and—due to regulatory quoting obligations that typically require they post a very large number of bids and offers simultaneously—risk significant losses” from high frequency trading firms engaging in latency arbitrage); Letter from HAP Trading LLC (June 16, 2025) (commenting that market makers in the options markets are faced with “latency arbitrage which can only be mitigated with the costliest latency solutions” and that the “latency arbitrage mitigation tool IEX proposes would cause us to reinvest in our competitive quoting efforts, set more new NBBOs, and spend more time using our quote to augment available liquidity at the inside of the market.”); Letter from Maven Securities (August 14, 2025) (commenting that the options market-making business is made “much more complex and costly” due to the presence of predatory latency arbitrage and that market makers face “pickoffs” that “happen very frequently and in aggregate, are very costly.”).

¹⁸ Letter from Virtu Financial, Inc. (July 9, 2025) (noting that “similar to other options exchanges, IEX Options plans to offer market maker quote protection mechanisms” and “Virtu believes the ORP could be an innovative risk management and execution efficiency tool in the execution toolbox”); Letter from Volant Trading (Aug 8, 2025) (commenting that IEX’s ORP “offers a precise, event-driven tool that reduces the need for broad cancels and encourages market makers to quote more continuously and competitively,” and strongly urging the Commission to approve this proposal); Letter from HAP Trading LLC (June 16, 2025) (explaining that options market structure already allows for activity-based risk controls, such as exchanges will automatically purge all of a market maker’s quotes due to certain triggers and that “IEX’s design is simply an extension of this existing framework.”); Letter from Select Vantage (July 8, 2025) (commenting that “the options markets are full of risk tools exchanges use to automatically cancel orders based on a variety of settings controlled by the market maker” and that IEX’s proposal is “more narrowly drawn than other risk tools to focus specifically on risks from latency arbitrage.”)

¹⁹ See Letter from CTC, LLC (June 28, 2025) (commenting that the ORP is “far from novel or unprecedented,” describing the market maker risk controls (MMRCs) that already exist in the market that are comparable to the ORP, and stating that “[i]t is unclear how these existing MMRCs on a Nasdaq exchange, which the SEC has approved and which are to our knowledge currently offered in various flavors on all Nasdaq markets, ‘honor displayed quotes’ in some way that the IEX Proposal does not”).

Citadel's assertion that a quote affected by the ORP will be able to "jump the queue" over other orders is baseless.²⁰

Citadel also argues that an activity-based limit will not result in a cancel until trades have been executed but this is not the case for quotes using the ORP. Yet, all quotes subjected to the ORP, as is the case for activity-based risk controls, are readily accessible unless and until there is a triggering event. It is only after that triggering event, a quote instability determination in the case of the ORP or a risk breach in the case of activity-based risk controls, that any action is taken. Further, activity-based controls are essentially blunt instruments that can cancel all interest in all instruments in a class based on as little as a *single* execution in a given series. In contrast, ORP evaluates on a series-by-series basis and impacts an individual instrument. Finally, the potential for an IEX quote to cancel or reprice because of the ORP, which is governed by a disclosed mathematical formula, will be far more transparent than these other mechanisms, which depend on individual market maker decisions that are not publicly disclosed.²¹

There simply is no rational basis to conclude that all existing mechanisms exchanges offer market makers to manage their risks are permissible, but the ORP alone is prohibited.

Allegations about "Unfair Discrimination" Are Unfounded

Citadel generally argues that the availability of the ORP would "unfairly discriminate" in favor of market makers that use it, but Citadel is not always clear who it claims would be unfairly impacted. The argument that retail investors will be unfairly treated was rebutted in our first letter, is flatly contradicted by the record and the clear views expressed by market participants,²² and is addressed further below. To the extent Citadel argues that liquidity takers in general would be unfairly treated, the Commission addressed that point conclusively in its D-Limit decision:

²⁰ Rule Filing, at 35-36. IEX Proposed Rule 22.170, avail. at [Exhibit 5](#).

²¹ See Letter from J.W. Verret (June 24, 2025) (commenting that IEX's ORP operates on similar principles as the sophisticated risk tools on options exchanges currently in the market that allow market makers to automatically cancel or modify quotes but that IEX's ORP has "greater transparency and precision"); Letter from SelectVantage, note 19, *supra*.

²² See, e.g., Letter from Open to the Public Investing, Inc. (July 9, 2025) (commenting that IEX's proposed options innovations "offers several important benefits for retail investors," including tighter spreads and improved liquidity, and that the "data in IEX's filings show that these benefits should translate directly into better execution for retail options orders."); Letter from HAP Trading LLC (June 16, 2025) ("HAP believes the SEC has a unique opportunity to enable this innovation and substantially reduce costs for retail investors and we urge the Commission to approve the proposal without delay"); Letter from All Options USA LLC (June 12, 2025) (commenting that "IEX's proposal...would particularly benefit retail investors, who gain from more robust and transparent liquidity and are currently caught in an increasingly oligopolistic market structure, driven by PFOF and ultra-low latency strategies").

“Thus, the proposal is not unfairly discriminatory because it makes available a benefit that any liquidity provider can readily access and provides a narrowly focused protection that is calibrated to impact only the small number of liquidity takers that engage in latency arbitrage in order to incentivize liquidity providers to post orders for the benefit of all market participants.”²³

Many market makers have written detailed letters supporting the Options Proposal, including some that trade options today and some that would like to have a chance to compete in the future. These firms explain in their comments the headwinds they face and how they believe the Options Proposal could help promote tighter quotes or more liquidity at the best price.²⁴ They recognize that the ORP is designed to lower the barriers and costs that deter many potential participants from entering the market altogether, thereby broadening access and fostering stronger competition.

Citadel argues in its latest letter that, if the ORP was in fact similar to other risk controls, there is no reason market makers using the ORP would be able to provide better quotes than displayed on other exchanges. It goes on to say this would amount to providing a benefit to acting as a market maker without a corresponding obligation. This makes no sense. Market makers set their quotes based on competitive pressures and the need to compensate for the various risks involved in their business. Existing risk controls can enable them to quote at tighter spreads or in greater size than if those controls did not exist. If the ORP reduces costs from latency arbitrage that other controls do not address, those reduced costs can also affect the instruments market makers quote in and how they set their quotes. Market makers on IEX, like those on other markets, will be subject to various obligations concerning the scope and frequency of their quoting

²³ Securities Exchange Act Release No. 89686, 85 FR, 54438, 54449 (September 1, 2020) (“D-Limit Decision”).

²⁴ See, e.g., Letter from Volant Trading (Aug 8, 2025) (explaining their exit from market making in options due to escalating costs as well as the arms race associated with maintaining latency competitiveness and strongly urging the Commission to approve the IEX proposal that supports a competitive, transparent, and fair trading environment); Letter from HAP Trading LLC (June 16, 2025) (commenting that the largest market participants desire to “increase their internalization, reduce existing competition, and prevent new entrants,” and “[t]he latency arbitrage mitigation tool IEX proposes would cause us to reinvest in our competitive quoting efforts, set more new NBOs, and spend more time using our quote to augment available liquidity at the inside of the market.”); Letter from CTC, LLC (June 28, 2025) (commenting that the proposal is “explicitly pro-competitive,” will “empower many existing liquidity providers to quote more aggressively,” and “support/encourage more upstart competitors in the market making space”); Letter from All Options USA LLC (June 12, 2025) (commenting that IEX’s proposal “enables new market maker entrants like ourselves to participate on the basis of price competitiveness rather than speed alone.”); see also Letter from Maven Securities (August 14, 2025) (commenting that the pool of market-makers is “shrinking” and “the current trajectory is turning the noble aim of making markets more efficient into a monopoly of the few, reducing competition and ultimately harming end users,” whereby retail and institutional end users pay the price in the form of “wider than necessary spreads and less available liquidity”).

activity. There is no logic to the argument that they must be subject to some new obligations to justify using a tool to limit costs that are not controlled by other measures.

Arguments about “Fading Liquidity” Are Baseless

Citadel continues to claim that quotes on IEX under our proposal will “fade” against incoming orders, particularly from retail investors. But after three tries, it is clear that its claim is based on sheer and unfounded speculation.

At the outset, this claim ignores the reality that the vast majority of all market maker quotes are canceled before they are executed and that any quote can be canceled or repriced after an investor order has been sent. As the Commission explained in approving the D-Limit order, the “ability of any market participant to successfully execute against any particular displayed quote is subject to a number of factors and is not guaranteed on any market, as at any time any market participant can be seeking to execute against an order that is being repriced, changed, cancelled, or executed by a different market participant.”²⁵ Commenters directly refute Citadel’s claims about quote availability, explaining that in the options market, quotes are routinely updated by constant repricing and canceling (including bulk cancellations) after an order is sent to an exchange.²⁶

Moreover, IEX has provided an extensive analysis based on public options data estimating the ORP would not affect quotes during more than a miniscule proportion of the trading day. Consistent with the analysis used by the SEC and court in reviewing D-Limit, we estimate that, mathematically, it is extraordinarily unlikely an incoming investor order seeking to access a quote would fail to trade because the ORP has been triggered.²⁷

²⁵ D-Limit Decision, 85 FR at 54445.

²⁶ See Letter from CTC, LLC (June 28, 2025) (commenting that it has “always been true” that quotes can be canceled and repriced after an investor order is sent to an exchange and explaining that “[o]rders sent to every exchange are always at risk of missing the market due to a cancel that may already be in flight (but not yet reflected in the published quote), or an order update from another participant with a faster connection (for example, microwave vs. fiber) or a shorter geographical distance to the relevant exchange data center.”); Letter from Volant Trading (Aug 8, 2025) (commenting that critics arguing that the speed bump and quote protections will reduce quote availability “misrepresent the actual functioning of today’s markets” where quote availability “is already limited by frequent and necessary bulk cancellations” and market makers regularly cancel entire swaths of quotes” and noting that it is “also important to dispel the notion that quotes are always fully available across exchanges”); see also Letter from J.W. Verret (June 24, 2025) (commenting that “the concept of absolute quote firmness that opponents invoke simply does not reflect current practice in modern options markets” where “[o]ptions markets are characterized by ubiquitous repricing and canceling” with market makers employing sophisticated risk controls for significant quote updating that are widely used across exchanges); Letter from Joseph Michael Gugino (July 21, 2025) (commenting that the critics “ignore massive pieces of reality” as “[t]he notion of individual investors facing canceled quotes during fast-moving markets exists within the current options markets.”).

²⁷ Rule Filing, at 86-87. At times in its letter, Citadel describes its liquidity fading allegation by saying that IEX is providing a “last look” advantage. As IEX explained in response to the same comment on the D-

A basic understanding of the latencies involved in sending investor orders confirms this conclusion. For example, as Maven Trading commented, the minimum latency for an order sent by an investor through a broker's systems is orders of magnitude larger than IEX's speed bump: "The extra de minimis delay should have no impact on any real end user's orders, but it will have a big impact on the market maker's ability to provide liquidity."²⁸

To further validate our analysis, IEX reviewed the potential impact of the ORP for the last full week of July 2025 (July 21 through July 25), using the same parameters and assumptions used in the earlier evaluation.²⁹ For this latest analysis, we considered the potential impact for all classes and series relating to equities and ETFs for each day (over 5,500 classes and over 1,400,000 series). This latest review confirmed the Exchange's evaluation from February, as the July results were within all of the previously stated values for each category outlined in the February analysis.

Citadel's only response is that IEX *and the SEC* must provide different data, including how many quotes will be cancelled because of the ORP, how many incoming orders will be prevented from trading, and even what the effect of the ORP will be on market maker profitability. Citadel's claim that additional data that it prescribes must be produced by IEX or the Commission is unsupported by fact or precedent and untethered from market realities. Citadel well knows that projecting how many quotes would cancel would require assumptions about multiple unknowable variables, including how many and how often market makers use the ORP. In considering the claim that retail or other investor orders will fail to trade with a quote solely because the ORP has triggered, the relevant data and analysis is the data and analysis we have provided. And it is the same type of data and analysis the Commission and court used to conclude that D-Limit quotes would be accessible by investors.

Further, to our knowledge, no data of any type has ever been required to justify the use of existing risk controls. Finally, the transparency of the ORP formula means that market participants, if they choose, can analyze public data to make their own assessments.

Limit order type, a "last look" advantage allows individual participants to enter orders, receive a contra-side matching order, and then decide whether to accept or reject it. The operation of the ORP like D-Limit is based on an objective and transparent methodology managed by IEX, and participants have no discretion to decide whether to accept it. The comparison is baseless. See <https://www.sec.gov/comments/sr-iex-2019-15/sriex201915-6808882-208490.pdf>, at 4. See also Letter from Volant Trading (Aug 8, 2025) (commenting that IEX's ORP is "symmetric and nondiscretionary – it does not provide any special access or 'last look' privileges. Rather, it ensures that market participants are not executed at stale prices due solely to physical or processing delays outside their control.").

²⁸ Letter from James Cosentino, Maven Securities, to Vanessa Countryman, SEC, dated August 14, 2025; see also Letter from All Options USA LLC (June 12, 2025) ("Quotes are not fading. They are available to liquidity takers whose strategies are not focused on sniping option quotes" within microsecond moves in the underlying asset.).

²⁹ See Rule Filing, at 86.

This ability of others to offer data based on the transparency of the indicator was also a factor that the Commission cited in approving the D-Limit proposal.³⁰

Citadel's unsupported allegation is further contradicted by an understanding of market maker incentives. Retail orders in general are considered highly desirable to trade with, which is why options exchanges provide special mechanisms for firms to compete to trade with them. If the ORP in practice actually caused quotes to fade against retail orders, the argument would be moot, because *no one would use it*.

Citadel's claim is further undercut by IEX's experience with D-Limit, as extensively laid out in our previous response letter. Citadel responds by claiming that only market makers benefit from D-Limit and that IEX has increased its quote presence but not its share of trading. Once again, there is no basis for that assertion, and the evidence flatly contradicts it. Public data shows that IEX's trading volume has grown in tandem with its increase in quote presence. Specifically, IEX's trading volume resulting from displayed quotes grew more than ten times from July 2020, just before D-Limit was introduced, to an average of 243 million shares per day in July 2025.

Further, public data from the securities information processors ("SIPs") shows the proportion of revenue from quoting compared to trading that IEX earned, compared to the same ratio on other exchanges. Based on the most recent data for Q1 2025, IEX's quote-to-trade revenue ratio was right in line with the same ratio across all exchanges and all three SIP data tapes (volume-weighted) of approximately 2:1. The two exchanges with the highest ratio were LTSE, with a ratio of 69:1 across all three tapes, and NYSE Texas (formerly Chicago), at 14.5:1.³¹ In fact, seven exchanges had ratios substantially higher than IEX's, further exposing the falsity of this line of attack.

Citadel disputes, again without basis, data IEX supplied in its earlier response letter showing that its quotes remained equally accessible before and after D-Limit was introduced, purportedly relying on its experience in "sweeping top-of-book liquidity" across exchanges. But Citadel fails to provide any actual data to support its charge, or any information about how it routes to IEX, or to offer evidence that it cannot access IEX's liquidity on a par with other markets. In attacking D-Limit, Citadel made essentially

³⁰ D-Limit Decision, 85 F.R. at 54448. Citadel in passing asserts that the "speed of cancellation" is not a relevant metric because when the ORP is triggered, those quotes will be "gone forever" to subsequent orders, but this is also an empty critique. The data IEX has supplied concerns how long the ORP could be in effect, and it is this, not the speed of cancellation, that is relevant to quote access. Further, any quote that is canceled for any reason, including those that are canceled because of activity-based controls or purge ports, are obviously unavailable to liquidity seekers until the market maker replaces them.

³¹ Although SIP revenue is divided equally based on quotes and trades, the ratio for exchanges is skewed because all of the FINRA/TRF revenue, which is a substantial part of the revenue total, is based solely on transactions given that FINRA does not operate a trading market. See [Q1 2025 CTA Quarterly Revenue Disclosure.pdf](#); [UTP Revenue Disclosure Q12025.pdf](#).

the same unsupported argument about its purported difficulties in routing to IEX, which was discredited by the Commission and the D.C. Circuit Court.³²

Finally, but by no means least, Citadel's speculation is belied by comments from hundreds of individual investors who understand that lowering competitive barriers will serve their interests.³³

Procedural Arguments Lack Merit

Having come up short with other arguments, Citadel now resorts to arguing that the Commission should have required IEX to withdraw and resubmit its proposal as a new filing, with a new comment period. It bases this astonishing assertion on nothing more than its own view of the "materiality" and timing of the amendment and pure conjecture about whether commenters might have commented differently if the amendment had been filed at a different time.

This is another case where Citadel is spinning arguments based on standards that it is making up. There is nothing – absolutely nothing – in the statute, rules, or interpretations, that would require IEX to make a new rule filing, and self-regulatory organizations routinely file amendments with varying levels of impact without doing so. In Amendment No. 3, IEX clarified how certain variables of the ORP formula would be applied, committed to making rule filings to change any of these variables, and supplied other supplemental data.³⁴ For what it's worth, the substance of Citadel's comments in all three of its letters would not appear to hinge on any of those changes. The implication that Citadel in particular has not had a fair chance to weigh in on this proposal is not credible.

Conclusion

Markets thrive and are most dynamic when regulations do not predetermine economic winners and losers but instead allow innovation and competition to drive how markets evolve to best serve investors. IEX's Options Proposal squarely advances these goals. By providing a voluntary risk protection tool, IEX seeks to provide market makers with

³² D-Limit Decision, 85 FR at 54441.(finding techniques for accounting for geographic and technological latency differences among exchanges are "commonplace"); D.C. Circuit Decision at 11, note 5 and accompanying text.

³³ See *supra* n. 5.

³⁴ Citadel also advances an odd claim that the SEC somehow "reversed" an initial decision to require IEX to withdraw and resubmit its original filing, relying on nothing more than a transient screenshot Citadel captured from the SEC's website. In reality, Citadel appears to be referring to Amendment No. 2, which was withdrawn solely to correct a clerical pagination error and immediately replaced with Amendment No. 3. As the Commission's July 17, 2025 Order makes clear: "*On June 13, 2025, the Exchange filed Amendment No. 2 to the proposed rule change, which it withdrew to correct a non-substantive pagination issue and refiled as Amendment No. 3 on June 17, 2025.*" This routine administrative process is standard Commission practice.

an optional, narrowly-tailored tool to mitigate the harmful effects of latency arbitrage, a tax on liquidity that deters market-maker participation and ultimately increases costs for investors. We believe that offering this tool will help to increase competition and liquidity on our options market, benefitting investors and furthering the goals of the Exchange Act.

Citadel's objections, by contrast, rest on speculation and amount to an effort to stifle innovation to preserve its own commercial advantages. Its arguments don't withstand scrutiny or contradict the facts that IEX's Options Proposal is fully consistent with Commission precedent, has been validated by judicial decision and by actual experience on our equities market, is further justified by substantial data and analysis, and is supported by a wide range of market participants and investors who recognize its value in broadening competition and improving market quality.³⁵ Consistent with the general principle that market forces, not regulatory blocks or mandates, should guide market progress, we believe the basis for the Commission to allow our proposal to go forward is not just sufficient but is overwhelming.

Sincerely,



John Ramsay
Chief Market Policy Officer, IEX

³⁵ Letter from American Securities Association (ASA) (July 9, 2025) ("The strength of U.S. capital markets lies in the ability of private sector innovation, not regulatory micro-management. Markets must be allowed to evolve to meet the needs of all the participants that depend on them."); Letter from CTC, LLC (June 28, 2025) ("We respectfully submit that blocking the Proposal by regulatory mandate would be irreconcilable with the statement of then-Commissioner, now-Chair Atkins that, according to Congress, 'competitive forces, rather than unnecessary regulation' should guide market development"); Letter from Virtu Financial, Inc. (July 9, 2025) (commenting that IEX's proposal represents a well-intentioned effort to advance the objectives of enhancing transparency, fostering robust competition among market participants, and ensuring the high quality of the retail investor experience").

Admin Record No.

127

September 3, 2025

Vanessa Countryman
Secretary
U.S. Securities and Exchange Commission
100 F Street N.E.
Washington, D.C. 20549-1090

Re: **Investors Exchange LLC; Notice of Filing of a Proposed Rule Change to Adopt Rules to Govern the Trading of Options on the Exchange for a New Facility Called IEX Options; Release No. 34-102190; File No. SR-IEX-2025-02**

Dear Ms. Countryman,

The undersigned public pension plans, institutional investors, and fiduciaries are writing to support a proposal by Investors Exchange LLC (“IEX”) to introduce a new U.S. market for trading options (the “Options Proposal”). The undersigned are responsible for, among other assets, investing retirement funds on behalf of millions of individuals and their families in aggregate assets under management exceeding \$1.7 Trillion.

A key feature of the Options Proposal is an optional risk setting that market makers could use to help protect them from the costs of latency arbitrage strategies. When a market maker elects this protection, IEX would use its existing 350-microsecond “speed bump,” in conjunction with a transparent mathematical formula, to identify incremental moments of the day when individual market maker quotes have not yet been updated to reflect the latest equities prices, and either cancel or reprice those quotes at the election of the market maker. All other options exchanges provide risk tools that protect market makers from trading risks by automatically cancelling quotes in various circumstances. IEX’s proposal provides similar risk protection specifically targeted to latency arbitrage costs, but in a way that is more transparent than some of these other tools.

Technological and regulatory changes in the U.S. capital markets have created efficiencies, but they have also brought increasing complexity and unintended consequences. One such consequence is the ability of a small number of the fastest trading firms to leverage increasingly small speed advantages to determine execution outcomes. When liquidity providers are systematically “picked off” by firms with the most sophisticated technology, the costs act as a tax on liquidity. This naturally reduces the incentive of firms to act as liquidity providers, and can lead to those that remain to widen their bid-offer spreads. These added costs ultimately fall on investors.

IEX has innovated on behalf of investors in equities markets to counter the effects of latency arbitrage, by providing narrowly targeted protection to liquidity providers in the microsecond time increments when they are the most vulnerable to these strategies. This includes IEX’s D-Limit order type, which many other members of the institutional investor community have supported in

comment letters to the SEC. Experience with their use shows that these innovations have provided clear benefits, by offering superior execution quality for liquidity providers and a significant new source of displayed liquidity that is available to investors and other market participants.

Investors accessing the options markets rely almost exclusively on competition amongst market makers to supply liquidity, which can be extremely challenging given the high number of individual options instruments (up to 1.5 million) that may be quoted at any given time. As with equities markets, latency arbitrage strategies in the options markets can impose high costs on market makers, affect bid-offer spreads, and depress competition. In fact, the number of firms competing in the options markets has steadily declined, a point punctuated by the recent departure of one of the most well-capitalized financial services firms in this space.

In crafting its Options Proposal, IEX is following the proven path that it has used to combat latency arbitrage on its equities market. As with D-Limit, IEX would use the same delay and type of transparent, objective formula that would affect quotations in moments when they are the most vulnerable. IEX has also provided data showing that, as in the case of D-Limit, such protection would be selectively applied in moments that add up to less than 1% of the trading day.

The undersigned believe that the Options Proposal is a carefully tailored, positive innovation by IEX that will result in fairer and more efficient options markets by reducing barriers to entry and encouraging price competition. This innovation directly benefits investors by ensuring that the markets they rely upon place the investors' interests first. We urge the Commission to approve its adoption.

[Signatories Attached]

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-IEX-2025-02.

Signature: 

Name: Jaime Llano

Title: Managing Director - Head of Multi-Asset Trading

Firm: Teacher Retirement System of Texas

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-IEX-2025-02.

Signature: 

Name: Sam Masoudi_____

Title: Chief Investment Officer_____

Firm: Wyoming Retirement System_____

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-LEX-2025-02.

Signature: 

Name: Zachary Paris

Title: Head of US Equity Trading

Firm: Janus Henderson Investors

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-LEX-2025-02.

Signature: Michael C. Viteri

Name: Michael C. Viteri

Title: Chief Investment Officer

Firm: Arizona State Retirement System

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-IEX-2025-02.

Signature:

A handwritten signature in black ink, appearing to read 'Anthony W. Godonis', is written over a horizontal line.

Name: Anthony W. Godonis

Title: Principal, Director of Trading

Firm: Copeland Capital Management, LLC

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-IEX-2025-02.

Signature: 

Name: ADAM HJ CONN

Title: DIRECTOR, HEAD OF TRADING

Firm: BAILLIE GIFFORD OVERSEAS LTD

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-IEX-2025-02.

Signature: Myriam Deslandes

Name: Myriam Deslandes
Title: Vice-President, Strategy, Execution and Portfolio Solutions
Firm: La Caisse¹

¹La Caisse is both an equity shareholder in IEX Group and, through our brokers, a customer of its trading platform.

The following firm supports the approval by the Securities and Exchange Commission of a new IEX Options exchange as proposed by Investors Exchange LLC, including the use on that exchange of an optional risk tool for market makers to counter the effects of latency arbitrage. This endorsement is authorized by way of signature below, as an appendix to a comment letter submitted by a group of public pension plans, institutional investors, and fiduciaries, referencing File No. SR-LEX-2025-02.

Signature: _____

Name: Kevin Duggan

Title: Senior Managing Director, Global Trading and Beta

Firm: Ontario Teachers' Pension Plan

Admin Record No. 129



T.RowePrice

Filed Electronically

September 18, 2025

Ms. Vanessa Countryman
Secretary, U.S. Securities and Exchange Commission
100 F Street, NE Washington, DC 20549-1090

Re: Investors Exchange LLC; Notice of Filing of a Proposed Rule Change to Adopt Rules to Govern the Trading of Options on the Exchange for a New Facility Called IEX Options; Release No. 34-102190; File No. SR-IEX-2025-02

T. Rowe Price is a global investment management organization with \$1.73 trillion in assets under management.¹ We serve a wide range of clients, from individual savers to large institutions and funds. We appreciate the opportunity to comment on the above-referenced proposal (the "IEX Proposal") by Investors Exchange LLC ("IEX") to launch a listed options market named IEX Options. We applaud IEX's efforts to provide an innovative solution for trading listed options that facilitates market making, promotes price formation, and enhances liquidity provision. We recommend the SEC approve the rule changes in the IEX Proposal.

As proposed, IEX Options would utilize two mechanisms intended to mitigate latency arbitrage. Specifically, IEX Options would feature: (1) a 350-microsecond delay or "speed bump" on all incoming options order messages, and (2) an Options Risk Parameter (ORP) function that market makers could use to adjust their quotes under limited circumstances. The ORP would operate based on a formula that identifies when a market maker's quote is substantially mispriced compared to recent price changes in the underlying securities. Market makers electing to use the ORP choose whether the ORP cancels or reprices the quote to better align it with the underlying security's price. A key purpose of the speed bump is to provide IEX with sufficient time to process the vast amount of data from the equity markets that factor into the ORP's formula.

We support the IEX Proposal because we believe:

- It promotes competition and facilitates market making by reducing latency arbitrage, which should result in tighter spreads on quotes and increased frequency of quoting.
- The ORP is carefully designed to not impede the ability of investors like T. Rowe Price to access displayed quotes.
- The ORP complements the broader risk controls historically used by options exchanges that allow market makers to manage their exposures.

¹ As of August 31, 2025.

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SA93

Ms. Vanessa Countryman
 September 18, 2025
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More Competition and Market Making for the Benefit of Investors. In recent years, competition in options markets has declined and quoted spreads have widened. Undoubtedly, latency arbitrage and adverse selection risk contribute to this phenomenon. Market makers experience scenarios where they must race to reprice their displayed quotes to reflect changing market prices before certain market participants targeting misalignments “pick off” stale quotes. This environment discourages firms from acting as market makers and necessitates they spend significant amounts on the fastest technology and connectivity.

The negative impacts of latency arbitrage are well-documented in the US equity markets. By comparison, the US listed options market is even more vulnerable to these risks. The number of listed options series dwarfs the universe of listed equity securities, and options market makers are constantly trying to keep up with price changes in equities so that they can refresh their options quotes.²

The ORP’s ability to automatically re-price or cancel the market maker’s quote during these moments of dislocation should give market makers greater confidence their quotes will not be adversely selected. Like IEX and many other commenters, we believe the ORP’s risk reducing features will encourage more firms to act as market makers, tighten their spreads, increase their frequency of quoting, and quote for a broader range of options. Each of these shifts would benefit the funds and accounts using listed options that T. Rowe Price manages as a fiduciary on behalf of institutional and retail investors.

Continued Accessibility of Displayed Quotes. For background, T. Rowe Price and many other large asset managers submit orders through their brokers based on a fundamental need for liquidity or an investment decision to adjust exposures. This means we are not targeting a particular quote on an options exchange to execute with based on pricing misalignments with the underlying equity security. From this standpoint and as a practical matter, the ORP’s repricing or cancellation mechanism would not reduce our ability to access quotes.

In addition, IEX’s ORP is thoughtfully calibrated. IEX modeled data for a designated period and determined the ORP would on average per series apply only .001% of the time over such period.³ The IEX Proposal also gives IEX the ability to exclude certain options series from the ORP and IEX factors into its calculations how sensitive the option is to price changes in the underlying equity. In addition to these elements that fine tune the application of the ORP, modifications to the ORP’s variables would be subject to filing with the SEC as a rule change, which promotes sound governance, oversight, and continued transparency to market participants.

² In its June 19, 2025 letter, IEX states there are approximately 1.5 million listed options series compared to approximately 11,000 listed equity securities (see page 8).

³ See IEX’s Amendment No. 3 filed June 17, 2025 for additional details and information on related statistics (pages 97-98).

Ms. Vanessa Countryman
September 18, 2025
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Overall, we do not view the ORP as a form of “quote fading” and believe IEX’s approach will maintain the integrity and reliability of the NBBO, while improving market quality as previously discussed.

ORP Complements Broader Risk Controls Used by Options Exchanges and Market Makers. Given market makers’ unique obligations to affirmatively publish quotes, existing options exchanges utilize various risk controls to protect market makers. These regulatory-approved tools include “activity-based risk limits” and “purge ports.”⁴ These tools give market makers the ability to cancel quotes within a given class or series of options, or across multiple classes/series. IEX’s proposed options market would also feature these tools. Market makers have discretion in how and when they deploy these tools, and the circumstances under which they are used are typically not transparent to the broader market.

The ORP complements these tools because it is based on a transparent mathematical formula that provides predictability to the marketplace. The historical risk controls noted above are also a blunt instrument as compared to the narrowly tailored ORP. The ORP may impact how, and how often, these blunter risk controls are used. If market makers can better control latency arbitrage risks with the ORP, they may rely less on activity limits and other tools that result in cancellations of large numbers of quotes. If that is the case, market quality and resilience will be further improved.

We greatly appreciate the SEC considering our feedback and are happy to further discussion our views.

Sincerely,

/S/ Mehmet S. Kinak
Mehmet S. Kinak
Vice President & Global Head of Equity Trading for T. Rowe Price Associates, Inc.

/S/ Jonathan D. Siegel
Jonathan D. Siegel
Vice President & Managing Legal Counsel - Legislative & Regulatory Affairs

/S/ Tamara P. Wiggs
Tamara P. Wiggs
Vice President & Head of Trading for T. Rowe Price Investment Management, Inc.

⁴ These tools, among others, are further described in IEX’s letter dated June 19, 2025 (see page 9).